

Proposal for a typological and ecological classification of
watercourses in Austrian bioregion 14 (Grazer Feld and
Grabenland) based on their macrozoobenthos colonization

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Master's thesis

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Abstract

The EU Water Framework Directive (WFD) requires an examination of the ecological status of water bodies, including the biological quality element macrozoobenthos. However, the results of the assessment of the ecological status in the region "Grazer Feld und Grabenland" (ecoregion 14) within the framework of the "Ordinance on the Monitoring of the Quality of Water Bodies" are not reasonable.

The aim of this research is to examine the adaptation of the assessment method to ecoregion 14, as well as to analyze seasonal differences in the colonization of benthos and their influence on the assessment. Furthermore, the special type "intermittent stream" should be identified by means of indicator species.

In April 2016, ten morphologically near-natural sampling sites were defined as reference sites and examined according to the "Guidance on the monitoring of the biological quality elements, part A2 macrozoobenthos." To conduct the analyses, additional data from ecoregions 14 "Grazer Feld und Grabenland," 13 "Östliche Flach- und Hügelländer," and 3 "Bergrückenlandschaft" was used.

Ecoregion 14 has been identified as a distinct typological region, with seasonal differences in the structure of biological communities within the ecoregion. The designation of intermittent streams as a special type was conducted by means of an "indicator species analysis" using indicator species for temporary water bodies. Suitable metrics could be found for the ecological assessment of intermittent and perennial water bodies.

Taking into account the sampling time in spring, a recommendation for an adjustment of the assessment could be given. For these adaptations as well as the designation of reference sites, further investigations regarding the hydrology and physico-chemical conditions are needed.

In addition, this research contributed to the establishment of a reference database for genetic taxonomic aids (ABOL barcoding, eDNA) by sequencing the COI gene of selected species, for example rare indicator species for intermittent streams.

Summary

The EU Water Framework Directive (WFD) requires the ecological status of our waters to be determined, including the biological quality element macrozoobenthos. However, the results of the ecological status assessment under the Water Status Monitoring Ordinance (GZÜV) in the Graz field and grabenland (bioregion 14) are often implausible.

The aim of this work is to examine the adaptation of the assessment method to Bioregion 14 and to analyze seasonal differences in benthic colonization and their influence on the assessment to . Furthermore, , with the help of from indicator species of the special type "intermittent stream."

In April 2016, ten morphologically near-natural study sites (reference sites) were defined and investigated in accordance with the "Guidelines for the assessment of biological quality elements, Part A2 Macrozoobenthos." Data from bioregions 14 "Grazer Feld and Grabenland," 13 "Eastern flat and hilly country," and 3 "Mountain ridge landscape" were also used for the analyses.

Bioregion 14 was identified as a separate typological region, with seasonal differences in the community structure within the bioregion. Intermittent streams were identified as a special type using "indicator species analysis" based on indicator species for temporary water bodies. Suitable metrics were found for the ecological assessment of intermittent and perennial water bodies.

Taking into account the sampling time in spring, a recommendation for an adjustment of the assessment was made. Further investigations into the hydrology and physicochemical conditions are necessary for these adaptations and the designation of reference sites.

In addition, a contribution was made to the development of a reference database for genetic taxonomic tools (ABOL barcoding, eDNA) by sequencing (COI gene) of selected species, including rare indicators for intermittent streams.

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1 Introduction

1.1 Introduction

"The new Water Rights Act (WRG, as amended by Federal Law Gazette I No. 82/2003) requires, in accordance with the provisions of the EU Water Framework Directive (WFD, Directive 2000/60/EC of the European Parliament and of the Council), a holistic view of water systems, which is expressed in the term 'ecological status'." (QZV - Ecology OG, 2010).

In addition to the effects of material pollution, other anthropogenic interventions, especially hydromorphological ones (residual water, hydropeaks, morphology, impoundment and migration barriers or disruption of the continuum), which alter the function of water bodies as habitats, are also assessed (OFENBÖCK et al., 2016).

Within the framework of the Water Status Monitoring Ordinance (GZÜV BGBl. II No. 479/2006, as amended) for watercourses, the four biological quality elements fish, macrozoobenthos, phytobenthos, and macrophytes are assessed on the basis of the Quality Objectives Regulation for Surface Water Ecology (QZV-Ökologie OG, 2010) and, based on the differences between the actual status and the reference status, determine the ecological status of a water body.

According to the WFD, Directive 2000/60/EC, the typological classification of watercourses is based on the ecoregions according to ILLIES (1978). This coarse-scale division into European ecoregions was subdivided for implementation in Austria in accordance with the WFD into finer typological (MOOG et al. 2001) and biogeographical regions verified on the basis of benthic communities (SCHMIDT-KLOIBER 2002). Fifteen river bioregions were designated for Austria, one of which is bioregion 14, "Grazer Feld und Grabenland," which also includes the ten study sites analyzed in this study.

Within the framework of the Water Status Monitoring Ordinance (GZÜV) and the state's internal monitoring system, investigations were carried out in the "Grazer Feld und Grabenland" bioregion 14 between 2010 and 2015.

"Grazer Feld und Grabenland" were carried out between 2010 and 2015.

"This showed that the current typological classification of the ditchland streams for classification according to the specifications the EU Water Framework Directive and the associated

associated Austrian methodological regulations for determining ecological status, particularly with regard to the quality element macrozoobenthos, is often associated with difficulties in interpretation (plausibility check), so that a review of the current classification appears sensible" (Mag. Thomas Battisti, Province of Styria, Water Authority).

The question arises as to whether the current assessment system is appropriate for the Grabenland region and its characteristics, as the results are often implausible. Furthermore, there is a suspicion that seasonal differences in fauna in this bioregion have a significant impact on the saprobic index (SI) and multimetric index 1 and 2 (MMI1, MMI2) parameters. The aim of the investigations is to analyze the ditchland streams typologically in order to better interpret the results obtained from the GZÜV.

Furthermore, there is a possibility that water bodies in the Graz field and Grabenland may dry up partially or completely in summer. These intermittent water bodies would then be classified as special types and, according to OFENBÖCK et al. (2010), cannot be assessed using the current method. For a future ecological classification of intermittent streams in Austria, it is necessary to detect metrics that respond to intermittent water bodies in addition to faunal differences.

DNA barcoding is increasingly being used as an effective taxonomic tool for identifying species that are difficult to differentiate morphologically. This work aims to contribute to the development of a genetic reference database (ABOL), primarily for indicator species of intermittent water bodies in Austria.

1.2 Objective

Due to the issues mentioned in the introduction, ten morphologically natural routes were selected in 2016 and sampled on April 4 and 5. The aim of this sampling is to conduct a faunal analysis of the winter aspect (OFENBÖCK et al., 2010) in the Graz field and ditch landscape with regard to different water types, as well as to identify indicator species for intermittent streams and possibly new reference sites for the bioregion

14 "Grazer Feld and Grabenl a n d ." If the assumption that

intermittent water bodies based on indicator species is confirmed, this will lead to the designation of the special type "intermittent water bodies."

Furthermore, using multivariate analysis methods and a larger data set, including the ten study sites, the bioregion 14 "Grazer Feld and Grabenland" will be characterized using various metrics and analyzed with the surrounding bioregion 3

"Bergrückenlandschaft" and bioregion 13 "Östliche Flach- u. Hügelländer."

The taxonomic identification level is an essential part of assessing the ecological status of a water body based on the quality element macrozoobenthos. Since the identification of morphologically difficult-to-differentiate species is not always clear, a genetic analysis is also carried out to identify different taxa within the genera *Ecdyonurus* and *Siphonurus* using the taxonomic tool

"barcoding." The genetic "barcodes" are entered into the ABOL database and ultimately contribute to the development of a reference database for the assessment and monitoring of European water bodies using new genetic methods.

1.3 Research questions and objectives

1.3.1 WFD-relevant questions

- Can the Graz field and Grabenland be identified as a separate bioregion 14 on the basis of macrozoobenthos?
- Is the current assessment system for streams in the Graz field and Grabenland plausible, or does it need to be adapted to the specific conditions of this bioregion?
- Can the designation of watercourse landscapes and watercourse nature areas according to FINK et al. (2000) be used for a typological classification of Bioregion 14 based on macrozoobenthos?
- Are there seasonal differences in the community structure within bioregion 14 (Grazer Feld and Grabenland) due to hydrological differences?
- Can indicators for intermittent streams be identified?
- Is it possible to develop a data-based and plausible assessment methodology that includes the characteristics and typology of the streams in the Graz Field and Grabenland bioregion?

1.3.2 Objectives and questions of the genetic analyses

- Sequencing of indicator species of intermittent water bodies (*Isonychia dubia*, *Siphonurus armatus*) to create genetic "barcodes."
- Contributing to the development of a genetic reference database (ABOL barcoding, e-DNA) by sequencing (COI gene) a few selected species.
- Verify or falsify morphological determinations of species that are difficult to differentiate using the taxonomic tool "barcoding."
- In the case of *Ecdyonurus subalpinus*, are cryptic species subsumed due to different color morphs?

1.4 General definitions

1.4.1 Typology of Austrian watercourses

1.4.1.1 Ecoregions, watercourse habitats

Based on the European zoogeographical division of Limnofauna Europaea according to ILLIES (1978), in order to comply with the Water Framework Directive, the areas that were only sketchily marked on maps were precisely delineated and the boundaries of the aquatic ecoregions for Austria were specified (MOOG & OFENBÖCK 2000, MOOG et al. 2001). This resulted in Austria's shares in six ecoregions (**Figure 1**): Italy, the Alps, the Dinaric West Balkans, the Central Low Mountains, the Carpathians, and the Hungarian Plain. This "a priori" designation, based primarily on expert analyses of environmental factors, was further verified or falsified by NMS analyses using data from benthic biocoenoses (SCHMIDT-KLOIBER et al. 2001).

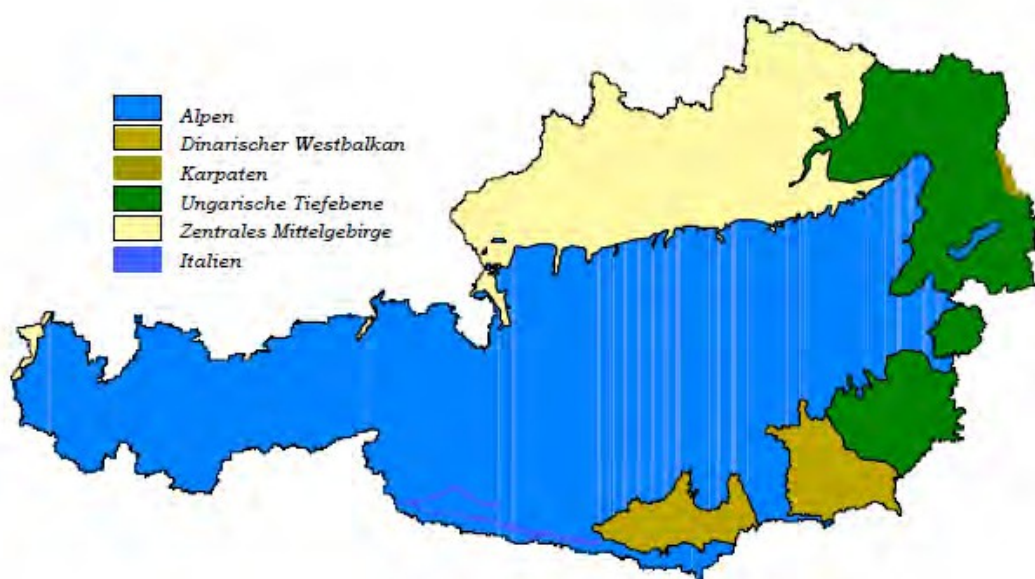


Figure 1: Zoogeographical ecoregions of Austria according to MOOG et al. (2001)

In order to comply with the WFD for Austria, the Austrian landscape was further divided into watercourse nature areas. This

This is an important preliminary step in classifying the diversity of watercourses in our federal territory. The ecoregions are too large for this purpose. Watercourse natural areas are landscape units that have typologically uniform elements in terms of watercourse ecological characteristics (FINK et al., 2000).

According to FINK et al. (2000), the following are watercourse ecological and typologically uniform environmental factors:

- Ecoregion
- Geological subsoil
- Climate: climatic factors, vegetation, soils
- Relief: Physiogeographical conditions, geomorphological aspects, altitude, landscape forms
- Hydrological characteristics
- Other sources according to local significance

The 40 designated watercourse nature areas (**Figure 2**) can be divided into six major areas according to FINK et al. (2000): Northern Alps (predominantly flysch and limestone), crystalline Central Alps, Southern Alps, foreland and marginal Alpine basins (Alpine foreland, Pannonian-Illyrian climate zone), granite and gneiss highlands, and inner Alpine basin landscapes.

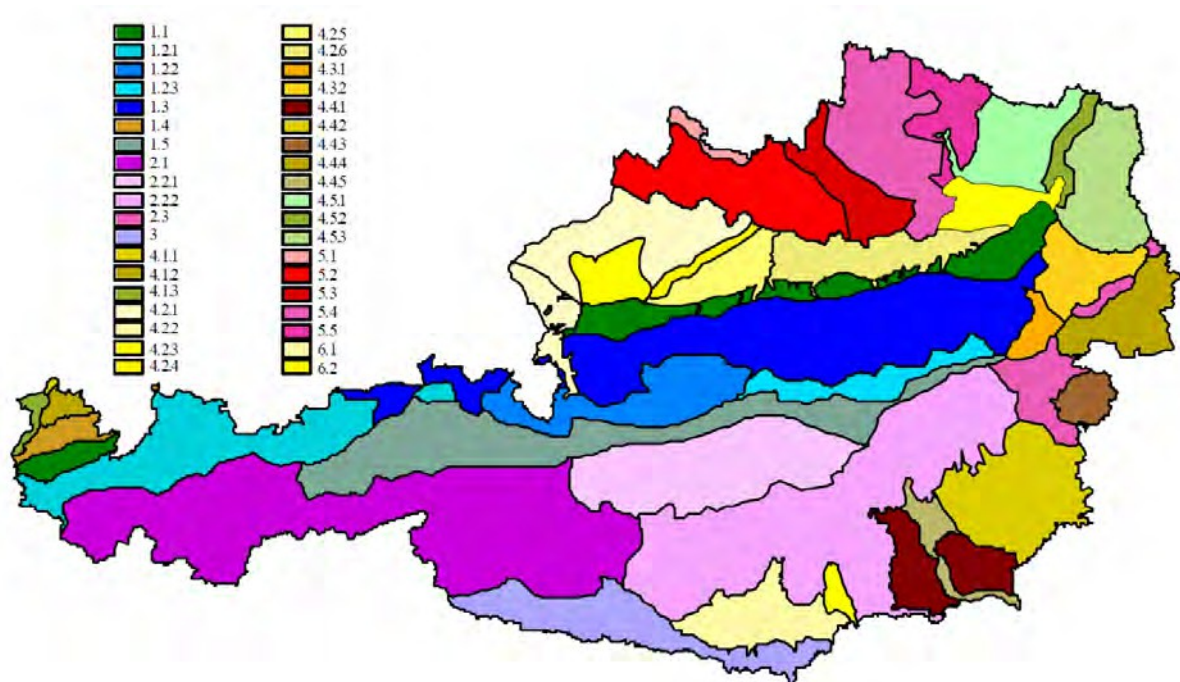


Figure 2: Natural watercourse areas in Austria according to FINK et al. (2000)

1.4.1.2 Basic types of watercourses and watercourse bioregions

In accordance with Annex II of the WFD, watercourses were classified based on purely abiotic, typological criteria (WIMMER et al., 2000 a). These are the zoogeographical regions according to ILLIES (1978), altitude, size based on the catchment area, and geology. Seventeen watercourse type regions and nine special types ("large rivers") were defined.

These abiotic components were then linked to the biotic component, the macrozoobenthos communities, and biocenotically similar watercourse natural areas or watercourse type regions were grouped together as "watercourse bioregions" (MOOG et al. 2001) (**Figure 3**).

3585 water body sections have been identified, but there is no suitable assessment method for all of these sections and for all biological quality elements due to a lack of data or excessive variability within the reference conditions (WIMMER et al., 2012).

Two sections with the special type "gorges and ravines" are currently listed as special types for Bioregion 14, Grazer Feld and Grabenland.

1.4.3 Reference conditions

The Water Framework Directive (WFD) sets a clearly defined objective for every water body in Austria, namely to achieve or maintain good ecological status, which is defined as a slight deviation from the reference condition typical for that water body. At the same time, further deterioration must be prevented (OFENBÖCK et al., 2016).

According to the Quality Objectives Ordinance - Ecology OG (2010), the reference value is the reference value for calculating the ecological quality ratio (EQR). The reference value is derived statistically from the range of values measured at the reference points of a water body type.

Reference conditions therefore exist when near-natural and water-type-specific conditions prevail in a water body.

1.4.4 Saprobic baseline condition

The saprobic basic condition (**Table 1**) is defined according to the (QZV - Ecology OG, 2010) as the reference condition for a water body type with regard to organic pollution. The saprobic basic conditions were assigned to the individual bioregions by STUBAUER (2002) in combination with altitude classes, catchment area size classes, and biological macrozoobenthos data.

In the Grazer Feld and Grabenland bioregion, the saprobic base condition in water bodies with a catchment area of more than 10 km² is 1.75. Since the current assessment of ecological status based on macrozoobenthos is only available for water bodies with a catchment area

>10 km² (OFENBÖCK et al., 2010), the saprobic baseline condition of 1.50 is not relevant for water bodies with a catchment area of less than 10 km².

Table 1: Basic saprobic conditions depending on bioregion, altitude, and catchment area from OFENBÖCK et al. (2010)

ÖKOREGION																		
ALPEN																		
ZENTRALES MITTELGEBIRGE																		
UNGARISCHE TIEFEbene																		
DINARISCHER WESTBALKAN																		
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15				
Vergletscherte Zentralalpen	Unvergletscherte Zentralalpen	Bergückenlandschaft und Ausläufer der Zentralalpen	Flysch oder Sandsteinvoralpen	Kalkvoralpen	Kalkhochalpen	Südalpen	Helvetikum	Alpine Molasse	Voralberger Alpenvorland	Bayerisch-Ostereichsches Alpenvorland	Granit- und Gneisgebiet der Böhmisches Masse	Östliche Flach- und Hügelländer der Ungarischen Tiefebene	Grazes Feld und Grabenland	Südliche Inneralpine Becken				
												Winter						
												Sommer (01.06-31.08)						

*bei hohem natürlichen organischen Anteil 1,75

1.4.5 Biotic and abiotic parameters

A metric is a biological measure (in the form of a characteristic value or index) used to describe communities that react clearly, directionally, and predictably to stressors (QZV Ecology OG, 2010). To calculate **biotic parameters** or core metrics (number of EPT taxa, total number of taxa, % oligochaeta & diptera taxa, etc.) and their further conversion into multi-metric indices, all taxa of the communities (biocenoses) of a water body are used (BARBOUR et al., 1999, OFENBÖCK et al., 2004). The catalog for the autecological classification of aquatic organisms in Austria, FAUNA AQUATICA AUSTRIACA (MOOG, Ed., 2017), serves as a basis. The catalog contains fauna lists of all taxa known in Austria and their classifications with regard to saprobity, biocenotic region, and feeding types.

Based on these classifications, it is possible to draw conclusions about the ecological functionality of a watercourse depending on the biocoenosis. According to the QZV Ökologie OG (2010), the ecological functionality of a watercourse is influenced by the living conditions in and around the water. These are water quality, discharge volume and discharge dynamics (hydrology) as well as structural features (morphology) (QZV Ökologie OG, 2010).

The classification of watercourse types according to WIMMER et al. (2000a) is based on the following **abiotic parameters**:

- Ecoregions (ILLIES, 1978)
- Watercourse natural environment (FINK et al., 2000)
- Geology
- Discharge regime (MADER et al., 1996)
- River order number (STRAHLER, 1957; WIMMER & MOOG, 1994)
- Catchment area (classes 1-4)
- Altitudes (classes 1-5)

1.4.6 Grazer Feld and Grabenland bioregion

The Grabenland, which together with the Grazer Feld forms Bioregion 14, lies in the Dinaric West Balkan Ecoregion and encompasses the two riverine natural areas "Weststeirisches Hügelland and Grabenland" and "Grazer Feld including Leibnitzer, Murecker and Radkersburger Feld," and surrounds the lower reaches of the Austrian Mur River (MOOG, 2001). The altitude of

Bioregion 14 is between 200 and 500 m, the geology consists of Tertiary and Quaternary sediments, and the drainage regime is pluvial (WIMMER et al., 2000a).

Four types of watercourses have been defined in the Graz Field and Grabenland bioregion (WIMMER et al., 2012). Watercourses with a percentage of less than 1% of the total length of all watercourses in bioregion 14 were excluded from the classification.

Table 2 shows the water body types (marked in green) in relation to bioregion, altitude class, and catchment area size (EZG size). In addition, the percentages of watercourse lengths, the saprobic basic conditions for macrozoobenthos (MZB) and phytozoobenthos, the trophic basic condition, and the fish regions (see legend in **Table 2**) are also shown.

Table 2: Types of watercourses in bioregion 14 – Graz field and Grabenland (WIMMER et al., 2012)

Seehöhe		EZG-Größe		TYP	0 5 10				
m	Klasse	km²	Klasse						
> 1.600	5	< 10	1						
		10–100	2						
		101–1.000	3						
		1.001–10.000	4						
800–1.599	4	< 10	1						
		10–100	2						
		101–1.000	3						
		1.001–10.000	4						
500–799	3	< 10	1		< 1 %				ER
		10–100	2		< 1 %				ER
		101–1.000	3		< 1 %				
		1.001–10.000	4		< 1 %				
200–499	2	< 10	1	14-2-1	50 %	1,50	II	me2	MR/ER/HR
		10–100	2	14-2-2	33 %	1,75	II	me2	HR/MR/EP
		101–1.000	3	14-2-3	10 %	1,75	II	me2	EP/HR
		1.001–10.000	4	14-2-4	6 %	1,75	II	me2	EP
< 200	1	< 10	1				II	me2	
		10–100	2		< 1 %	1,75	II	me2	EP
		101–1.000	3		< 1 %		II	me2	
		1.001–10.000	4		< 1 %		II	me2	

Legende:

-  prozentueller Anteil der Gewässerlänge am jeweiligen Typ, bezogen auf die Länge aller Fließgewässer der Bioregion
-  saprobieller Grundzustand Makrozoobenthos
-  saprobieller Grundzustand Phytozoobenthos
II = untere Hälfte Gewässergüteklasse II
-  trophischer Grundzustand
me2 = meso-eutroph 2
- Fischregion:

ER = Epirhithral

MR = Metarhithral

HR = Hyporhithral

EP = Epipotamal

2 Materials and methods

2.1 Sampling

Samples were taken by the Institute of Hydrobiology and Water Management (IHG), BOKU Vienna. The samplers were Thomas Huber and Roland Sumper. The chemical measurements were carried out by Patrick Leitner.

The sampling methodology is based on the "Guidelines for the assessment of biological quality elements Part A2 – Macrozoobenthos (OFENBÖCK et al., 2010). The sampling sites must be selected so that they are morphologically representative of a larger water body (at least 500 m). The 20 individual samples (sample units) according to the multi-habitat sampling method (MHS) (MOOG, 2004) must be taken in such a way that they reflect the percentage frequency of the habitats (**Table 3, Table 4** within the sampling site (**Figure 4**).

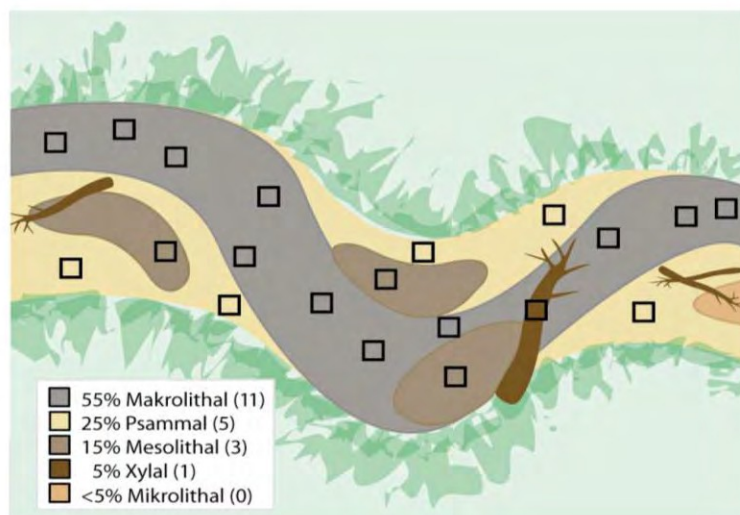


Figure 4: Conceptual representation of the distribution of individual samples (AQEM Consortium, 2002)

The hand net, with an edge length of 25 cm and a mesh size of 500 μm , is placed on the riverbed at the sampling site against the current and an area of 25x25 cm on the bottom is stirred up, swirled around, and collected. If possible, depending on the substrate, samples are taken up to a depth of 15-20 cm into the interstitial zone.

Table 3: Description of abiotic subhabitats (choriontopes) according to MOOG et al. (1999)**Abiotische Choriotope:**

Abkürzung	Substrat-bezeichnung	Verbale Beschreibung	Durchmesser
HYG	Hygropetrische Stellen	dünner Wasserfilm über steinigem Substrat	
MGL	Megalithal	Große Steine, Blöcke und anstehender Fels	> 40 cm
MAL	Makrolithal (Blöcke)	Grobes Blockwerk, etwa kopfgroße Steine bis maximal 40 cm Durchmesser vorherrschend mit variablen Anteilen von Steinen, Kies und Sand	20 - 40 cm
MSL	Mesolithal (Steine)	Faust- bis handgroße Steine mit variablem Kies- und Sandanteil	6,3 - 20 cm
MIL	Mikrolithal (Grobkies)	Grobkies (Taubenei- bis Kinderfaustgröße) mit Anteilen von Mittel- und Feinkies sowie Sand	2 - 6,3 cm
AKL	Akal (Kies)	Fein- und Mittelkies	0,2 - 2 cm
PSM	Psammal (Sand)	Sand	0,063 - 2 mm
PSP	Psammopelal	Sandiger Schlamm	
PEL	Pelal	Schlick, Schluff und Schlamm	< 0,063 mm
ARG	Argillal	Tonfraktion	

Table 4: Description of biotic subhabitats (choriontopes) according to MOOG et al. (1999)**Biotische Choriotope:**

Abkürzung	Substratbezeichnung	Verbale Beschreibung
PHY	Phytal	Aufwuchsalgen
FIL	fädige Algen	Algenbüschel, Fadenalgen, Algenwatten
MAK	Makrophyten	Submerse Wasserpflanzen, inkl. Moose und Characeen
LEB	lebende Pflanzenteile	Wurzelbärte, Ufergrasbüschel etc.
XYL	Xylal	Totholz, Baumstämme, Äste etc.
CPO	CPOM	Grobes partikuläres organisches Material, Fallaub
FPO	FPOM	Feines partikuläres organisches Material, Detritus
SPH	Abwasserbakterien	Abwasserbakterien, -pilze (<i>Sphaerotilus</i> , <i>Leptomitus</i>), Schwefelbakterien (<i>Beggiatoa</i> , <i>Thiothrix</i>)
SAP	Sapropel	Faulschlamm
SON	Sonstiges	Nicht beschriebene organische Habitate

Coarse organic and inorganic material (branches, stones, large leaves, etc.) should be carefully removed from the sample on site to reduce the amount of foliage in the laboratory. Furthermore, with regard to laboratory work, it is advisable to

"slurry" to remove unnecessary substrate (sand, gravel) from the sample.

Prevent large, rare, and protected animals from being fixed (formalin—final concentration 4% or 70% ethanol). To do this, use "pre-picking" to record these animals from the sample and return them to the watercourse.

After sampling, ensure that the nets and containers are thoroughly washed out and free of sediment and organic material. The collection equipment must be sterilized or dried thoroughly for several hours before reuse to prevent the spread of fungal pathogens (crayfish plague, *Phytophthora*, etc.).

2.2 Laboratory work

The sorting of organisms and rough identification at family or genus level was carried out by Roland Sumper. Further species identification was carried out by Wolfram Graf, Thomas Huber, Patrick Leitner, Berthold Janecek, and Ferdinand Sporka.

In accordance with the screening method, further identification ("post-sorting") is carried out in the laboratory at the identification level of the screening taxa list (OFENBÖCK et al., 2010 – Appendix, point 19.5) to complete the taxonomic lists.

In accordance with the detailed macrozoobenthos method (OFENBÖCK et al., 2010), it is possible to use partial sampling techniques, e.g., in the event of massive occurrence of organic material or organisms, and then extrapolate these results. An area-based partial sample consists of at least 5 of 30 cells (total sample) or as many cells as necessary to obtain at least 700 individuals. For the preparation and processing of partial samples, it is essential to follow the steps in (OFENBÖCK et al., 2010 – p. 36 12.2.1).

2.3 Indices and distributions

In the course of evaluating a water body according to the detailed method described in the guidelines (OFENBÖCK et al., 2010), the indices length-zonal distribution, nutrition types, and saprobic valences are also evaluated graphically using the program ©ECOPROF 1995-2013 - Version 4.0. The sum of the valences of the indices for biocenotic regions (longitudinal), saprobic status, and trophic types, plotted on the ordinate in a graph, always equals ten.

2.4 Ecological status assessment systems in Austria

"Macrozoobenthos can be used to detect material pollution as well as the effects of various stressors (degradation of water morphology, impoundment, residual water, use in the catchment area)" (WAGNER et al., 2010). The "screening method" and the

"Detailed MZB method" developed, with varying taxonomic determination effort and different assessment resolutions for ecological status. Detailed specifications and information can be found in the "Guidelines for the assessment of biological quality elements Part A2 – Macrozoobenthos" (OFENBÖCK et al., 2010). Both assessment methods are applied using the program ©ECOPROF 1995-2013 - Version 4.0.

2.4.1 Screening method

The organisms are identified in the field using the screening taxa list, which contains 286 screening taxa. Abundances are estimated using a 5-step scale according to ÖNORM M 6232 (MOOG et al. 1999). To complete the taxal lists, based on the identification level of the screening taxa list, the samples are fixed and packed in the laboratory according to the

"screening" in the field, the samples are fixed and packaged (OFENBÖCK et al., 2010).

The ecological status class is determined using the screening method based on two evaluation steps (**Figure 5**). The evaluation step "Screening – General pollution" includes the metrics "Number of screening taxa," "Number of sensitive taxa," and

"degradation score." The evaluation step "screening - organic pollution" includes the metric "saprobic score." From both evaluation steps, an internationally comparable "ecological quality ratio" (EQR) is determined using the "observed value" and the "expected value" corresponding to the type of watercourse. The "expected values" are based on limit values for "still very good status" or "still good status." According to the

"worst case" procedure, the ecological status class (very good ecological status, good ecological status or action required) is determined (OFENBÖCK et al., 2010).

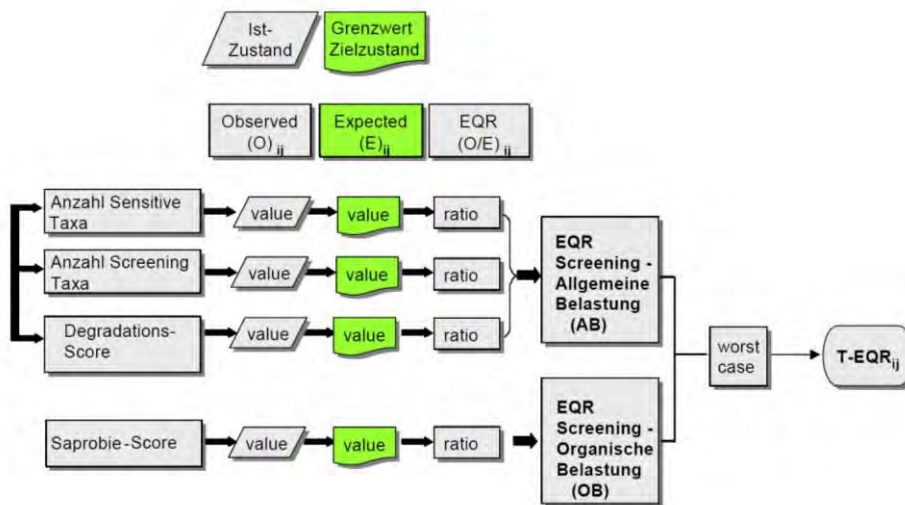


Figure 5: Flow chart of the procedure for the indicative assessment according to the screening method (OFENBÖCK et al., 2010)

2.4.2 Detailed MZB method

The detailed MZB method is based on precise taxonomic processing of the organisms, which is particularly important for the validity of the data collected and its evaluation. The determination must always be made down to species level; the minimum taxonomic requirements for the assessment follow ÖNORM M 6232 and the guideline for determining the saprobiological water quality of watercourses (MOOG et al. 1999).

For reasons of time and cost, partial sampling techniques are used in practice to identify organisms at the species level and then extrapolated. The rest of the sample is examined by means of "post-sorting" for taxa that did not occur in the partial sample to ensure that all taxa in the sample are recorded. Abundances are not estimated as in the screening method, but each individual is counted (OFENBÖCK et al., 2010).

The ecological status class of the detailed method is assessed based on the three modules saprobic, general degradation, and acidification (**Figure 6**). The calculated saprobic index or the multimetric indices (MMI) calculated from individual metrics are assigned to a status class and the ecological

status class (very good/high, good, moderate, poor, bad) (OFENBÖCK et al., 2010).

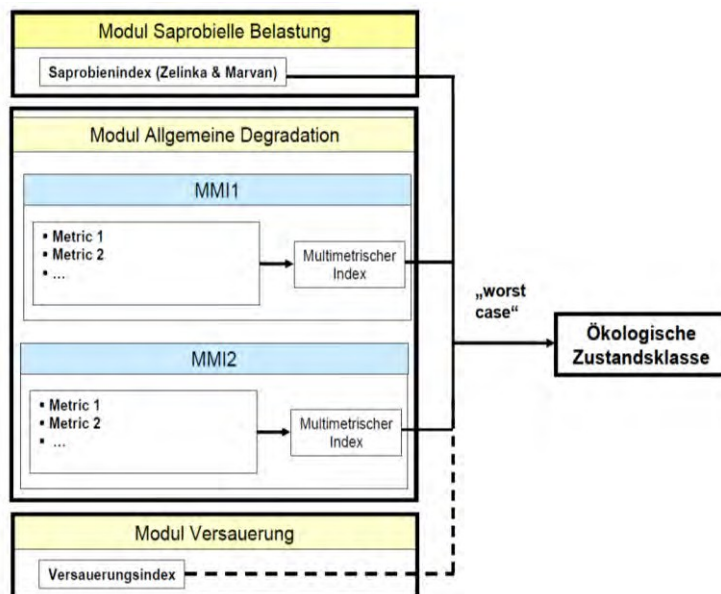


Figure 6: Determination of the ecological status class based on macrozoobenthos (OFENBÖCK et al., 2010)

2.4.2.1 River types

"Depending on the type of water body, different multimetric indices are used according to their relevance and significance" (OFENBÖCK et al., 2010, p. 62). Metrics and indices are applicable biological measures. Different combinations of multimetric indices are used depending on the type of water body. For bioregion 14 "Grazer Feld and Grabenland," the use of MMI1 is defined as primary and MMI6 as secondary (**Figure 7**).

	Degradationsindex	RETI	Gesamt taxa	EPT-Taxa	%EPT-Taxa	Litoralanteile	Litoral & Profundal Anteile	%Oligochaeta & Diptera Taxa	Regionsindex (LZI)	Diversitätsindex (Margalef)	Degradationsindex/ Gesamt taxa
MMI1	x	x	x	x		x		x		x	
MMI2	x		x	x						x	
MMI3	x		x	x		x		x			
MMI4	x		x	x							
MMI5	x	x			x				x		
MMI6	x			x	x	x					
MMI7				x	x	x					x
MMI8					x	x			x		x
MMI9	x				x	x			x		
MMI10											x
MMI11	x			x	x		x				

Figure 7: Multimetric indices and underlying metric combinations (OFENBÖCK et al., 2010)

The following metrics are used for the current assessment of bioregion 14 "Grazer Feld and Grabenland" (OFENBÖCK et al., 2010):

- Degradation index
- RETI (Rhithron nutrition type index)
- Total taxa
- EPT taxa
- % EPT taxa
- Littoral proportions
- % Oligochaeta & Diptera taxa
- Diversity index (Margalef)
- SI (Saprobic Index)

2.5 Data basis for the analyses and "best available sites"

A total of 65 data sets from the "Grazer Feld und Grabenland" bioregion were used for the MZB analyses, including the 10 hydromorphologically near-natural investigation sites from the 2016 sampling series. In addition, 133 selected data sets from bioregion 13 "Eastern flat and hilly lands" and 29 data sets from bioregion 3 "Ridge landscape" were analyzed (Appendix 8.5). All sampling sites are multi-habitat samples (AQEM Consortium, 2002).

In addition, a data set of the "best available sites" (bas) was analyzed, which excludes unsatisfactory and poor conditions.

2.6 Analysis of the zoenosis structure

The community structure is analyzed using **NMS analysis (non-metric multidimensional scaling)**. NMS analysis is a multivariate ordination. This means that objects are arranged in a coordinate system independently of synthetic axes, according to their natural distribution and their similarity (TER BRAAK, 1995). The data from multidimensional relationships is reduced to two dimensions. Individual points (benthoscenotic samples) that are closer together are more similar in their species composition, while points that are more dissimilar in their species composition are further apart.

With the help of **cluster analysis**, data sets (macrozoobenthos communities) are represented as a dendrogram and grouped into clusters based on their similarities. The similarity within a cluster is high. The similarities or dissimilarities are calculated exclusively on the basis of the species composition and abundance of the respective sample (GERRITSEN et al., 2000), without taking into account the ecological functions and interactions of the taxa (BOHLE, 2001).

The NMS analyses and cluster analysis are performed using the PCORD 5 program (MCCUNE & MEFFORD, 2006). The NMS setup consists of the following parameter settings

"Distance Measure=Sorensen (Bray-Curtis)", "Number of axes k =2 or 3", "Number of runs with real data=50", "Stability criterion=0.00001", "Iterations to evaluate stability=15", "Maximum number of iterations=250" and "Initial step length=.20". The cluster setup consists of the parameter settings "Distance Measure=Sorensen (Bray-Curtis)", "Group Linkage Method=Flexible Beta", "Dendrogram Width=Narrow Width", "Dendrogram Spacing=Single-spaced", "beta= -,25"

2.7 Indicator Species Analyses

This purely mathematical analysis method is performed using the PCORD 5 program (MCCUNE & MEFFORD, 2006). The ISA identifies indicator species for a group, e.g., a bioregion or cluster. A species is considered an indicator species if it is significantly represented in a group and occurs in the majority of the study sites within that group (DUFRENE & LEGENDRE, 1997). Indicator species are identified at an indicator value of 25 or higher.

2.8 Analysis of relevant metrics at different sampling dates (summer/winter – aspect or seasons)

The analyses of the metric values, which were calculated using the program ©ECOPROF 1995-2013 - Version 4.0, are performed using the program "IBM SPSS Statistics 25" with the non-parametric Kruskal-Wallis test, 1-way ANOVA (k samples). Post-hoc pairwise tests are performed with the Mann-Whitney U (2 samples) test.

Both tests compare whether the distributions of variables (metrics, indices) between groups (ecological status classes) are significantly ($p \leq 0.05$, Bonferroni-corrected) different or equally distributed.

2.9 Phylogenetic analysis and creation of "barcodes" for selected taxa of bioregion 14

The taxonomic tool DNA barcoding (HEBERT et al., 2003, 2005) is an approach that uses sequencing of short genetic markers, mitochondrial cytochrome c oxidase subunit 1 (COI) genes, and tissue samples to assign taxonomic categories, e.g., species.

The following morphologically determined taxa were included in the genetic analysis:

- | | |
|--|------------------------------|
| • 11 individuals of <i>Ecdyonurus cf. subalpinus</i> | (Heptageniidae), Wellingbach |
| • 20 individuals of <i>Ecdyonurus cf. subalpinus</i> | (Heptageniidae), Vocherabach |
| • 01 individuals of <i>Ecdyonurus torrentis</i> | (Heptageniidae), Vocherabach |
| • 05 individuals of <i>Siphonurus cf. armatus</i> | (Siphonuridae), Pfaffenbach |
| • 03 individuals of <i>Siphonurus cf. aestivalis</i> | (Siphonuridae), Pfaffenbach |
| • 08 individuals of <i>Ironoquia dubia</i> | (Limnephilidae), Pfaffenbach |

DNA extraction was performed according to the modified "Multichannel Ammonium Acetate Extraction" protocol by SAMBROOK, FRITSCH, & MANIATIS (1989). The amplified protein-coding gene is cytochrome c oxidase subunit I (COI). LCO1490 (-GGTCAACAAATCATAAAGATATTGG) was used as the forward primer (and) and as the reverse primer (-)(HCO2198 (-TAAACTTCAGGGTGACCAAAAAATCA-) according to FOLMER et al. (1994). The conditions for the polymerase chain reaction (PCR) were taken from GATTOLLIAT, JEAN-LUC, et al. (2015) and are as follows: 5 min. at 94 °C, then 30 sec. at 94 °C, 30 sec. at 48 °C (changed to 49 °C), and 60 sec. at 72 °C (40 cycles) as well as a final elongation phase of 10 min. at 72 °C. The amplified PCR products (7 µl) are amplified in a process with ExoSap-IT enzyme (Amersham Biosciences) (0.7 µl) and distilled H₂O (1.4 µl) at 37 °C for an incubation period

from 45 min. and a second step (inactivation of the enzyme) at 80 °C for 15 min. The end products are purified using the "BigDye Terminator v3.1 Cycle Sequencing Kit" (Applied Biosystems) (0.25 µl), "5x Buffer" (2 µl), distilled H₂O (5.625 µl) and forward or reverse primers (0.125 µl) under the following conditions: 94 °C for 5 min. followed by 32 cycles at 94 °C for 10 sec., 50 °C for 5 sec., 60 °C for 4 min. and a final extension phase of 7 min. at 60 °C. Finally, the sequencing products are purified again with Sephadex G-50 (Amersham Biosciences) and visualized using an ABI 3130xl Genetic Analyzer (Applied Biosystems).

The sequences are edited using the MEGA6 program (TAMURA et al., 2013) and aligned. Neighbor-joining trees based on p-distances (sequence differences in percent) are created to represent evolutionary relationships.

Types of color morphs within the species *Ecdyonurus cf. subalpinus*

 Type red	Wellingbach: <i>E. cf. subalpinus</i> 22 (photo) Vocherabach: <i>E. cf. subalpinus</i> 02 <i>E. cf. subalpinus</i> 05	 <u>Dark type</u>	Wellingbach: <i>E. cf. subalpinus</i> 24 (photo) remaining individuals Vocherabach: remaining individuals
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2.10 Study area

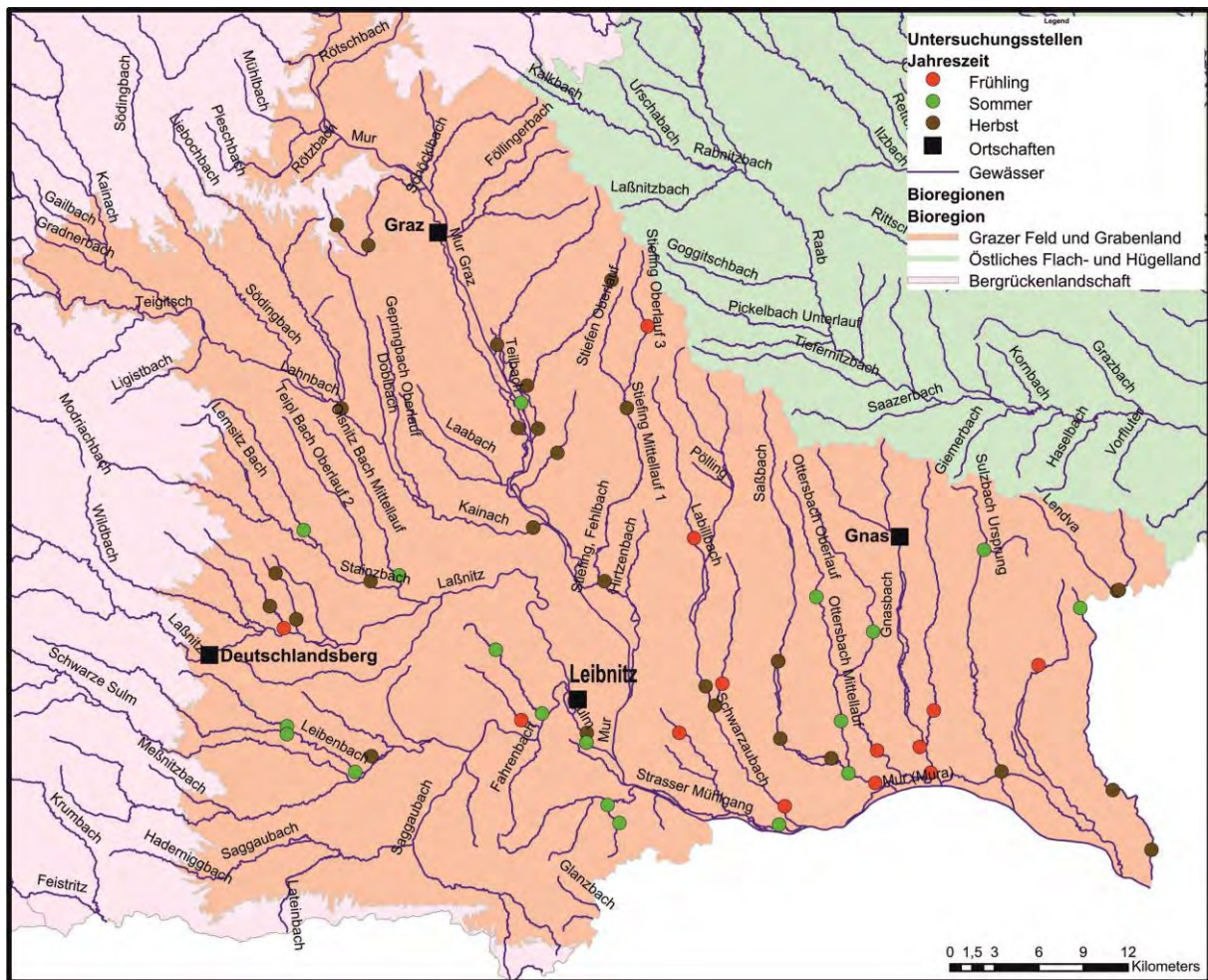


Figure 8: The study area "Grazer Feld and Grabenland" with the neighboring bioregions "Eastern flat and hilly country" and "Mountain ridge landscape"; red dots = study sites in spring; green dots = study sites in summer; brown dots = study sites in autumn

The Grabenland, which together with the Grazer Feld forms bioregion 14, is located in the Dinaric West Balkan ecoregion, comprises the two watercourse nature areas "Weststeirisches Hügelland und Grabenland" and "Grazer Feld including Leibnitzer, Murecker und Radkersburger Feld" and surrounds the lower reaches of the Austrian Mur river (MOOG, 2001 & SCHMIDT-KLOIBER, 2002). The altitude of Bioregion 14 is between 200 and 500 m, the geology consists of Tertiary and Quaternary sediments, and the drainage regime is pluvial (WIMMER et al., 2000a).

The area surrounding the streams in the Graz field and Grabenland is heavily dominated by agricultural land (pastures, cornfields) with scattered small wooded areas along the watercourses. In some places, there is riparian vegetation with at least one row of trees.

Table 5: Overview of the 10 sampling sites in the 2016 sampling series in the Graz Field and Grabenland bioregion with sampling date and type; MHS=Multi Habitat Sampling

No	Water	Coordinates	Coordinates	Date	Sampling
1	Lieberbach	46°47'20.13	15°40'2.80"E	06.04.2016	MHS
2	Glauningbach	46°44'41.47	15°48'6.70"E	06.04.2016	MHS
3	Labillbach	46°52'38.00	15°38'45.14"E	05.04.2016	MHS
4	Poppendorfer Bach	46°46'5.12"N	15°51'10.46"E	06.04.2016	MHS
5	Gnasbach (old arm)	46°44'47.66"N	15°50'26.61"E	06.04.2016	MHS
6	Aubachl (Fruttnerbach)	46°47'37.30	15°56'51.92"E	06.04.2016	MHS
7	Vocherabach	46°49'45.78	15°16'53.29"E	05.04.2016	MHS
8	Wellingbach	46°46'9.32"N	15°29'26.55"E	05.04.2016	MHS
9	Pfaffenbach	46°45'34.30	15°37'35.36"E	05.04.2016	MHS
10	Stiefing	47° 0'24.93"N	15°36'36.83"E	05.04.2016	MHS

2.11 Test sites for the 2016 sampling series

2.11.1 Lieberbach

The Lieberbach sampling site (46°47'20.13"N, 15°40'2.80"E) was sampled on a sunny and dry day on April 6, 2016. The Lieberbach is a meandering watercourse with a catchment area of 15 km², an altitude of ~290 m and a river order number of 2 (WIMMER & MOOG, 1994). The representative sampling site is located approx. 800 m east of the village of Hütt above the road bridge in the municipality of Sankt Veit am Vogau. Numerous rootstocks, dead wood, and accumulations of coarse particulate organic matter (CPOM) are present in places. There is at least one row of trees on both sides of the banks. The right bank is dominated by agriculture and the left bank by forestry. The study site is located in the middle of a forest area, so there is a high degree of shade.



Source: Amap

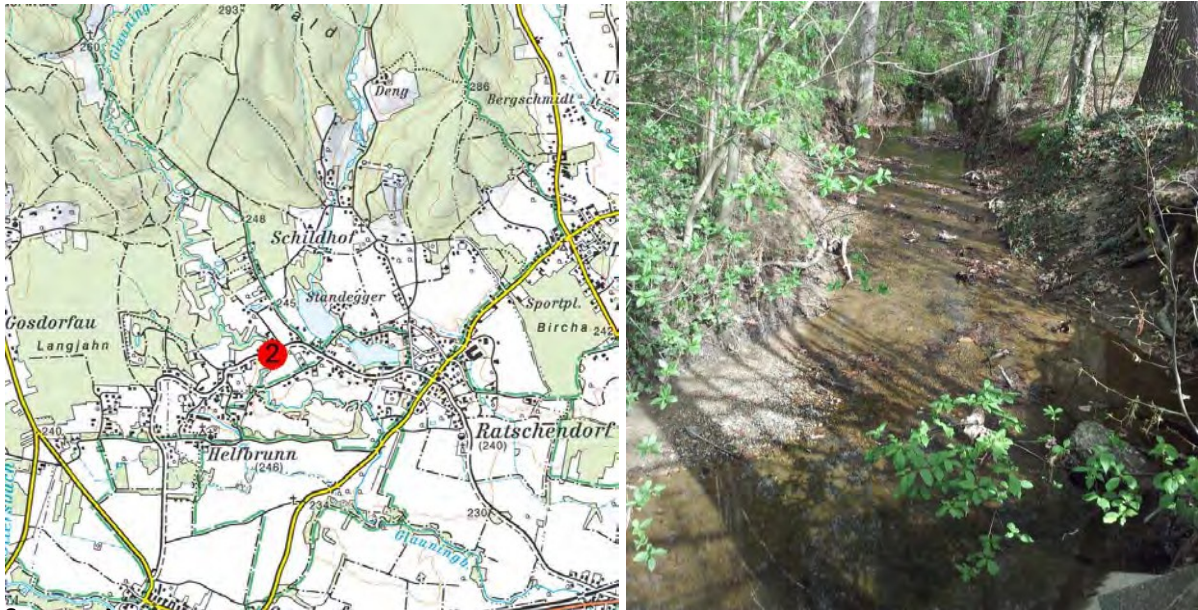
Figure 9: Investigation site 1, Lieberbach, photo: Patrick Leitner, BOKU

Table 6: Percentage distribution of choriotores and abiotic parameters of the Lieberbach stream on April 6, 2016

Pelal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		20	5	30	45	
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s] [m/s]	Average flow velocity [m/s]	Average water depth [m]
12.2	8.46	397	84.4	0.6	0.1	0

2.11.2 Glauningbach

The Glauningbach was sampled on a dry and sunny day on April 6, 2016. The sampling site (46°44'41.47"N, 15°48'6.70"E) is located approximately one kilometer west of the village of Ratschendorf in the municipality of Ratschendorf. The watercourse is meandering with straight sections. The sampling site is located at an altitude of 240 m above sea level with a catchment area of 11 km² and a river order number of 2 (WIMMER & MOOG, 1994). In addition to root beards and coarse particulate organic matter (CPOM), the section has natural riparian vegetation with at least one row of trees, which means that the proportion of shade is high. As the study site is located in a residential area, the surrounding land is mainly used for meadows and, in some cases, arable land.



Source: Amap

Figure 10: Study site 2, Glauningbach, photo: Patrick Leitner, BOKU

Table 7: Percentage distribution of choriotopes and abiotic parameters of the Glauningbach stream on April 6, 2016

Pelal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		20	5	30	45	
Temperature [°C]	pH	Conductivity [μS]	O2 saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s]	Average water depth [m]
12.9	8.2	189	94.5	0.3	0.1	0.15

2.11.3 Labillbach

The investigation site on the Labillbach (46°52'38.00"N, 15°38'45.14"E) is located approx. 230 m east of the village of Unterlabill in the municipality of Mitterlabill at an altitude of 300 m, with a catchment area of 25 km² and a river order number of 3 (WIMMER & MOOG, 1994). It was sampled on April 5, 2016, in dry and sunny weather. The long and deeply incised stream is surrounded by steep, natural banks with numerous washed-out rootstocks and scattered root beards. Except for the 1 to 2 rows of riparian woodland, the surrounding area is heavily dominated by agricultural land.



Source: Amap

Figure 11: Study site 3, Labillbach, photo: Patrick Leitner, BOKU

Table 8: Percentage distribution of choriotores and abiotic parameters of the Labillbach stream on April 5, 2016

Pelal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		75	20		5	
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s]	Average water depth [m]
15.2	7.6	410	105	0.6	0.15	0

2.11.4 Poppendorfer Bach

The Poppendorfer Bach is a pendulous watercourse with a catchment area of 30 km² and a river order number of 2 (WIMMER & MOOG, 1994). The sampling site (46°46'5.12"N, 15°51'10.46"E) was sampled on April 6, 2016, on a sunny day, at an altitude of 240 m and 550 m west of the village of Hart-Unterberg in the municipality of Hart. The stream bed is surrounded by steep, natural banks, with relatively little shade due to the fact that the trees are mostly single-row stands. In addition to dead wood and root beards, coarse particulate organic matter (CPOM) is also present. The surrounding land is used exclusively for agriculture.



Source: Amap

Figure 12: Study site 4, Poppendorfer Bach, photo: Patrick Leitner, BOKU

Table 9: Percentage distribution of choriotopes and abiotic parameters of the Poppendorfer Bach on April 6, 2016

Pelal	Psammo-pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
	25	5	25	45		
Temperature [°C]	pH	Conductivity [µS]	O2 saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s]	Average water depth [m]
13.8	8.29	669	104	0.7	0.3	0

2.11.5 Gnasbach - Old arm

The Gnasbach oxbow lake was sampled on April 6, 2016, in sunny weather near the Peterquelle company. The sampling site (46°44'47.66"N, 15°50'26.61"E) is located 1 km southeast of the village of Deutsch Goritz in the municipality of Deutsch-Goritz at an altitude of 230 m and has a river order number of 4 (WIMMER & MOOG, 1994). The banks of the long stream are close to their natural state and there is a lot of dead wood and coarse particulate organic matter (CPOM). The surrounding land is mainly used for agriculture and as meadows. As the Gnasbach oxbow lake had the hydromorphological characteristics of a standing water body on the day of the investigation, it cannot really be classified as a watercourse.



Source: Amap

Figure 13: Study site 5, Gnasbach (old arm), photo: Patrick Leitner, BOKU

Table 10: Percentage choriotope distribution and abiotic parameters of the Gnasbach (old arm) on April 6, 2016

Pelal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
100						
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s]	Average water depth [m]
17.6	7.5	641	23.8	~ 0	~ 0	0.05

2.11.6 Aubachl (Fruttnerbach)

The Aubachl study site (46°47'37.30"N, 15°56'51.92"E) is located 550 m northwest of the village of Pichla near Radkersburg in the municipality of Tieschen at an altitude of 250 m, with a catchment area of 10 km² and a river order number of 1 (WIMMER & MOOG, 1994). The meandering stream was sampled on April 6, 2016, on a sunny day. The banks are near-natural, but without continuous tree cover, although there are numerous root beards. With the exception of a small wooded area, the surrounding area is largely characterized by agricultural land.



Source: Amap

Figure 14: Study site 6, Aubachl (Fruttnerbach), photo: Patrick Leitner, BOKU

Table 11: Percentage distribution of choriotores and abiotic parameters of the Aubachl (Fruttnerbach) on April 6, 2016

Pelal	Psammo-pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		30	10	50	10	
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s]	Average water depth [m]
15.4	7.7	636	103.6	0.7	0	0.15

2.11.7 Vocherabach

The Vocherabach was sampled on a sunny and hazy (Sahara dust) day on April 5, 2016. The sampling site (46°49'45.78 N, 15°16'53.29"E) is located at an altitude of 318 m north of the village of Vochera on the Laßnitz in the municipality of Groß Sankt Florian. The pendulous watercourse drains a catchment area of 14 km² and has a river order number of 2 at the sampling site (WIMMER & MOOG, 1994). The Vocherabach has natural banks, but in places there is only a single row of trees on one side, which has a negative effect on shading. In some places, boulders have been used to stabilize the banks, but these do not affect the overall morphology of the stream. Root beards, moss, and

algae growth on the substrate. The surrounding land is used exclusively for agriculture.



Source: Amap

Figure 15: Study site 7, Vocherabach, photo: Patrick Leitner, BOKU

Table 12: Percentage distribution of choriotores and abiotic parameters of the Vocherabach stream on April 5, 2016

Pelal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		30			60	10
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s] [m/s]	Average flow velocity [m/s]	Average water depth [m]
12.1	8.2	236	92.1	0.8	0.15	0.2

2.11.8 Wellingbach

The Wellingbach was sampled on April 5, 2016, in sunny weather approximately 600 meters north of the village of Heimschuh in the municipality of Heimschuh. At the sampling site (46°46'9.32"N, 15°29'26.55"E) at an altitude of 280 meters, the Wellingbach has a catchment area of 10 km² and a river order number of 3 (WIMMER & MOOG, 1994). The banks are near-natural with mostly at least one row of riparian vegetation on both sides. The stream bed is oscillating with structures such as root beards, dead wood, and coarse particulate organic material

(CPOM). The surrounding land is heavily used for agriculture, although the study site is located in a small wooded area, which means that there is also a high degree of shade.



Source: Amap

Figure 16: Study site 8, Wellingbach, photo: Patrick Leitner, BOKU

Table 13: Percentage distribution of choriotypes and abiotic parameters of the Wellingbach stream on April 5, 2016

Pelal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		5	10	80	5	
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s] [m/s]	Average water depth [m]
12.8	7.89	364	102.6	0.6	0.25	0

2.11.9 Pfaffenbach

The sampling site (46°45'34.30"N, 15°37'35.36"E) is located between the villages of Rabenhof and Wagendorf in the municipality of Sankt Veit am Vogau at an altitude of 270 m and was sampled on April 5, 2016. The catchment area is approx. 10 km⁽²⁾ and the Pfaffenbach has a river order number of 2 (WIMMER & MOOG, 1994). The banks are partly steep and natural with riparian vegetation. The meandering stream has structures such as

Root beards, dead wood, mosses (*Fontinalis* sp.), algae growth, and coarse particulate organic matter (CPOM) are present. The surrounding land is used for forestry and agriculture. The study site is located in a young spruce forest, so the amount of shade is approximately 50%.



Source: Amap

Figure 17: Study site 9, Pfaffenbach, photo: Patrick Leitner, BOKU

Table 14: Percentage choriotope distribution and abiotic parameters of the Pfaffenbach stream on April 5, 2016

Argillal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
15		50	30	5		
Temperature [°C]	pH	Conductivity [μS]	O ₂ saturation [%]	max Flow velocity [m/s]	Average flow velocity [m/s] [m/s]	Average water depth [m]
15.3	8.44	68	70.7	0.2	0.05	0

2.11.10 Stiefing

The investigation site in Stiefing (47° 0'24.93"N, 15°36'36.83"E) was sampled on a sunny day 650 m north of the village of Klein-Empersdorf in the municipality of Empersdorf on April 5, 2016. The area is located at an altitude of 353 m and covers a catchment area of 10 km² with a river order number of 3 (WIMMER & MOOG, 1994). The sunken

stream bed is surrounded by near-natural banks. There are no continuous riparian trees, resulting in a low degree of shading. Root beards and coarse particulate organic matter (CPOM) are present in the stream bed. Upstream, the Stiefing is surrounded by a forest on the right bank and farmland on the left bank. Downstream, the surrounding land is dominated by agriculture.



Source: Amap

Figure 18: Study site 10, Stiefing, photo: Patrick Leitner, BOKU

Table 15: Percentage distribution of choriotypes and abiotic parameters of the Stiefing on April 5, 2016

Argillal	Psammo- pelal	Psammal	Akal	Microlithal	Mesolithic	Macrolithal
		40	15	45		
Temperature [°C]	pH	Conductivity [μS]	O2 saturation [%]	max Flow velocity [m/s] [m/s]	Average flow velocity [m/s]	Average water depth [m]
14.1	7.7	355	96.1	0.6	0.2	0

3 Results

3.1 Macrozoobenthos colonization of the 2016 sampling series

3.1.1 Species inventory and dominance structure

Only organisms from the families Chironomidae and Limnephilidae were found in all 10 investigations (**Table 16**). Other strongly represented families are

Lumbriculidae, Gammaridae, and Baetidae. The total number of individuals per m² varies between 670 and 7955 within the 10 study sites.

Within the good ecological status class (Lieberbach, Vocherabach, and Wellingbach), the orders Ephemeroptera and Diptera dominate (**Figure 19**). The Wellingbach also has a high dominance of the order Crustacea. Within the ecologically moderate sites, the orders Crustacea, Ephemeroptera and Diptera dominate, with the order Oligochaeta (70%) dominating over the order Ephemeroptera (9%) in the Pfaffenbach and, conversely, the order Ephemeroptera (50%) dominating over the orders Oligochaeta (14%) and Plecoptera (19%) in the Glauningbach. The ecologically unsatisfactory Poppendorfer Bach and Labillbach streams are very dominantly represented by taxa of the order Diptera. The Gnasbach, the only stream with a poor ecological status, is dominated by the order Bivalvia.

Further species and genus determinations of the total taxon list with frequencies, which are also outlined in the following benthic community descriptions of the 10 study sites in spring 2016, can be found in Appendix 8.1 "Total taxon list with frequencies of the 10 streams sampled in 2016."

Table 16 a: Families of different large groups at the investigation sites, including absolute [Ind./m²] and relative [%] abundance

	Lieberbach		Glauningbach		Labillbach		Poppendorferb.		Gnasb. Old arm		Aubachl		Voherabach		Wellingbach		Pfaffenbach		Stiefing	
	[Ind/m²]	[%]	[Ind/m²]	[	[Ind/m²]	[%]	[Ind/m²]	[%]	[Ind/m²]	[%]	[Ind/m²]	[%]	[Ind/m²]	[%]	[Ind/m²]	[%]	[Ind/m²]	[%]	[Ind/m²]	[%]
TURBELLARIA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4.8	<1	-	-	-	-
PLANARIIDAE	-	-	7.2	1	-	-	-	-	0.8	<1	-	-	-	-	-	-	-	-	-	-
NEMATODA	-	-	-	-	-	-	0.8	<1	20	1	-	-	0.8	<1	-	-	0.8	<1	-	-
GASTROPODA																				
LYMNAEIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.8	<1	-	-
PHYSIDAE	-	-	-	-	5.6	<1	-	-	-	-	1.6	<1	0.8	<1	-	-	-	-	-	-
PLANORBIDAE	1.6	<1	-	-	-	-	4.8	<1	-	-	-	-	-	-	1.6	<1	2.4	<1	-	-
BIVALVIA	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
PISIDIIDAE	-	-	-	-	16	<1	1.6	<1	1040.8	49	1.6	<1	16.8	1	2.4	<1	503.2	9	0.8	<1
OLIGOCHAETA	-	-	-	-	25.6	<1	6.4	<1	-	-	-	-	-	-	12.8	1	-	-	6.4	<1
HAPLOTAXIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	28	1	-	-
LUMBRICIDAE	0.8	<1	9.6	1	0.8	<1	0.8	<1	-	-	1.6	<1	-	-	4.8	<1	8.8	<1	0.8	<1
NAIDIDAE	-	-	-	-	627.2	8	-	-	6.4	<1	-	-	-	-	-	-	-	-	-	-
TUBIFICIDAE	14.4	2	55.2	8	27.2	<1	41.6	3	180.8	9	145.6	9	-	-	-	-	540	10	8	<1
ENCHYTRAEIDAE	-	-	-	-	0.8	<1	-	-	-	-	-	-	1.6	<1	-	-	89.6	2	-	-
PROPAPPIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6.4	<1
LUMBRICULIDAE	29.6	3	2	4	-	-	6.4	<1	40	2	3.2	<1	3.2	<1	20	1	3074.4	58	103.2	4
HIRUDINEA																				
GLOSSIPHONIIDAE	-	-	-	-	0.8	<1	-	-	32	2	-	-	-	-	-	-	-	-	-	-
HAEMOPIIDAE	-	-	-	-	-	-	-	-	0.8	<1	-	-	-	-	-	-	1.6	<1	-	-
ERPOBDELLIDAE	-	-	-	-	-	-	-	-	5.6	<1	1.6	<1	-	-	-	-	3.2	<1	-	-
CRUSTACEA																				
CYCLOPIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	25.6	<1	-	-
GAMMARIDAE	167.2	19	8.8	1	56	1	191.2	14	11.2	1	659.2	40	-	-	404.8	17	262.4	5	1022.4	38
ASELLIDAE	-	-	1	<1	-	-	-	-	289.6	14	-	-	-	-	-	-	51.2	1	-	-
HYDRACHNIDIA	1.6	<1	-	-	-	-	0.8	<1	-	-	0.8	<1	17.6	1	12.8	1	-	-	0.8	<1

Table 16 b: Families of different large groups at the study sites, including absolute [Ind./m²] and relative [%] abundance

	Lieberbach		Glauningbach		Labillbach		Poppendorferb.		Gnasb. Altarm		Aubachl		Vocherabach		Wellingbach		Pfaffenbach		Stiefing	
	[Ind./m ²]	[%]	[Ind./m ²]	[	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]
EPHEMEROPTERA																				
SIPHONURIDAE	-	-	19.2	3	-	-	-	-	-	-	1.6	<1	-	-	-	-	460.8	9	-	-
BAETIDAE	4	<1	0.8	<1	30.4	<1	173.6	1	-	-	265.6	16	371.2	21	124.8	5	0.8	<1	332.8	13
HEPTAGENIIDAE	302.4	35	0	<1	7.2	<1	1.6	<1	-	-	152.8	9	172	10	256.8	11	-	-	575.2	22
LEPTOPHLEBIIDAE	56.8	7	311.2	46	614.4	8	-	-	-	-	5.6	<1	73.6	4	478.4	20	25.6	<1	-	-
EPHEMERIDAE	-	-	-	-	-	-	-	-	-	-	0.8	<1	-	-	46.4	2	-	-	31.2	1
EPHEMERELLIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.8	<1	-	-	-	-
CAENIDAE	4	<1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
ODONATA																				
CALOPTERYGIDAE	-	-	-	-	-	-	-	-	-	-	-	-	1.6	<1	0.8	<1	-	-	2.4	<1
GOMPHIDAE	-	-	-	-	-	-	-	-	-	-	-	-	0.8	<1	-	-	-	-	-	-
CORDULEGASTRIDAE	-	-	-	-	1.6	<1	-	-	-	-	3.2	<1	-	-	-	-	-	-	1.6	<1
CORDULIIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2.4	<1	-	-	-	-
PLECOPTERA																				
PERLODIDAE	8.8	1	109.6	16	-	-	-	-	-	-	6.4	<1	23.2	1	10.4	<1	-	-	-	-
TAENIOPTERYGIDAE	-	-	-	-	-	-	-	-	-	-	-	-	10.4	1	16	1	-	-	-	-
NEMOURIDAE	-	-	14.4	2	12.8	<1	-	-	-	-	0.8	<1	37.6	2	190.4	8	144	3	1.6	<1
LEUCTRIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6.4	<1	-	-	-	-
HETEROPTERA																				
NEPIDAE	-	-	-	-	-	-	-	-	0.8	<1	-	-	-	-	-	-	-	-	-	-
VELIIDAE	-	-	0.8	<1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
GERRIDAE	-	-	-	-	-	-	-	-	0.8	<1	-	-	-	-	0.8	<1	-	-	-	-
MEGALOPTERA																				
SIALIDAE	-	-	-	-	-	-	-	-	4	<1	-	-	-	-	0.8	<1	-	-	-	-

Table 16 c: Families of different large groups at the study sites, including absolute [Ind./m²] and relative [%] abundance

	Lieberbach		Glauningbach		Labillbach		Poppendorferb.		Gnasb. Old arm		Aubachl		Vocherabach		Wellingbach		Pfaffenbach		Stiefing	
	[Ind./m ²]	[%]	[Ind./m ²]	[	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]	[Ind./m ²]	[%]
COLEOPTERA																				
DYTISCIDAE	-	-	-	-	-	-	0.8	<1	-	-	-	-	-	-	-	-	0.8	<1	-	-
ELMIDAE	11.2	1	3.2	<1	0.8	<1	3.2	<1	-	-	84.8	5	1.6	<1	64	3	-	-	6.4	<1
GYRINIDAE	-	-	-	-	-	-	-	-	-	-	-	-	2.4	<1	-	-	-	-	-	-
HYDRAENIDAE	22.4	3	-	-	-	-	6.4	<1	-	-	16	1	3.2	<1	253.6	11	0.8	<1	-	-
HYDROPHILIDAE	-	-	-	-	-	-	-	-	1.6	<1	-	-	-	-	-	-	1.6	<1	-	-
SCIRTIDAE	-	-	-	-	-	-	-	-	-	-	-	-	3.2	<1	-	-	-	-	-	-
TRICHOPTERA																				
RHYACOPHILIDAE	-	-	-	-	-	-	-	-	-	-	-	-	2.4	<1	14.4	1	-	-	2.4	<1
GLOSSOSOMATIDAE	53.6	6	2	3	-	-	-	-	-	-	0.8	<1	-	-	-	-	-	-	-	-
HYDROPSYCHIDAE	1.6	<1	-	-	-	-	-	-	-	-	0.8	<1	24.8	1	1.6	<1	-	-	2.4	<1
POLYCENTROPODIDAE	-	-	-	-	-	-	-	-	-	-	-	-	0.8	<1	1.6	<1	0.8	<1	-	-
PSYCHOMYIIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4	<1	-	-	-	-
PHRYGANEIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	0.8	<1	-	-
LIMNephilidae	10.4	1	6.4	1	21.6	<1	20.8	2	3.2	<1	26.4	2	8	<1	83.2	4	37.6	1	29.6	1
GOERIDAE	5.6	1	-	-	-	-	2.4	<1	-	-	1.6	<1	-	-	-	-	0.8	<1	0.8	<1
LEPTOCERIDAE	12	1	-	-	-	-	15.2	1	-	-	-	-	-	-	-	-	-	-	-	-
DIPTERA																				
PEDICIIDAE	36.8	4	14.4	2	84	1	41.6	3	-	-	28	2	49.6	3	13.6	1	-	-	5.6	<1
CHIRONOMIDAE	127.2	15	5	8	6419.2	81	799.2	61	461.6	22	222.4	14	799.2	44	307.2	13	31.2	1	509.6	19
SIMULIIDAE	-	-	0	<1	0.8	<1	-	-	-	-	-	-	131.2	7	8.8	<1	0.8	<1	2.4	<1
ATHERICIDAE	-	-	-	-	-	-	-	-	-	-	-	-	-	-	12.8	1	-	-	-	-
CERATOPOGONIDAE	-	-	3.2	<1	-	-	-	-	-	-	-	-	7.2	<1	0.8	<1	12.8	<1	-	-
LIMONIIDAE	-	-	-	-	1.6	<1	-	-	1.6	<1	4.8	<1	36	2	2.4	<1	0.8	<1	7.2	<1
PSYCHODIDAE	-	-	-	-	-	-	-	-	16	1	-	-	-	-	0.8	<1	-	-	-	-
TABANIDAE	-	-	-	-	0.8	<1	-	-	2.4	<1	1.6	<1	-	-	-	-	-	-	-	-
TIPULIDAE	0.8	<1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
Total number of individuals	872.8		670.4		7955.2		1,319.2		2,120.0		1,639.2		1,800.8		2,368.0		5312.0		2,660	

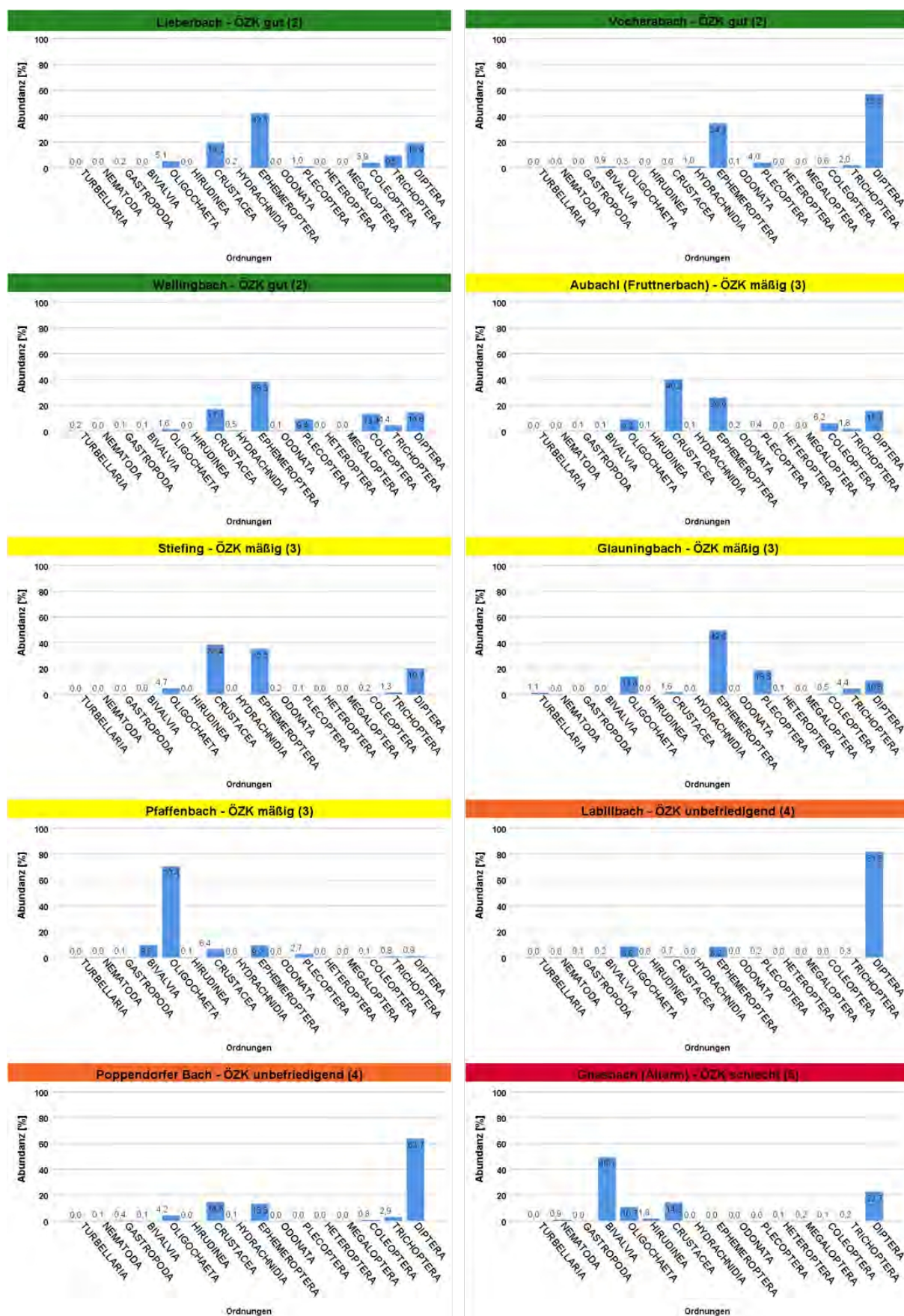


Figure 19: Dominance of the major groups at the 10 sampling sites in the 2016 sampling series in spring; ÖZK = ecological status class

3.1.1.1 Benthic community of the Lieberbach

The macrozoobenthic community consists of 48 taxa, which form 21 families, with a total number of individuals of approximately 870. The families Heptageniidae (35%), Gammaridae (19%) and Chironomidae (15%) (Table 16).

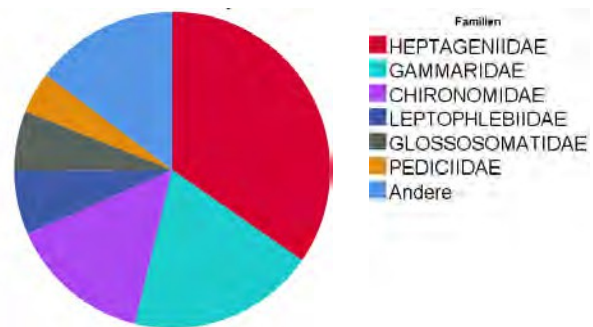


Figure 20: Dominance of families (abundance >4%) in the Lieberbach

Among the mayflies, the family Heptageniidae (35%) is dominant, with the genera *Ecdyonurus* and *Rhithrogena* and the species *Rhithrogena picteti* (>270 ind./m²). Stoneflies are only represented by the genus *Isoperla* (<10 ind./m²). The highest individual density among the caddisflies is found in the Glossosomatidae family (6%) with the species *Agapetus fuscipes* (>50 ind./m²). Species such as *Hydropsyche angustipennis*, *Anabolia furcata*, *Potamophylax rotundipennis*, and *Lithax obscurus* were also identified.

Among the beetles, the genera *Elmis*, *Limnius*, and *Hydraena* were identified. Among the diptera, the genera *Dicranota* and *Orthocladius* (>70 ind./m²) are most commonly represented. The family Chironomidae (midges) is represented by 16 taxa.

3.1.1.2 Benthic community of the Glauningbach

The community studied in the Glauningbach consists of 38 taxa within 20 families (Table 16). The total number of individuals is approximately 670. The highest abundances are found in the families Leptophlebiidae (46%), Perlodidae (16%), Chironomidae and Tubificidae, each with a relative abundance of 8%.

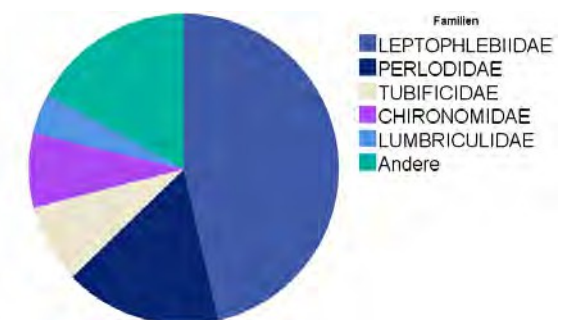


Figure 21: Dominance of families (abundance >4%) in the Glauningbach

Among the mayflies, the genus *Habrophlebia* dominates, with the species *Habrophlebia fusca* (4 ind./m²) and the majority identified as taxon *Habrophlebia* sp. juv. (approx. 300 ind./m²). The genus *Siphonurus* was identified with the species *Siphonurus aestivalis* and *Siphonurus armatus*. The stoneflies are dominated by the genus *Isoperla* (110

ind./m²) dominated, alongside the genera *Amphinemura* and *Nemoura*. The caddisfly fauna consisted mainly of the species *Agapetus fuscipes* (>20 ind./m²) and the species *Anabolia furcata*, *Isonychia dubia*, and *Potamophylax rotundipennis*.

The chironomid fauna consists of five taxa, which account for 8% of the total number of individuals and are most frequently represented by the genus *Orthocladus* (28 ind./m²). No representatives of the orders Gastropoda and Bivalvia were found.

3.1.1.3 Benthic fauna of the Labillbach

There are 46 taxa at the study site, comprising a total of around 7960 individuals from 21 families (Table 16). The Chironomidae family accounts for 80% of the total abundance. The families Naididae and Leptophlebiidae are each represented by 8%. The Pediciidae family reaches an abundance of 1%. The proportion of all other families is less than 1% of the total individual density.

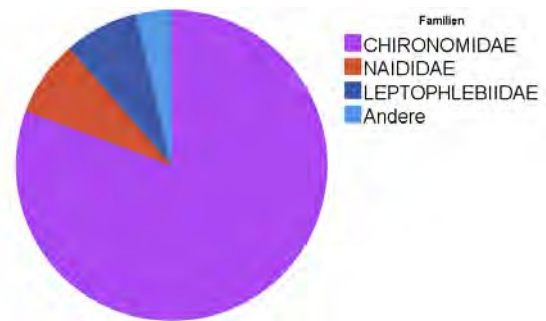


Figure 22: Dominance of families (abundance >4%) in the Labillbach

In the mayfly family, the taxon *Habrophlebia* sp. juv. dominates with 614 ind./m² alongside the species *Baetis rhodani* (30 ind./m²) and *Rhithrogena picteti* (4 ind./m²). Stoneflies are only represented by the genus *Amphinemura* (13 ind./m²). Among the caddisflies, the species *Anabolia furcata* (5 ind./m²) and *Potamophylax rotundipennis* (14 ind./m²) from the family Limnephilidae and the genus *Limnephilus* (2 ind./m²) were identified.

The Chironomidae family (midges) is represented in the coenosis by 21 taxa. The genus *Orthocladus* (5471 ind./m²) is represented by the species *Paratrichocladus rufiventris* (182 ind./m²), the tribe *Orthocladini* COP (115 ind./m²), the species *Rheocricotopus fuscipes* (105 ind./m²), *Orthocladus rivicola* (102 ind./m²) and *Cladotanytarsus vanderwulpi* (102 ind./m²) and the genus *Corynoneura* with 102 ind./m² . Beetles are represented by the genus *Elmis* (1 ind./m²) and the family Simuliidae with the genus *Simulium*.

3.1.1.4 Benthos community of Poppendorfer Bach

The Poppendorfer Bach study site comprises 44 taxa, which results in a total number of individuals of around 1320. The most dominant of the total 17 families is Chironomidae with a relative abundance of 81% (Table 16). Other strongly represented families are Gammaridae (14%) and Baetidae (13%).

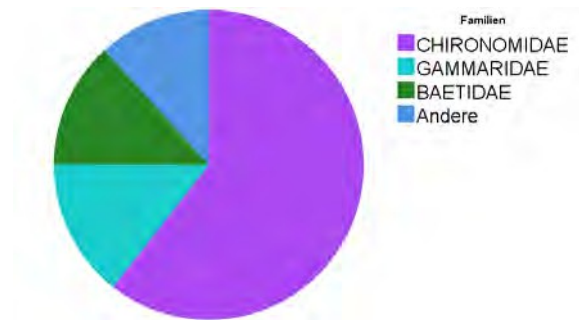


Figure 23: Dominance of families (abundance >4%) in the Poppendorfer Bach

Mayflies consist of the species *Baetis rhodani* (39 ind./m²) and the genera *Baetis* and *Ecdyonurus*. Stoneflies are not present in the sampled community. Caddisflies are represented by the species *Anabolia furcata* (6 ind./m²) and *Isonychia dubia* (2 ind./m²).

Among the chironomids, 18 taxa were identified, including the genus *Orthocladius* (500 ind./m²) and the species *Polypedilum convictum* (66 ind./m²), *Diamesa insignipes* (36 ind./m²), *Orthocladius rivicola* (33 ind./m²) and *Paratendipes cf. albimanus* (32 ind./m²).

The beetle group is represented in the coenosis by the species *Platambus maculatus* (1 ind./m²), *Oulimnius tuberculatus* (3 ind./m²) and the genus *Hydraena* (6 ind./m²). In addition to the species *Gammarus fossarum* with 170 ind./m², the species *Gammarus roeseli* was also sampled at this study site. No black flies were detected at the study site.

3.1.1.5 Benthic community of the Gnasbach (old arm)

A total of 34 taxa from 20 families with a total of 2,120 organisms were sampled. The families Pisidiidae (49%), Chironomidae (22%), Asellidae (14%) and Tubificidae (9%) dominate the coenosis (Table 16).

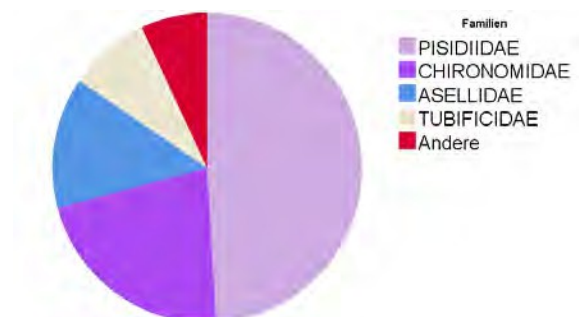


Figure 24: Dominance of families (abundance >4%) in the Gnasbach

Among the EPT taxa (Ephemeroptera, Plecoptera, Trichoptera), no taxa of mayflies and stoneflies were found. Caddisflies are sparsely represented with 3 ind./m² from the family Limnephilidae.

The Chironomidae family dominates with the genera *Procladius* (*Holotanypus*) (316 ind./m²), *Micropsectra* (64 ind./m²), and *Chironomus* (41 ind./m²).

The genus *Pisidium* has a very high number of individuals at 1040/m². The Tubificidae family is represented by the species *Aulodrilus japonicus*, *Tubifex tubifex*, *Potamotheix moldaviensis*, and *Limnodrilus hoffmeisteri*. In addition to the species *Asellus aquaticus* (290 ind./m²), several taxa of the order Hirudinea were also found, as well as one individual of the species *Nepa cinerea cinerea*.

3.1.1.6 Benthic community of the Aubach (Fruttnerbach)

The biocoenosis consists of 56 taxa, which form 25 families with a total of around 1640 individuals per square metre (Table 16). The Gammaridae family accounts for the largest share of the total abundance with 40%, followed by Baetidae () (21%), (Chironomidae) () (14%) and Tubificidae with 8%.

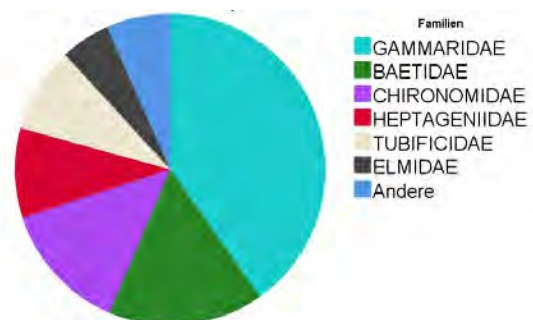


Figure 25: Dominance of families (abundance >4%) in the Aubach

Among the mayflies the taxon *Baetis* sp. juv. (170 ind./m²) and the species *Baetis rhodani* (96 ind./m²) dominate. Other species include *Electrogena ujhelyii* (69 ind./m²), *Rhithrogena picteti* (12 ind./m²), *Rhithrogena semicolorata* (1 ind./m²) and *Ecdyonurus subalpinus* (10 ind./m²). The genus *Habrophlebia* and the species *Ephemera danica* and *Siphonurus armatus* were also detected. Stoneflies are represented by the genera *Isoperla* sp. (6 ind./m²) and *Protonemura* sp. (1 ind./m²). The caddisfly fauna consists of the species *Lithax obscurus* (2 ind./m²), *Potamophylax rotundipennis* (6 ind./m²), *Limnephilus* cf. *stigma* (2 ind./m²), *Hydropsyche angustipennis*, and *Agapetus fuscipes*, each with one individual.

The midges comprise 15 taxa, with the genus *Orthocladius* (*Orthocladius*) (131 ind./m²), the species *Polypedilum* (*Tripodura*) *pullum/scalaenum* (21 ind./m²) and the subfamily Tanytarsini (20 ind./m²) are the dominant taxa. Among the beetles, the genera *Elmis* (71 ind./m²), *Limnius* and the species *Oulimnius tuberculatus* (13 ind./m²) as well as the genus *Hydraena* are present. In the family Tubificidae (Oligochaeta), the dominant species are *Tubifex tubifex*, *Limnodrilus hoffmeisteri*, and *Rhyacodrilus coccineus*.

3.1.1.7 Benthic community of the Vocherabach

The Vocherabach contains 83 taxa from 27 families with a total individual density of 1800 per square meter (Table 16). The dominant families are Chironomidae (44%), Baetidae (21%), Heptageniidae (10%) and Simuliidae (7%).

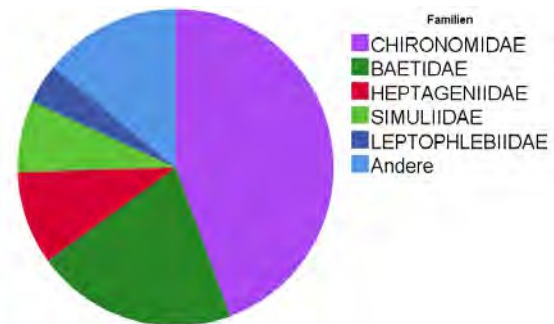


Figure 26: Dominance of families (abundance >4%) in the Vocherabach

Mayflies are represented by the species *Baetis rhodani* (230 ind./m²), *Baetis niger* (13 ind./m²), and *Centroptilum luteolum* (1 ind./m²) from the Baetidae family. Other species found include *Ecdyonurus* cf. *subalpinus* (34 ind./m²), *Electrogena ujhelyii* (9 ind./m²) and *Rhithrogena picteti* (72 ind./m²) as well as the species *Paraleptophlebia submarginata* (21 ind./m²) and the genus *Habrophlebia* (53 ind./m²). Among the stoneflies, the species *Isoperla tripartita* (21 ind./m²) and the genus *Isoperla*, the species *Brachyptera risi* (10 ind./m²) and *Nemoura cinerea* (2 ind./m²) and the genera *Amphinemura* and *Protonemura*. The caddisfly fauna consists of the taxon *Rhyacophila* s. str. (2 ind./m²) and the species *Hydropsyche saxonica* (12 ind./m²), *Hydropsyche bulbifera* (1 ind./m²), *Polycentropus irroratus* (1 ind./m²) and *Potamophylax rotundipennis* (3 ind./m²).

Chironomids are represented in the zoenosis with 39 taxa and a total population of around 800 individuals. The most abundant taxa are *Orthocladius* (*Orthocladius*) sp. (162 ind./m²), *Orthocladiini* COP (53 ind./m²) and *Conchapelopia* sp. (49 ind./m²). The family Simuliidae is represented by the genera *Simulium* and *Prosimulium*. No taxa from the orders Hirudinea, Amphipoda and Isopoda could be sampled. In addition to the genus *Limnius*

, the species *Orectochilus villosus* (Gyrinidae) and a representative of the family Scirtidae were identified among the beetles.

3.1.1.8 Benthic community of the Wellingbach

The study site comprises 70 taxa from 34 families with an absolute abundance of 2368 individuals (Table 16). The highest densities are found in the families Leptophlebiidae (20%), Gammaridae (17%), Chironomidae (13%), and Hydraenidae and Heptageniidae, each with an 11% share.

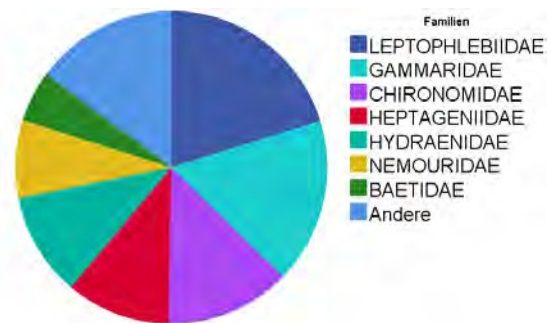


Figure 27: Dominance of families (abundance > 4%) in the Wellingbach stream

The species *Baetis rhodani* (54 ind./m²), *Rhithrogena picteti* (172 ind./m²), *Habroleptoides confusa* (426 ind./m²) and *Ephemera danica* (46 ind./m²) dominate the mayfly fauna alongside the species *Ecdyonurus subalpinus*, *Electrogena ujhelyii*, *Habrophlebia lauta* and *Ephemerella mucronata*. Stoneflies are dominated by the genus *Protonemura* (184 ind./m²). Furthermore, the species *Isoperla tripartita*, *Brachyptera risi* and the genera *Amphinemura* and *Leuctra* can be found in the sample. The taxa *Rhyacophila s. str.*, *Hydropsyche sp. juv.*, *Plectrocnemia sp.*, *Lype sp.*, *Anabolia furcata*, *Halesus sp.*, and *Potamophylax rotundipennis* form the caddis fly fauna, among others.

Chironomids are represented by 17 taxa, with the dominant taxa being *Tanytarsus sp.* (70 ind./m²), *Orthocladius (Euorthocladius) thienemanni* (38 ind./m²), *Paratritissocladius excerptus* (19 ind./m²) and *Conchapelopia sp.* (19 ind./m²). The Simuliidae family is represented by 4 taxa.

3.1.1.9 Benthic community of the Pfaffenbach

The benthos community of the Pfaffenbach reaches a total individual density of 5312 animals from 29 families and comprises 46 taxa (Table 16). The dominant families are Lumbriculidae (58%), Tubificidae (10%), Pisidiidae (9%), and Siphonuridae (9%).

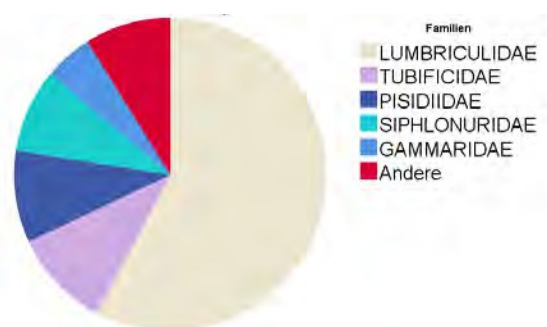


Figure 28: Dominance of families (abundance > 4%) in the Pfaffenbach stream

The mayflies are mainly represented by the species *Baetis rhodani* (1 ind./m²) and the genus *Habrophlebia* (26 ind./m²), the mayflies consist mainly of the species *Siphonurus aestivalis* (199 ind./m²) and *Siphonurus armatus* (262 ind./m²). Stoneflies are represented by the species *Nemoura cinerea* (144 ind./m²). The dominant species among caddisflies is *Ironoquia dubia* with 34 ind./m². Other species recorded are *Plectrocnemia conspersa* (1 ind./m²), *Oligostomis reticulata* (1 ind./m²), *Glyphotaelius pellucidus* (2 ind./m²), *Limnephilus rhombicus* (1 ind./m²), *Potamophylax rotundipennis* (1 ind./m²) and *Lithax obscurus* (1 ind./m²).

The chironomid fauna consists of eight taxa, none of which is dominant. Among the beetles, one taxon from the family Dytiscidae, one taxon from the genus *Hydraena*, and two individuals from the family Hydrophilidae are represented. The oligochaetes (Oligochaeta) are represented by the families Tubificidae (540 ind./m²) and Lumbriculidae (3074 ind./m²), or rather the species *Emboloccephalus velutinus* (450 ind./m²) and *Rhyacodrilus coccineus* (90 ind./m²) as well as *Stylodrilus heringianus* (2018 ind./m²) originating from these families. The species *Gammarus fossarum* and *Asellus aquaticus* were also detected.

3.1.1.10 Benthic community of the Stiefing

The biocoenosis of the Stiefing consists of 22 families. The total of 53 taxa form an absolute abundance of 2660 animals (Table 16). The main proportions are contributed by the families Gammaridae (38%), Heptageniidae (22%), Chironomidae (19%) and Baetidae (13%).

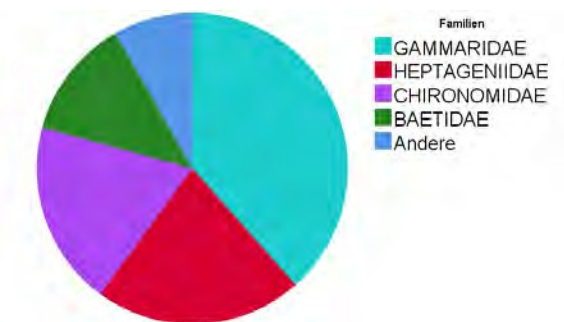


Figure 29: Dominance of families (abundance >4%) in the Stiefing

Mayflies are dominant, with the species *Baetis rhodani* (127 ind./m²) alongside *Rhithrogena picteti* (123 ind./m²) and *Electrogena ujhelyii* (104 ind./m²). The species *Centroptilum luteolum*, *Ecdyonurus subalpinus*, *Rhithrogena semicolorata*, and *Ephemera danica* were also identified. Only one stonefly taxon was sampled, belonging to the genus *Amphinemura* (2 ind./m²). Caddisflies are represented by the taxa *Rhyacophila* s. str. (2 ind./m²), *Hydropsyche* sp. juv. (2 ind./m²) and the species *Potamophylax rotundipennis* (2 ind./m²). Furthermore, the species *Anabolia furcata*, *Lithax obscurus* and

the genus *Halesus* are present with an absolute abundance of 1 individual per square meter each.

The midges make up a total population density of 510 individuals from 16 taxa. The largest proportions are accounted for by the species *Rheocricotopus (Rheocricotopus) fuscipes* (215 ind./m⁽²⁾) and the genera *Orthocladius (Orthocladius)* (134 ind./m⁽²⁾) and *Rheotanytarsus* with 79 ind./m⁽²⁾.

In addition to the beetle genus *Elmis* and the species *Limnius volckmari*, 1022 individuals of the species *Gammarus fossarum* were also sampled.

3.1.2 Overview of the zoenotic structure of the 2016 sampling series

The Lieberbach, Vocherabach, and Wellingbach (good ecological status) are dominated by approximately 50% of taxa from the EPT families (**Figure 30**). Within the moderate study sites, this dominance declines to the Glauningbach, where the Leptophlebiidae family accounts for almost 50% of the abundance. Taxa from the Siphonuridae family are present in the Pfaffenbach and Glauningbach streams. The Chironomidae family dominates in the unsatisfactory and poor survey sites. In general, the Ephemeroptera families dominate over the Plecoptera and Trichoptera families.

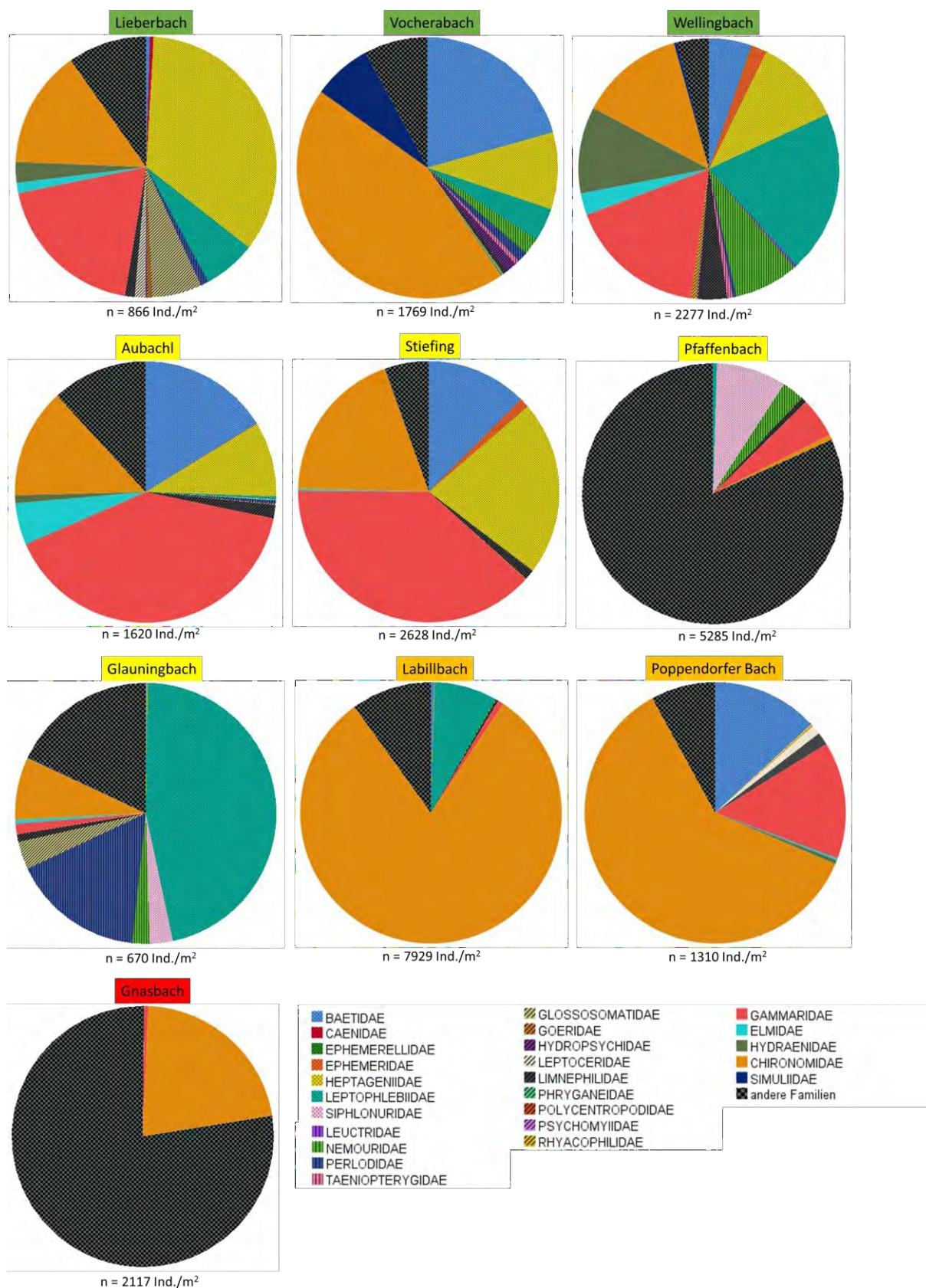


Figure 30: Individual dominance (families) at the 10 sampling sites in the 2016 sampling series in spring, with the ecological status highlighted in color (green = good, yellow = moderate, orange = unsatisfactory, red = poor).

3.2 Indices and distributions of the 2016 sampling series according to the detailed MZB assessment method

3.2.1 Length-zonal distribution according to biocenotic regions

Based on the macrozoobenthos fauna, this distribution reflects the niche occupation of the taxa present in the sample. The Lieberbach, Wellingbach, Vocherabach, Aubachl, and Stiefing show a strong distribution bias toward rhithral (epi- to hyporhithral) (**Figure 31**). The Lieberbach shows a clear faunistic epirhithral peak, and the Aubachl and Stiefing have a relatively high littoral proportion. In the Poppendorfer Bach and Labillbach, a clear potamal, but above all epipotamal influence can be seen from the distribution, although the rhithral proportions still predominate. In addition, the littoral/profundal proportion is relatively high in both streams. The length-zone distribution curve for the Pfaffenbach shows a non-specific focus from metarhithral to epipotamal. The littoral/profundal proportion is also very high. In the Glauningbach and Gnasbach, the littoral proportions predominate, with the Glauningbach still characterized by rhithral and the Gnasbach by epi- to hypopotamal.

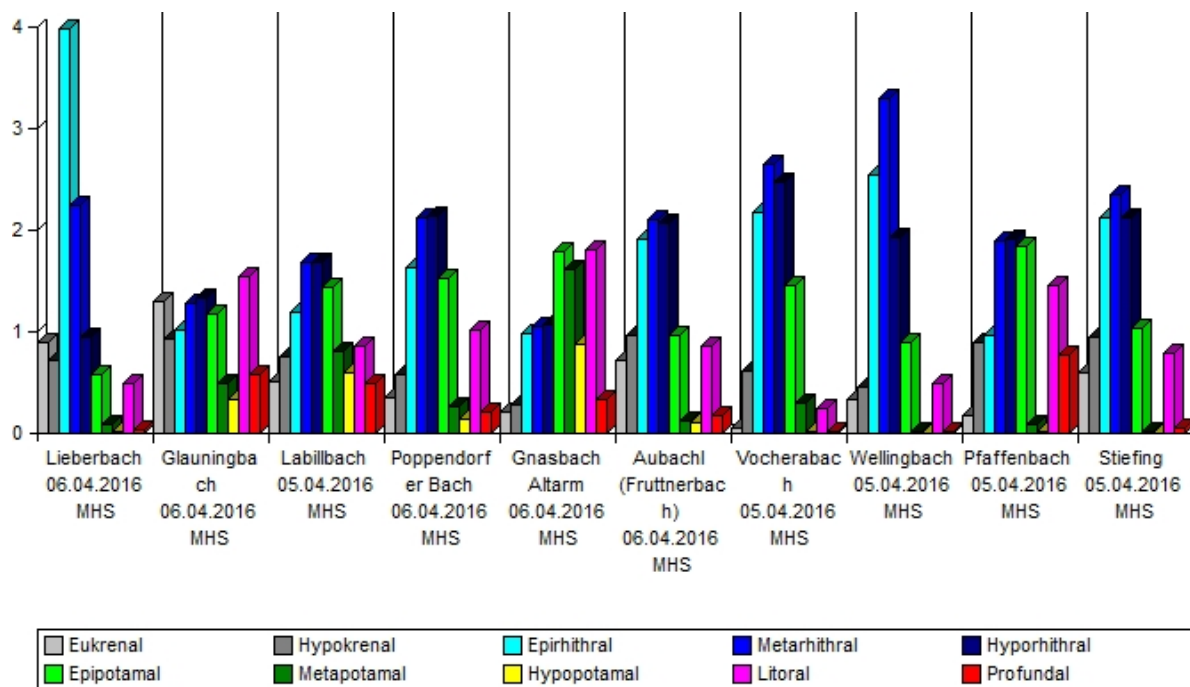


Figure 31: Length distribution by biocenotic regions (see legend) based on macrozoobenthos. ©ECOPROF 1995-2013 - Version 4.0 MHS=Multi Habitat Sample

3.2.2 Functional feeding types

Grazers and detritivores dominate within the 10 study sites, although grazers are very low in the Pfaffenbach with a valence of 0.10 (**Figure 32**). In the Glauningbach, Gnاسبach, and Pfaffenbach, the proportions of shredders and grazers are very low. Detritivores are highly dominant in the Glauningbach and Pfaffenbach. Predators are present in all streams, although their proportion is very low in the Stiefing (0.11) and Pfaffenbach (0.04). Active filter feeders dominate in the Gnاسبach, while the valence of shredders is relatively high in the Aubachl and Stiefing.

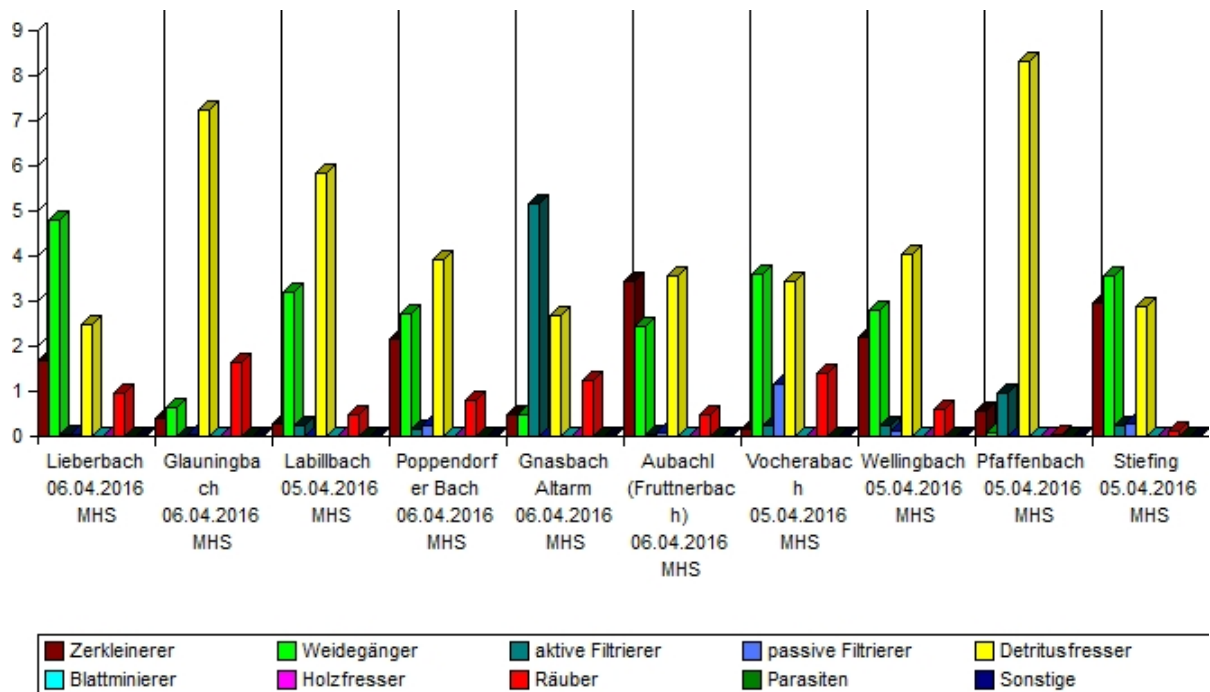


Figure 32: Composition of feeding types (see legend) based on macrozoobenthos. ©ECOPROF 1995-2013 - Version 4.0 MHS=Multi Habitat Sample

3.2.3 Saprobic valences

Apart from the Gnasbach, the beta-mesosaprobe component dominates in the streams (Figure 33). The Lieberbach has the highest xenosaprobe content at 1.75. The Gnasbach (valence 1.64), Glauningbach (valence 1.57), Aubachl (valence 1.15), and Labillbach (valence 0.70) have elevated polysaprobe contents. The Labillbach and Gnasbach also have very low xenosaprobe proportions, with the Gnasbach also having very low oligosaprobe proportions. The Gnasbach is also the only stream with an alpha-mesosaprobe peak. The Lieberbach has a very high oligosaprobe valence in addition to its high xenosaprobe proportion. The Wellingbach has the highest oligosaprobe content with a valence of 4.03.

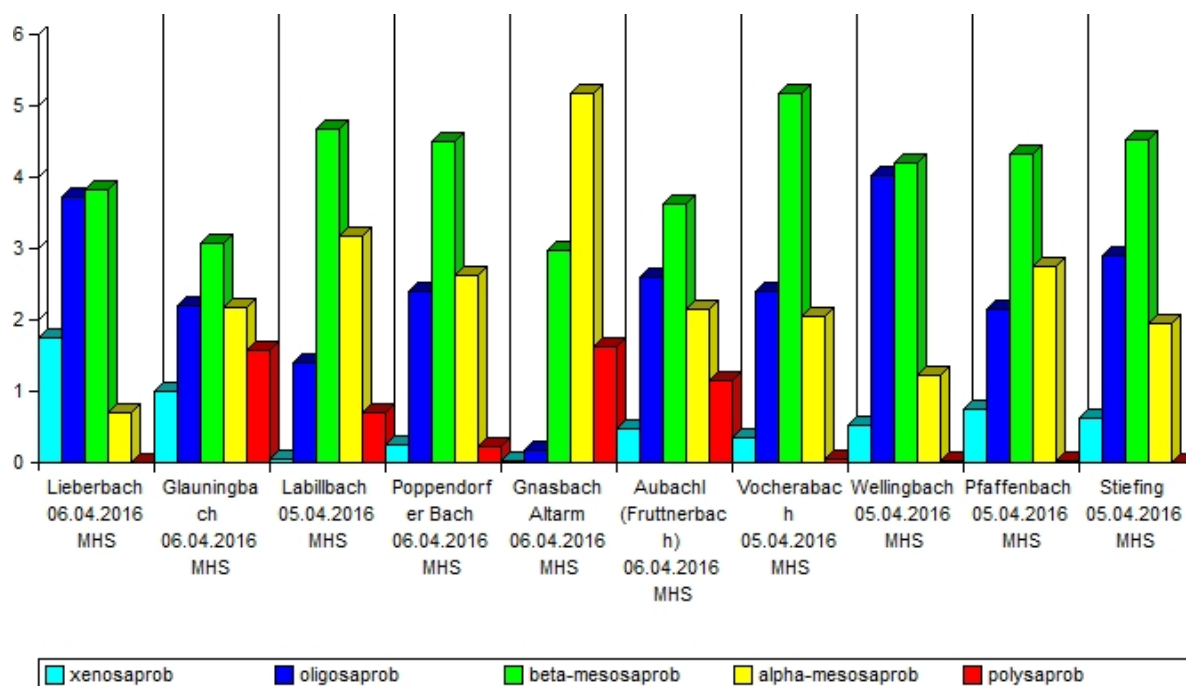


Figure 33: Distribution of saprobic valences or organic pollution (see legend) based on macrozoobenthos. ©ECOPROF 1995-2013 - Version 4.0 MHS=Multi Habitat Sample

3.3 Ecological status assessment according to the detailed MZB method

The Lieberbach, Vocherabach, and Wellingbach were classified as having "good" ecological status (Table 17). The ecological status of the remaining streams was rated as

"moderate," "unsatisfactory," and "poor." In the case of the Gnasbach (old arm), it should be noted that the EPT taxa and degradation index scores are "zero," which is unusual for a watercourse. The saprobic index was rated very good or good except for the Gnasbach (unsatisfactory) and the Labillbach (moderate).

Table 17: Results of ecological status class according to the detailed method using EcoProf 4.0; SI = saprobic index, SZK = saprobic status class, MMI_1,2 = multi-metric index 1 and 2, IND = individuals/m², EC=ecological status, GTZ=total taxon count, EPT-T=Ephemeroptera-Plecoptera-Trichoptera taxa, %OD-T= % Oligochaeta & Diptera taxa, DIV=diversity index (Margalef), DEG=degradation index, RETI=Rhithron feeding type index, LIT=littoral; *=values correspond to scores – for detailed breakdowns, see Appendix 8.2

Water	SI/SZK	MMI_1	MMI_2	IND	ÖZ	GTZ*	EPT-T*	% EPT-T*	% OD-T*	DIV*	DEG*	RETI*	LIT*
Lieberbach	1.35/very good	0.63	0.6	866.4	good	0.57	0.49	0.69	0.72	0.65	0.39	0.81	0.78
Glauningbach	2.11/good	0.44	0	670.4	moderate	0.44	0.38	0.7	0.74	0.5	0.26	0.14	0
Labillbach	2.31/moderate	0	0	7928.8	unconfirmed	0.52	0.22	0.34	0.46	0.44	0.04	0.41	0.71
Poppendorfer Bach	2.02/good	0.46	0.36	1310.4	moderate	0.48	0.22	0.37	0.61	0.47	0.14	0.6	0.69
Gnasb. Old arm	2.82/unf.	0	0	2116.8	poor	0	0	0	0.57	0.39	0	0.13	0
Aubachl (Fruttnerb.)	2.09/good	0	0.53	1620	moderate	0.65	0.47	0.58	0.69	0.66	0.3	0.7	0.71
Vocherabach	1.91/good	0.69	0.62	1768.8	good	0.91	0.58	0.51	0.55	0.95	0.5	0.5	0.82
Wellingbach	1.62/very good	0.75	0.71	2276.8	good	0.86	0.68	0.64	0.78	0.87	0.66	0	0.77
Pfaffenbach	1.92/good	0	0	5284.8	moderate	0.53	0.33	0.5	0.74	0.46	0.07	0.08	0.61
Stiefing	1.78/good	0.59	0.53	2628	moderate	0.61	0.41	0.54	0.62	0.6	0.4	0.75	0.72

3.4 Ecological status assessment using the screening method

The screening method indicated a "good" ecological status for the Aubachl (Fruttnerbach), Vocherabach, and Wellingbach. According to the screening method, "action is required" for all other streams (Table 18). More detailed lists of scores and EQRs can be found in Appendix 8.2.

Table 18: Results of the ecological status class according to the screening method with EcoProf 4.0; SCR-T=screening taxa, SEN-T=sensitive taxa, DEG=degradation score, SAP=saprobic score, ÖZ=ecological status, AB-EQR I/II = general pollution-ecological quality ratio for "still very good" status, OB=organic pollution, II/III= "still good" status, HB=action required

	SCR-T	SEN-T	DEG	SAP	ÖZ	AB-EQR I/II	OB-EQR I/II	AB-EQR II/III	OB-EQR II/III
Lieberbach	29	9	64	82.74	HB	0.53	0.99	0.84	0.72
Glauningbach	25	7	41	99.25	HB	0.37	1.19	0.58	0.86
Labillbach	24	5	13	103.64	HB	0.19	1.24	0.29	0
Poppendorfer Bach	24	4	30	96.64	HB	0.24	1.16	0.38	0.84
Gnasb. Old arm	2	1	0	123.11	HB	0.03	1.47	0.04	1.07
Aubachl (Fruttnerb.)	36	15	56	94.6	good	0.64	1.13	1.01	0.82
Vocherabach	36	13	78	78.33	good	0.7	0.94	1.1	0
Wellingbach	49	19	101	76.03	good	0.96	0.91	1.51	0.66
Pfaffenbach	32	5	13	110.42	HB	0.19	1.32	0.29	0.96
Stiefing	30	10	59	82.66	HB	0.53	0.99	0.84	0.72

3.4.1 Comparison of the assessment systems with the "best available sites" from the 2016 sampling series

The sampling sites in the 2016 sampling series are the best possible near-natural sites in terms of hydromorphology. Nevertheless, due to the detailed method used, the limit values for very good ecological status or good ecological status are only achieved by three streams (**Table 19**).

Table 19 Comparison of the required limit values and the limit values achieved for the detailed method and for the screening method of the "best available sites" in 2016, colored according to the assessed ecological status (blue = very good, green = good, yellow = moderate, orange = unsatisfactory or action required (screening), red = poor; EPT = Ephemeroptera, Plecoptera, Trichoptera; RETI = Rithron feeding type index;

pool, EPT (Ephemeroptera, Plecoptera, Trichoptera, RETI) (non-feeding type max)																						
Metrics Detailed method	Reference value		Lieberbach		Vocherabach		Wellingbach		Aubachl		Stiefing		Glauningbach		Pfaffenbach		Labillbach		Poppen- dorfer Bach		Gnasbach	
Total taxa	77		44		70		66		50		47		34		41		40		37		30	
EPT taxa	36.5		18		21		25		17		15		14		12		8		8		0	
EPT taxa [%]	58.96		40.91		30		37.88		34		31.91		41.18		29.27		20		21.62		0	
Oligochaeta & Diptera [%]	7		54.55		41.4		59.0		52		46.81		55.88		56.1		35		45.95		43.33	
Diversity index	9.45		6.16		8.96		8.18		6.21		5.68		4.76		4.32		4.13		4.46		3.68	
Degradation index	175.5		68		88		115		53		70		45		12		7		25		0	
RETI	0.88		0.71		0.44		0.53		0.61		0.66		0.13		0.07		0.36		0.53		0.11	
Littoral	5.81		4.51		4.75		4.5		4.13		4.21		3.46		3.54		4.14		3.99		3.19	
Metrics Screening Method	Threshold (still in very good condition)	Limit (still good condition)	Lieberbach		Vocherabach		Wellingbach		Aubachl		Stiefing		Glauningbach		Pfaffenbach		Labillbach		Poppendorf- er Bach		Gnasbach	
Screening taxa	-	-	29		36		49		36		30		25		32		24		24		23	
Sensitive taxa	20	13	9		13		19		15		10		7		5		5		4		1	
Degradation score	105	65	64		78		101		56		59		41		13		13		30		0	
Saprobie score (1.75 SGZ)	83.5	115	82.74		78.33		76.0		94.6		82.66		99.25		110.42		103.64		96.64		123.11	

3.5 Similarity analyses of the cenoses

This chapter presents the results of the coenotic analyses in graphical form. These are based on all available data and on the data set limited to the "best available sites (bas)" (see Chapter 2.5).

3.5.1 Identification of bioregions 14, 13, and 3 based on the coenotic structure

The mountain ridge landscape (3) is clearly separated from the adjacent bioregions, the eastern flat and hilly lands (13) and the Graz field and trench land (14). The latter bioregions cluster somewhat apart from each other, but there is a clear overlap between bioregions 13 and 14 (**Figure 34a**).

The clustering of bioregion 3 and bioregion 14 "Grazer Feld and Grabenland" is also confirmed by the "best available sites," with a clear overlap between bioregions 13 and 14 (**Figure 34b**).

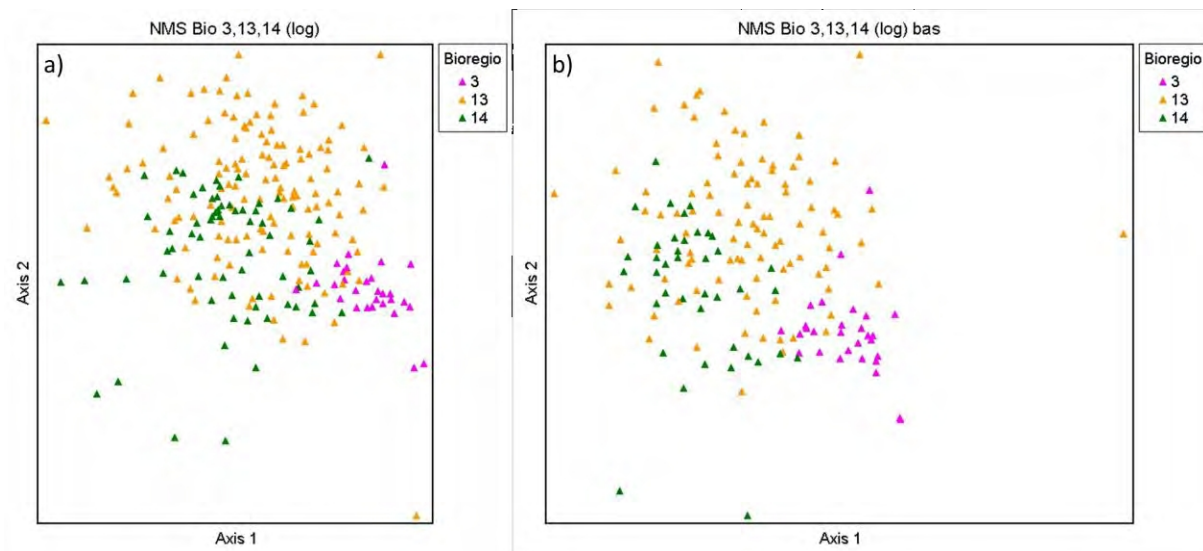


Figure 34: a) NMS scatter plot of the bioregions Bergrückenlandschaft (3), Östliche Flach- und Hügelländer (13) and Grazer Feld und Grabenland (14); $\log=\log(n+1)$ transformed abundances; $n=227$ study sites; 17.60129=final stress for 3-dimensional solution; 2-D view axes 1 and 2 **b)** NMS scatter plot of the bioregions mountain ridge landscape (3), eastern flat and hilly lands (13) and Graz field and trench land (14); $\log=\log(n+1)$ transformed abundances; bas="best available sites"; $n=167$ investigation sites; 16.75981=final stress for 3-dimensional solution; 2-D view axes 1 and 2

3.5.2 Influence of the season within bioregion 14 "Grazer Feld and Grabenland" and bioregion 13

An analysis of bioregions 13 and 14 using data from spring shows a separation of the 10 investigation sites (bioregion 14) from 2016. The five sites (blue circle) (bioregion 14), which form a common cluster with the study sites in bioregion 13, are based on data from 2007 (**Figure 35a**). These relate to water bodies (Sulm, Kainach, Laßnitz) which, in terms of fauna, do not have the characteristics of the streams in

"Grazer Feld and Grabenland" , da sie ihren Ursprung in der Bioregion 3

"mountain ridge landscape." These three investigation sites and two others (Saßbach, Schwarzaubach) near the confluence with the Mur are excluded from further analysis from the "best available sites" data set.

The analysis using the "best available sites" data set reveals a clear separation. The Pfaffenbach and Glauningbach (red arrow) separate benthically within bioregion 14 (**Figure 35b**).

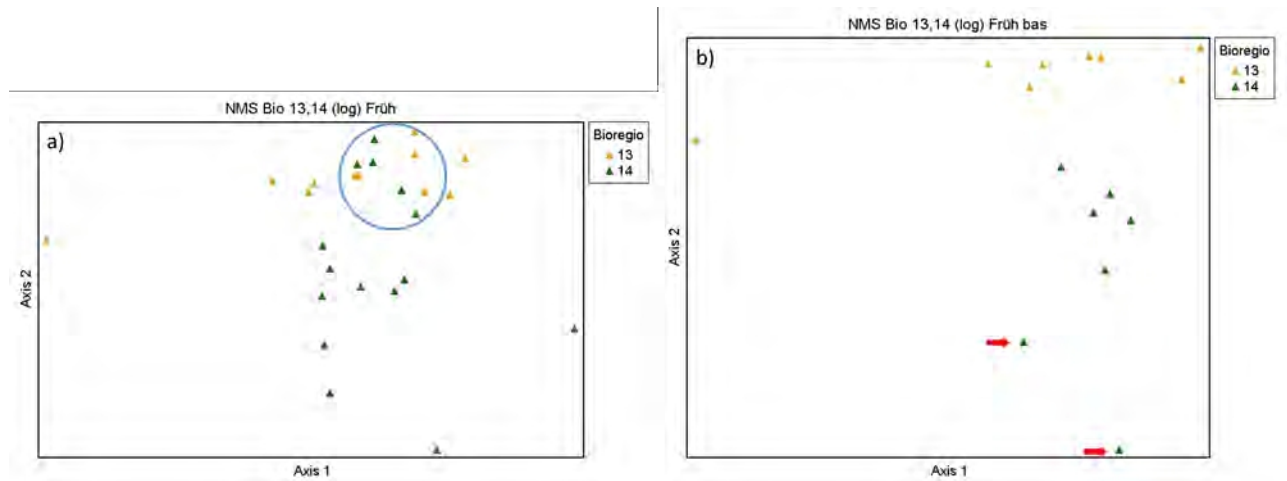


Figure 35: a) NMS scatter plot of the bioregions Grazer Feld and Grabenland (14, green) and Östliche Flach- und Hügelländer (13, orange) with overlay "Bioregions"; log=log(n+1) transformed abundances; Früh=survey sites in spring; n=26; 9.19964=final stress for 3-dimensional solution; 2-D view axes 1 and 2 **b)** NMS scatter plot of the bioregions Grazer Feld and Grabenland (14, green) and Östliche Flach- und Hügelländer (13, orange) with overlay "Bioregions"; log=log(n+1) transformed abundances; Spring=test sites in spring; bas="best available sites"; n=15; 4.74091=final stress for 3-dimensional solution; 2-D view axes 1 and 2

A clear separation results from an analysis of bioregions 13 and 14 based on all available data from the summer (**Figure 36a**) and the best available sites (**Figure 36b**).

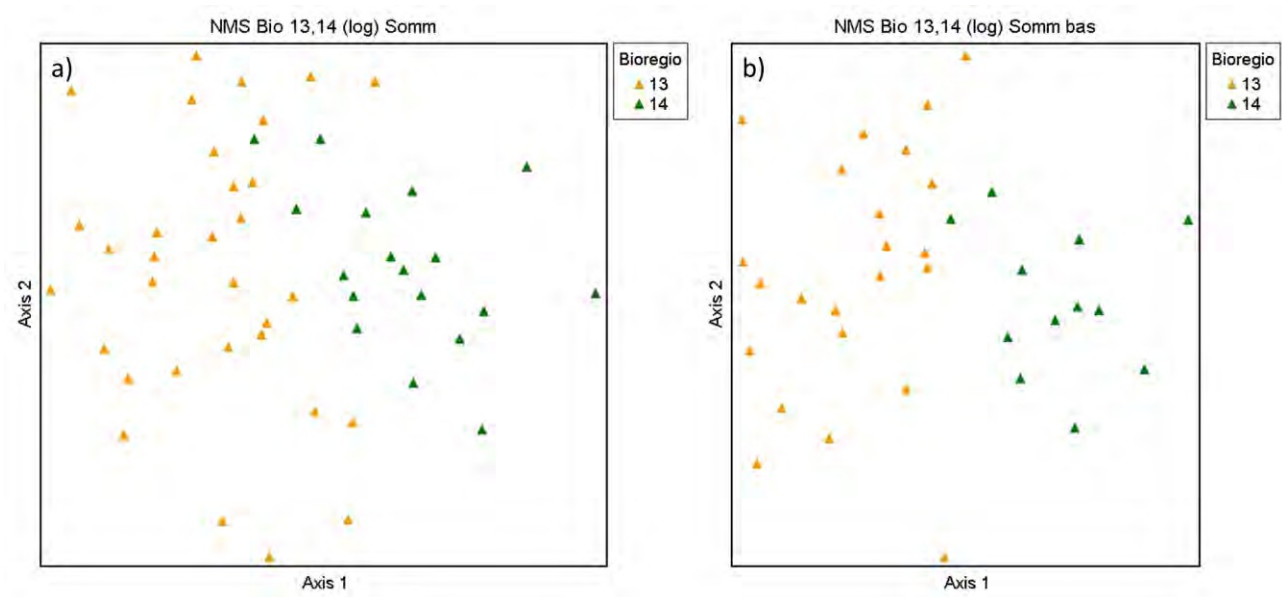


Figure 36: a) NMS scatter plot of the Grazer Feld and Grabenland bioregions (14, green) and Östliche Flach- und Hügelländer (13, orange) with "bioregions" overlay; $\log=\log(n+1)$ transformed abundances; Somm=survey sites in summer; $n=50$; 14.11888=final stress for 3-dimensional solution; 2-D view axes 1 and 2 b) NMS scatter plot of the bioregions Grazer Feld and Grabenland (14, green) and Östliche Flach- und Hügelländer (13, orange) with overlay "Bioregions"; $\log=\log(n+1)$ transformed abundances, Somm=test sites in summer; bas="best available sites"; $n=35$; 13.51827=final stress for 3-dimensional solution; 2-D view axes 1 and 2

When analyzing autumn data from bioregions 13 and 14, clear overlaps are visible. Both bioregions form a common point cloud (**Figure 37** a, b).

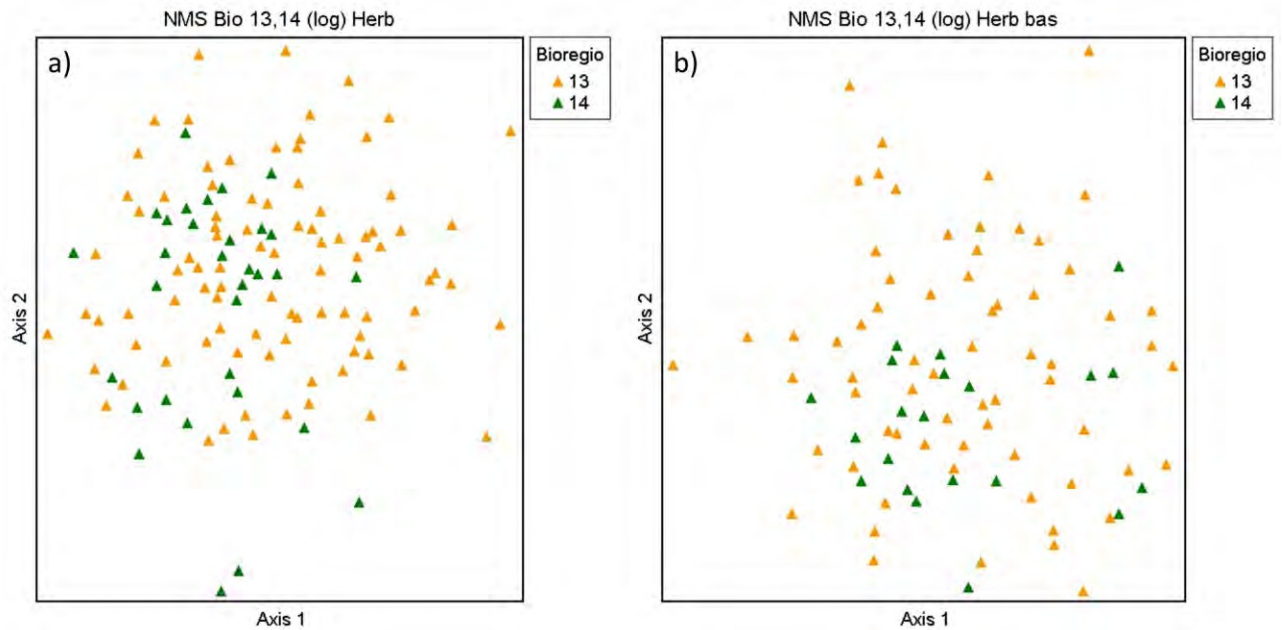


Figure 37: a) NMS scatter plot of the bioregions Grazer Feld and Grabenland (14, green) and Östliche Flach- und Hügelländer (13, red) with overlay "Bioregions"; log=log(n+1) transformed abundances; Herb=survey sites in autumn; n=122; 17.95290=final stress for 3-dimensional solution; 2-D view axes 1 and 2 **b)** NMS scatter plot of the bioregions Grazer Feld and Grabenland (14, green) and Östliche Flach- und Hügelländer (13, red) with overlay "Bioregions"; log=log(n+1) transformed abundances, Herb=investigation sites in autumn; bas="best available sites"; n=88; 16.87932=final stress for 3-dimensional solution; 2-D view axes 1 and 2

A clear biocenotic separation or clustering of the sampling sites with a summer aspect (June 1 to September 23) compared to sampling sites with a winter aspect (September 24 to May 31) is shown in **Figure 38a**, with the exception of two outliers, which are autumn samples. Analysis of the data for bioregion 14 by season reveals a clear clustering of the sampling sites in spring compared to summer and autumn, which lies above the points for spring and summer. Furthermore, within spring, the sampling sites Pfaffenbach and Glauningbach (red arrow) cluster separately from the others (**Figure 38a**). An analysis using the "best available sites" shows these distinctions between summer and winter and between the seasons even more clearly (**Figure 38b**).

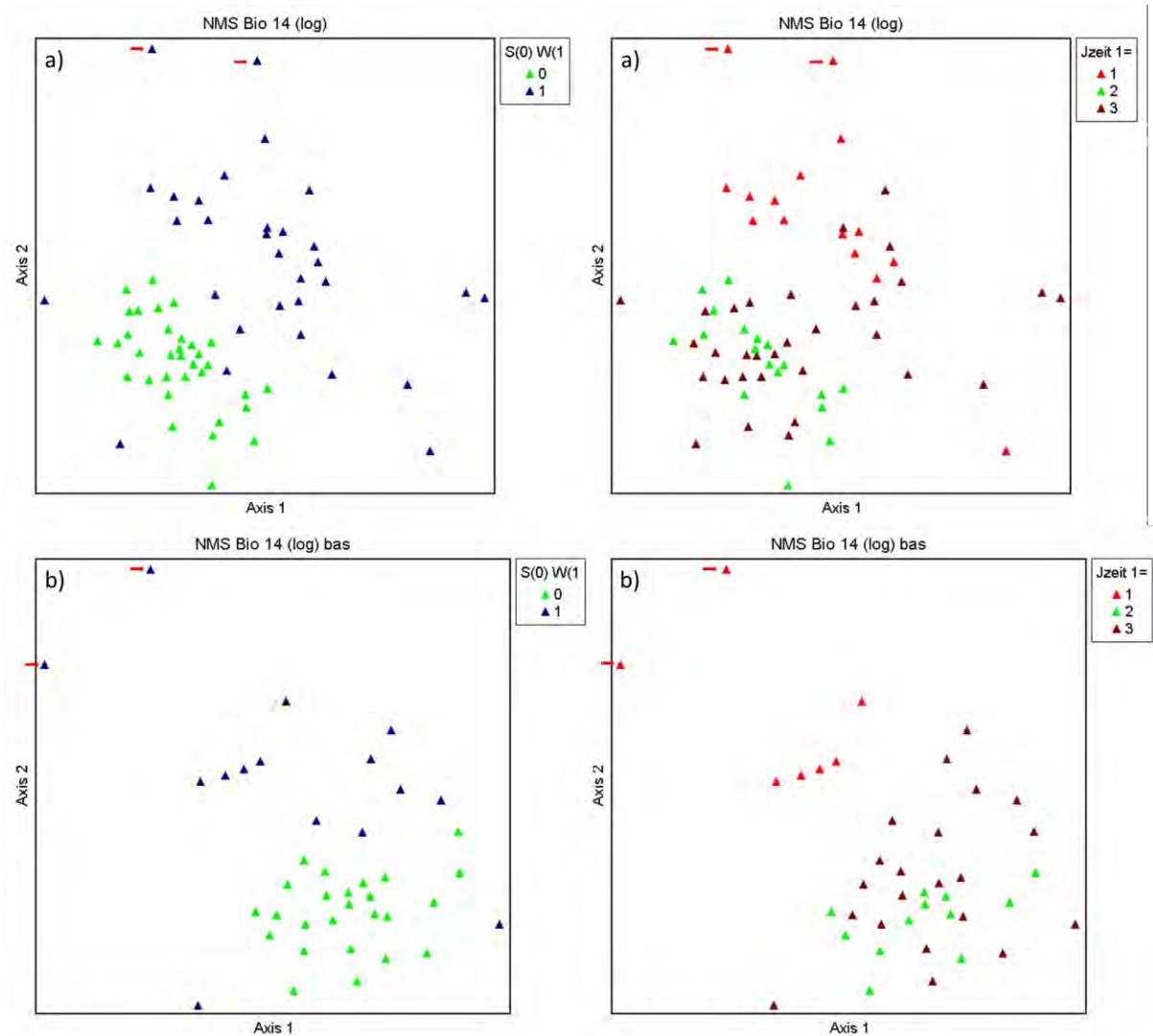


Figure 38: a) NMS scatter plot of the Grazer Feld and Grabenland bioregion (14) with "Season" overlay (0= summer aspect (June 1–September 23), 1= winter aspect (September 24-May 31) and with overlay "seasons" (1=spring, 2=summer, 3=autumn); 15.95133=final stress for 3-dimensional solution; 2-D view axes 1 and 2; log=log(n+1) transformed abundances; n=65; **b)** NMS scatter plot of the Grazer Feld and Grabenland bioregion (14) with overlay "Season" (0= summer aspect (June 1 to September 23), 1= winter aspect (September 24 to May 31); bas=best available sites and with overlay "seasons" (1=spring, 2=summer, 3=autumn); log=log(n+1) transformed abundances; bas=best available sites; 13.49975=final stress for 3-dimensional solution; 2-D view axes 1 and 2; n=40

A cluster analysis also shows that the spring sampling sites form a separate cluster, with sampling sites 59171 (Glauningbach) and 59178 (Pfaffenbach) splitting off (**Figure 39**, cluster 4). One cluster consists of pure

autumn samples (cluster 1), and a large cluster is formed from a mixture of summer and autumn investigation sites (cluster 2).

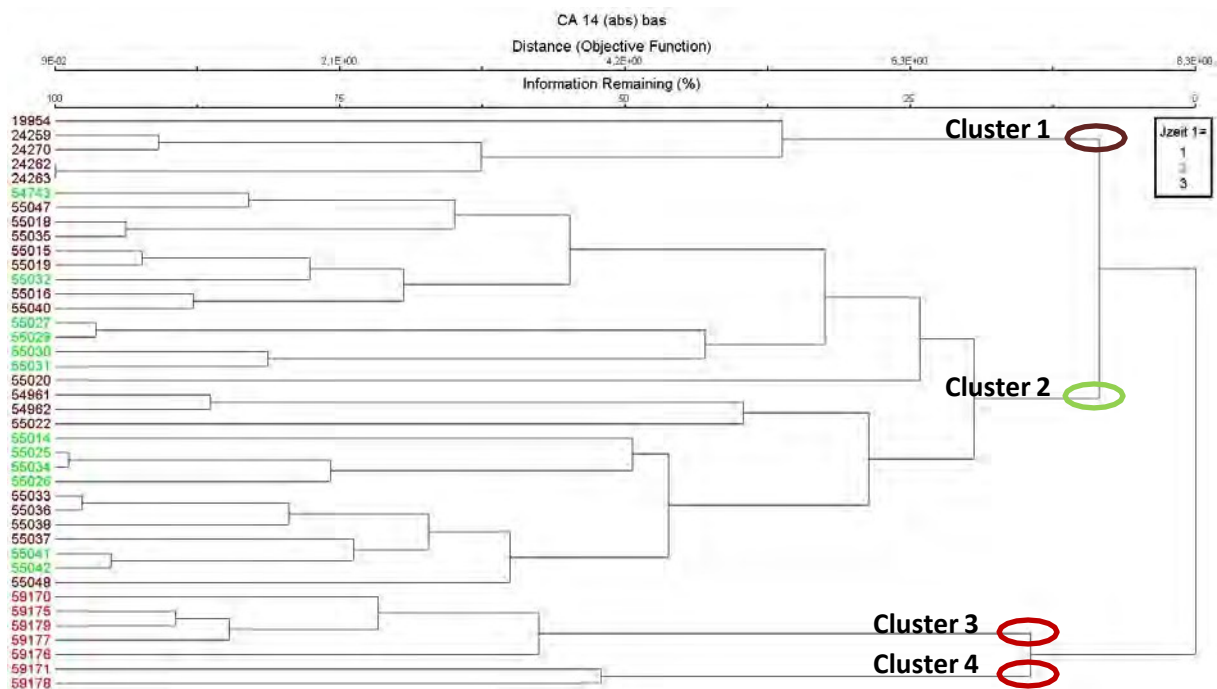


Figure 39: Cluster analysis of bioregion 14 "Grazer Feld and Grabenland" with seasonal overlay; 1 = spring, 2 = summer, 3 = fall; bas = best available sites; clusters 1-4 = manual classification into faunal types; abs = absolute abundances; bas = best available sites

3.5.3 Watercourse natural areas or watercourse landscapes as possible typological subdivision based on macrozoobenthos

A breakdown of the study sites into river landscapes (**Figure 40**) and river nature areas (**Figure 41**) does not reveal any clear clustering based on macrozoobenthic communities. The classification of the study sites into river nature areas shows a weak grouping of the river nature area

4.4.1 (Western Styrian Hills and Eastern Muric Grabenland).

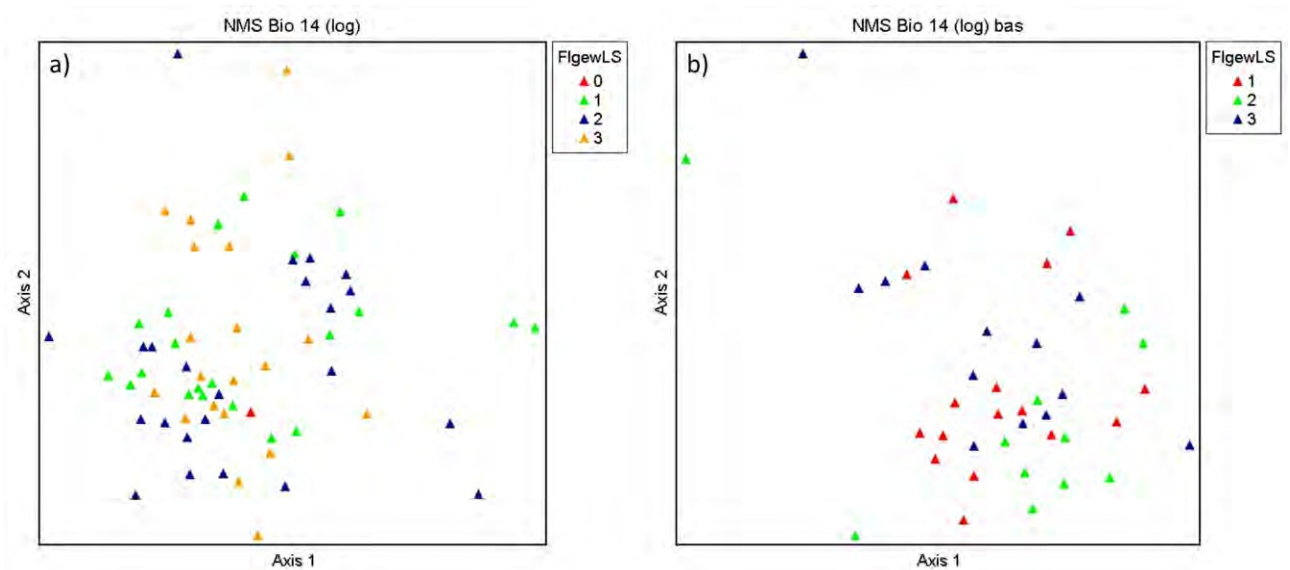


Figure 40: a) NMS scatter plot of the Grazer Feld and Grabenland bioregion (14) with overlay "watercourse landscapes"; 1=Western Styrian Hills, 2=Grazer Feld, 3=Eastern Murian Grabenland, 0=not clearly assignable; 15.95133=final stress for 3-dimensional solution; 2-D view axes 1 and 2; log=log(n+1) transformed abundances; n=65 **b)** NMS scatter plot of the Graz Field and Grabenland bioregion (14) with overlay "watercourse landscapes"; 1=Western Styrian hill country, 2=Grazer Feld, 3=Eastern Murian Grabenland; 13.49975=final stress for 3-dimensional solution; 2-D view axes 1 and 2; log=log(n+1) transformed abundances; bas=best available sites; n=40

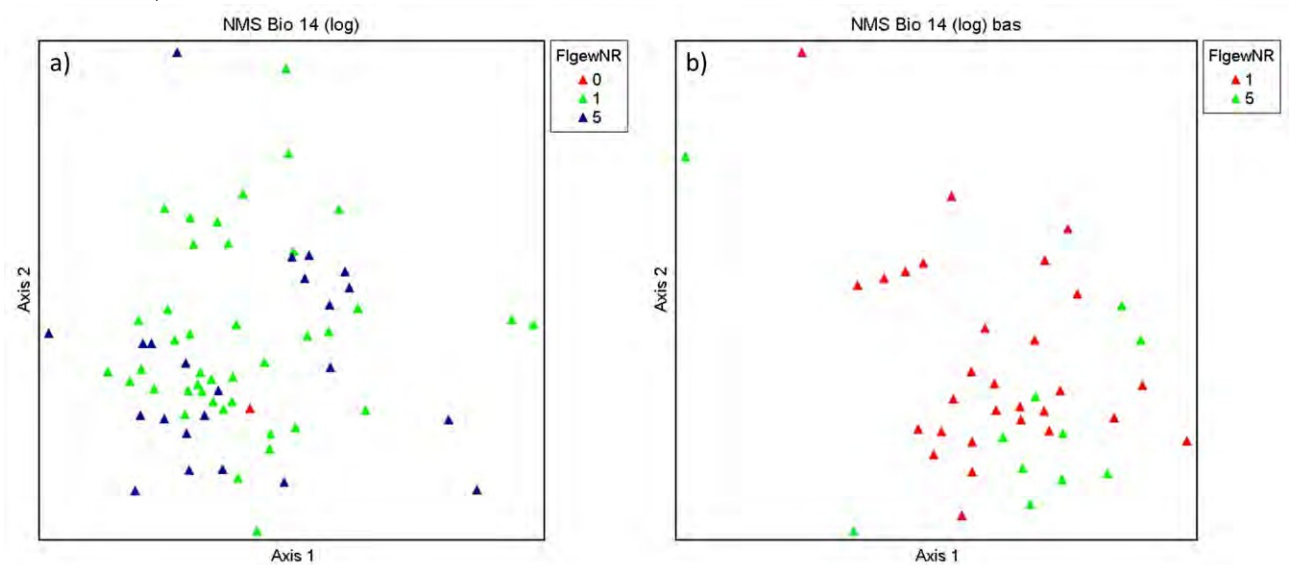


Figure 41: a) NMS scatter plot of the Graz Field and Grabenland bioregion (14) with overlay "watercourse natural areas"; 1=watercourse natural area 4.4.1 (Western Styrian Hills and Eastern Murian Grabenland), 5=watercourse natural area 4.4.5 (Grazer Feld including Leibnitzer, Murecker and Radkersburger Feld), 0=not clearly assignable; 15.95133=final stress for 3-dimensional solution; 2- D view axes 1 and 2; log=logarithmic abundances; n=65 **b)** NMS scatter plot of the Graz Field and Grabenland bioregion (14) with overlay "watercourse natural areas"; 1=watercourse natural area 4.4.1 (Western Styrian Hills and Eastern Murian Grabenland), 5=watercourse natural area 4.4.5 (Grazer Feld including Leibnitzer, Murecker and Radkersburger Feld); 13.49975=final stress for 3-dimensional solution; 2-D view axes 1 and 2; log=log(n+1) transformed abundances; bas=best available sites; n=40

3.5.4 Analysis of the 2016 sampling series in the Grazer Feld and Grabenland

The Stiefing and Aubachl streams mix cenotically in the group of streams with "good ecological status." Within the streams with moderate ecological status, the Glauningbach and Pfaffenbach streams stand out. The Pfaffenbach and Gnasbach clearly separate themselves from the rest of the study sites. Both show higher littoral proportions, with the Gnasbach exhibiting lower temperatures, increased fine sedimentation, and low diversity (**Figure 42**).

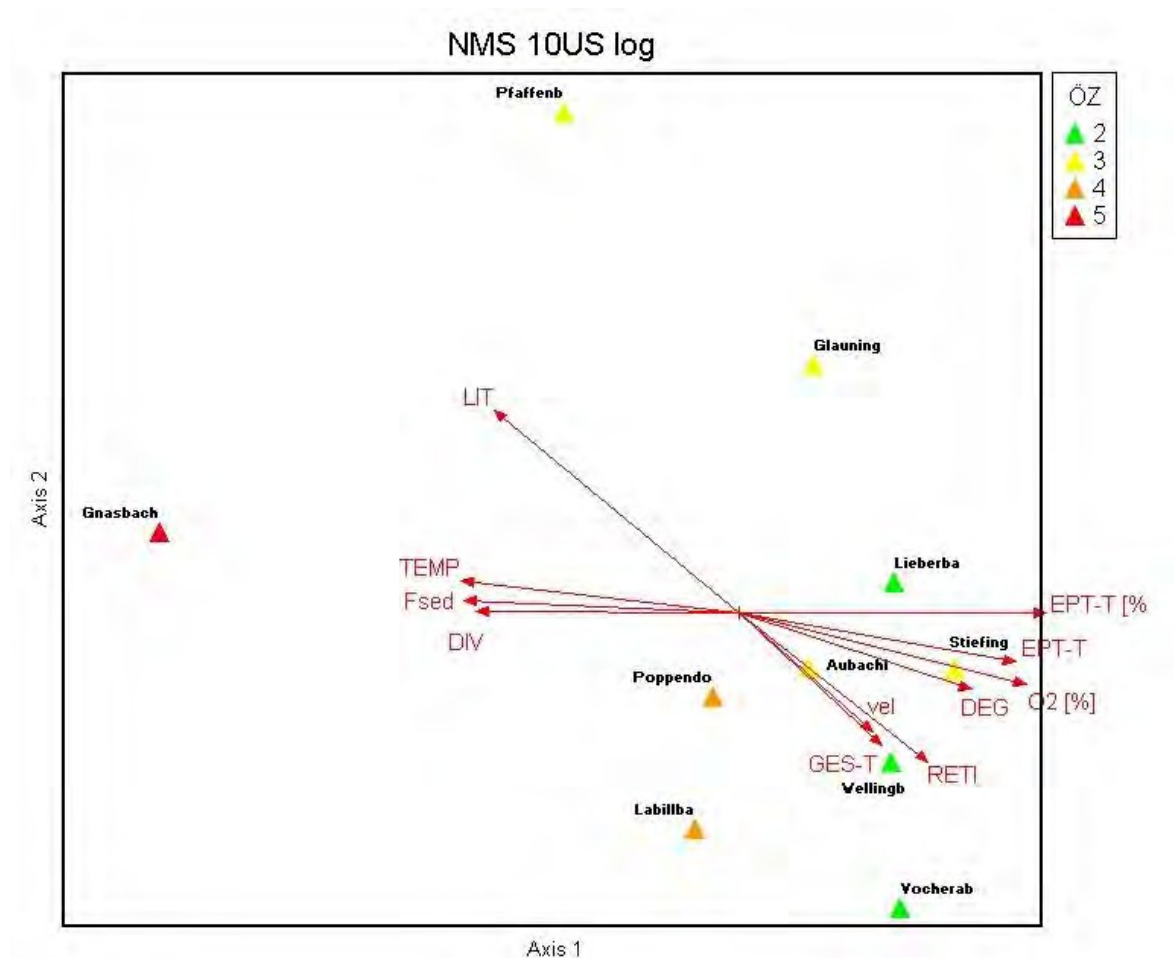


Figure 42: NMS scatter plot of the 10 study sites in the Grazer Feld and Grabenland bioregion (14) with "ecological status" overlay; 7.90906=final stress for 2-dimensional solution, vector cut-off value: $r^2=0.3$; ÖZ=ecological status, EPT-T=Ephemeroptera-Plecoptera-Trichoptera taxa, O2=oxygen saturation, DEG=degradation, vel=velocity, RETI=rhithral feeding type index, GES-T=total taxa, DIV=diversity, Fsed=fine sediment, TEMP=temperature, LIT=littoral; $r^{(2)} \geq 0.3$

The NMS analysis without the Gnasbach (old arm), which indicated the character of a standing water body at the time of the investigation, shows "groupings" of streams with "good" and "unsatisfactory" ecological status. In general, there is no compact distribution

of the points. The "outliers" – the Glauningbach and Pfaffenbach – show increased littoral proportions. Increased EPT-T, degradation index, total taxa, RETI, velocity, and oxygen saturation indicate a "good" or, in the case of the Stiefing, a "moderate" ecological status (**Figure 43**).

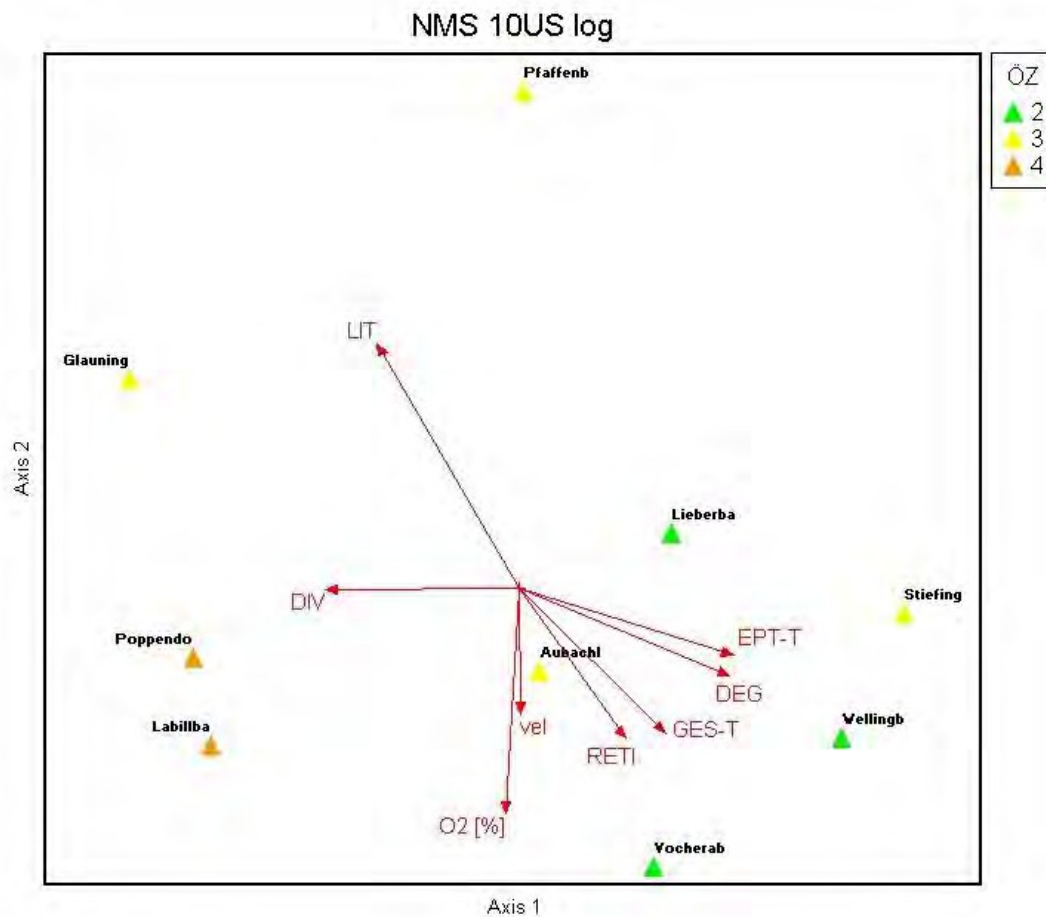


Figure 43: NMS scatter plot of the 10 study sites in the Grazer Feld and Grabenland bioregion (14) with overlay "ecological status" without outliers Gnasbach (old arm) (standing water at the time of the study), 7.10510=final stress for 2-dimensional solution, vector cut-off value: $r^2=0.3$; ÖZ=ecological status, EPT-T=Ephemeroptera-Plecoptera-Trichoptera taxa, DEG=degradation, GES-T=total taxa, RETI=rhithral feeding type index, vel=velocity, O2 [%]=oxygen saturation [%], DIV=diversity, LIT=littoral; $r^{(2)} \geq 0.3$

3.6 Indicator species analyses

The seasonal demarcation of bioregion 13 "eastern flat and hilly country" and bioregion 14 "Graz field and trench land" (**Figure 35, Figure 36**) represent indicator taxa in spring (**Table 20**) and summer (**Table 21**).

Table 20: Indicator species (indicator value >20) of bioregions 13 (eastern flat and hilly country) and 14 (Graz field and trench country) in spring

Bioregion 13			Bioregion 14		
Major group	Family	Genus/species	Large group	Family	Genus/species
Diptera	CHIRONOMIDAE	Odontomesa fulva	Ephemeroptera	LEPTOPHLEBIIDAE	Habrophlebia sp.
Diptera	CHIRONOMIDAE	Prodiamesa olivacea	Lumbricida	LUMBRICIDAE	Eiseniella tetraedra
Diptera	CHIRONOMIDAE	Parametriocnemus sp.	Ephemeroptera	HEPTAGENIIDAE	Rhithrogena picteti
Coleoptera	GYRINIDAE	Orectochilus villosus	Lumbriculida	LUMBRICULIDAE	Stylodrilus sp.
Trichoptera	HYDROPSYCHIDAE	Hydropsyche incognita/pellucidula	Ephemeroptera	HEPTAGENIIDAE	Ecdyonurus subalpinus
Tubificida	NAIDIDAE	Nais elinguis	Veneroida	PISIDIIDAE	Pisidium sp.
Trichoptera	HYDROPSYCHIDAE	Hydropsyche sp.	Coleoptera	HYDRAENIDAE	Hydraena sp.
Diptera	CHIRONOMIDAE	Brillia flavifrons	Diptera	CHIRONOMIDAE	Stictochironomus sp.
Tubificida	TUBIFICIDAE	Limnodrilus hoffmeisteri	Ephemeroptera	HEPTAGENIIDAE	Ecdyonurus sp.
Tubificida	TUBIFICIDAE	Tubifex tubifex	Ephemeroptera	HEPTAGENIIDAE	Electrogena ujhelyii
Diptera	CHIRONOMIDAE	Chironomini Gen. sp.	Plecoptera	NEMOURIDAE	Amphinemura sp.
Ephemeroptera	HEPTAGENIIDAE	Rhithrogena semicolorata group	Trichoptera	GOERIDAE	Lithax obscurus
Lumbricida	PROPAPPIDAE	Propappus volki	Tubificida	TUBIFICIDAE	Rhyacodrilus coccineus

Table 21: Indicator species (indicator value >20) of bioregions 13 (eastern flat and hilly country) and 14 (Graz field and trench country) in summer

Bioregion 13			Bioregion 14		
Major group	Family	Genus/species	Large group	Family	Genus/species
Diptera	SIMULIIDAE	Simulium sp.	Veneroida	PISIDIIDAE	Pisidiidae Gen. sp.
Ephemeroptera	BAETIDAE	Baetis sp.	Diptera	TABANIDAE	Tabanidae Gen. sp.
Ephemeroptera	EPHEMERELLIDAE	Ephemerella ignita	Coleoptera	ELMIDAE	Elmis maugetii
Ephemeroptera	BAETIDAE	Baetis fuscatus	Diptera	CHIRONOMIDAE	Microtendipes sp.
Diptera	CHIRONOMIDAE	Tvetenia bavarica/calvescens	Coleoptera	ELMIDAE	Limnius volckmari
Diptera	ATHERICIDAE	Atherix ibis	Diptera	CHIRONOMIDAE	Prodiamesa olivacea
Trichoptera	HYDROPSYCHIDAE	Hydropsyche modesta	Coleoptera	ELMIDAE	Oulimnius tuberculatus
Diptera	CHIRONOMIDAE	Saetheria reissi	Gordiida	GORDIIDAE	Gordiidae Gen. sp.
Ephemeroptera	BAETIDAE	Baetis buceratus	Ephemeroptera	EPHEMERIDAE	Ephemera danica
[Kl:Nematoda]	[Kl:Nematoda]	Nematoda Gen. sp.	Coleoptera	HYDRAENIDAE	Hydraena riparia
Diptera	SIMULIIDAE	Simulium equinum	Trichoptera	LEPTOCERIDAE	Athripsodes sp.
Ephemeroptera	OLIGONEURIIDAE	Oligoneuriella rhenana	Trichoptera	LEPTOCERIDAE	Leptoceridae Gen. sp.
Diptera	CERATOPOGONIDAE	Bezzia sp.	Tubificida	TUBIFICIDAE	Psammoryctides albicola

In autumn, there is no clear seasonal separation between bioregion 13 and bioregion 14 (**Figure 37**). No significant indicator species for the respective bioregions could be identified in autumn.

Within Bioregion 14 (Grazer Feld and Grabenland), the suspicion of intermittent streams can be confirmed based on cluster classification 1 to 4 (**Figure 39**). For the Glauningbach and Pfaffenbach streams, the Ephemeroptera indicator species

Siphonurus armatus, *S. aestivalis* and the taxon *Habrophlebia* sp. and Trichoptera indicator species *Ironoquia dubia*, *Oligostomis reticulata* and *Glyphotaelius pellucidus* (**Table 22**).

Table 22: ISA - significant ($p \leq 0.05$ or indicator value (IV) ≥ 50) indicator taxa per cluster (**Figure 39**)

Major group	Family	Genus/species	Cluster	IV	p
Trichoptera	LIMNephilidae	Drusinae Gen. sp.	1	1	0.0002
Trichoptera	Beraeidae	Beraeodes minuta		10	0.0002
Coleoptera	Dytiscidae	Platambus maculatus		83	0.0002
Ephemeroptera	Ephemeridae	Ephemera sp.		83.4	0.0028
Coleoptera	Elmidae	Elmis rietscheli		80	0.0028
Diptera	Chironomidae	Polypedilum scalaenum-Gr.		80	0.0036
Tubificida	Tubificidae	Limnodrilus sp.		60	0.0054
Tubificida	Tubificidae	Aulodrilus plurisetia		79.8	0.0064
Coleoptera	Elmidae	Limnius sp.		70	0.011
Ephemeroptera	Leptophlebiidae	Leptophlebiidae Gen. sp.		58.2	0.018
Ephemeroptera	Leptophlebiidae	Habrophlebia lauta		57	0.0218
Tubificida	Tubificidae	Tubifex tubifex		59	0.0282
Arhynchobdellida	Erpobdellidae	Erpobdella octoculata		51.2	0.0336
Ephemeroptera	Leptophlebiidae	Paraleptophlebia submarginata		56.1	0.0342
Diptera	Chironomidae	Natarsia sp.		65.9	0.0354
Ephemeroptera	Caenidae	Caenis sp.		57.3	0.0368
Diptera	Chironomidae	Brillia flavifrons		51.4	0.0372
Odonata	Calopterygidae	Calopteryx virgo		62.5	0.0386
Plecoptera	Nemouridae	Nemoura sp.		61	0.0386
Ephemeroptera	Heptageniidae	Ecdyonurus sp.		72.3	0.049
Tubificida	Tubificidae	Tubificidae Gen. sp.	2	92	0.0002
Coleoptera	Elmidae	Elmis maugetii		89.9	0.0004
Veneroida	Pisidiidae	Pisidiidae Gen. sp.		82.1	0.0012
Ephemeroptera	Baetidae	Baetis vernus		67.9	0.0162
Coleoptera	Elmidae	Limnius volckmari		74.3	0.033
Amphipoda	Gammaridae	Gammarus sp.		64.6	0.054
Ephemeroptera	Heptageniidae	Rhithrogena picteti	3	100	0.0002
Trichoptera	Limnephilidae	Limnephilidae Gen. sp.		87.8	0.0006
Trichoptera	Limnephilidae	Halesus sp.		60	0.007
Plecoptera	Nemouridae	Protonemura sp.		58.9	0.011
Diptera	Chironomidae	Rheocricotopus fuscipes		62.7	0.031
Coleoptera	Hydraenidae	Hydraena sp.		73.3	0.033
Ephemeroptera	Heptageniidae	Electrogena ujhelyii		64.5	0.033
Ephemeroptera	Heptageniidae	Ecdyonurus subalpinus		49.5	0.041
Ephemeroptera	Baetidae	Baetis rhodani		66.9	0.048
Ephemeroptera	Siphonuridae	Siphonurus aestivalis	4	100	0.001
Lumbriculida	Lumbriculidae	Lumbriculus variegatus		10	0.001
Ephemeroptera	Siphonuridae	Siphonurus armatus		99.8	0.001
Trichoptera	Limnephilidae	Ironoquia dubia		99.1	0.001
Tubificida	Tubificidae	Emboloccephalus velutinus		98.6	0.0018
Isopoda	Asellidae	Asellus aquaticus		81.8	0.0058
Ephemeroptera	Leptophlebiidae	Habrophlebia sp.		86.9	0.0066
Tubificida	Tubificidae	Rhyacodrilus coccineus		82.5	0.0072
Haplotaxida	Haplotaxidae	Haplotaxis gordioides		49.3	0.0332
Lumbriculida	Lumbriculidae	Rhynchelmis limosella		48.6	0.0406
Diptera	Pediciidae	Pediciidae Gen. sp.		50	0.048
Heteroptera	Velidae	Velia sp.		50	0.0482
Lumbriculida	Lumbriculidae	Bythonomus lemani		50	0.0482
Seriata	Planariidae	Planaria sp.		50	0.0482
Diptera	Chironomidae	Pentaneurini Gen. sp.		43.1	0.048
Coleoptera	Hydrophilidae	Hydrophilidae Gen. sp.		50	0.0518
Diptera	Chironomidae	Orthocladus rivulorum		50	0.0518
Diptera	Chironomidae	Parachironomus sp.		50	0.0518
Diptera	Chironomidae	Psectrocladius psilopterus		50	0.0518
Pulmonata	Planorbidae	Planorbidae Gen. sp.		50	0.0518
Pulmonata	Lymnaeidae	Radix ovata/peregra		50	0.0518
Trichoptera	Phryganeidae	Oligostomis reticulata		50	0.0518
Trichoptera	Limnephilidae	Glyptotaelius pellucidus		50	0.051

3.7 Seasonal response of different metrics in bioregion 14 "Grazer Feld and Grabenland"

In order to show the reactions of selected metrics within an ecological status class to the seasons or within a season to the different ecological status classes, the investigation sites were analyzed using box plots based on all metrics included and not included in the current detailed assessment.

3.7.1 Analysis of all investigation sites

The degradation index shows no significant differences within the ecological status classes over the course of the year (**Figure 44**). In spring, there are significant differences between the status classes good and poor ($p=0.021$) and between good and unsatisfactory ($p=0.011$). In summer, there are significant differences between good and unsatisfactory ($p=0.009$) and significant differences between moderate and unsatisfactory ($p=0.029$) and between good and poor ($p=0.031$). In autumn, the condition class good differs significantly from moderate ($p=0.036$) and unsatisfactory ($p<0.001$). There are also significant differences between moderate and unsatisfactory ($p=0.017$). In spring, there is a very large variance within the moderate survey sites.

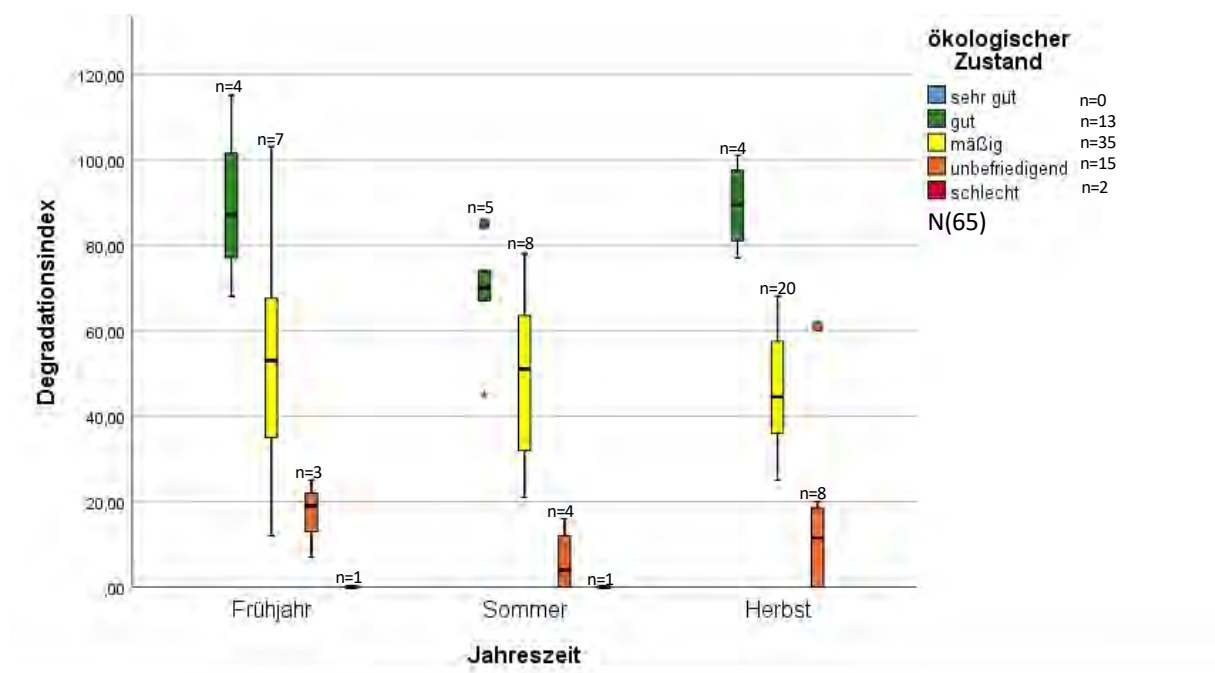


Figure 44: Box plot of the metric degradation index from different sampling sites in bioregion 14 with the assessed ecological status, grouped by sampling time (season).

The Rhithron nutrition type index shows seasonal stability with no significant differences (**Figure 45**). In autumn, there are significant differences between good and unsatisfactory ($p=0.009$) and between moderate and unsatisfactory ($p=0.019$). In all seasons, there are large variations within the sampling sites with moderate ecological status.

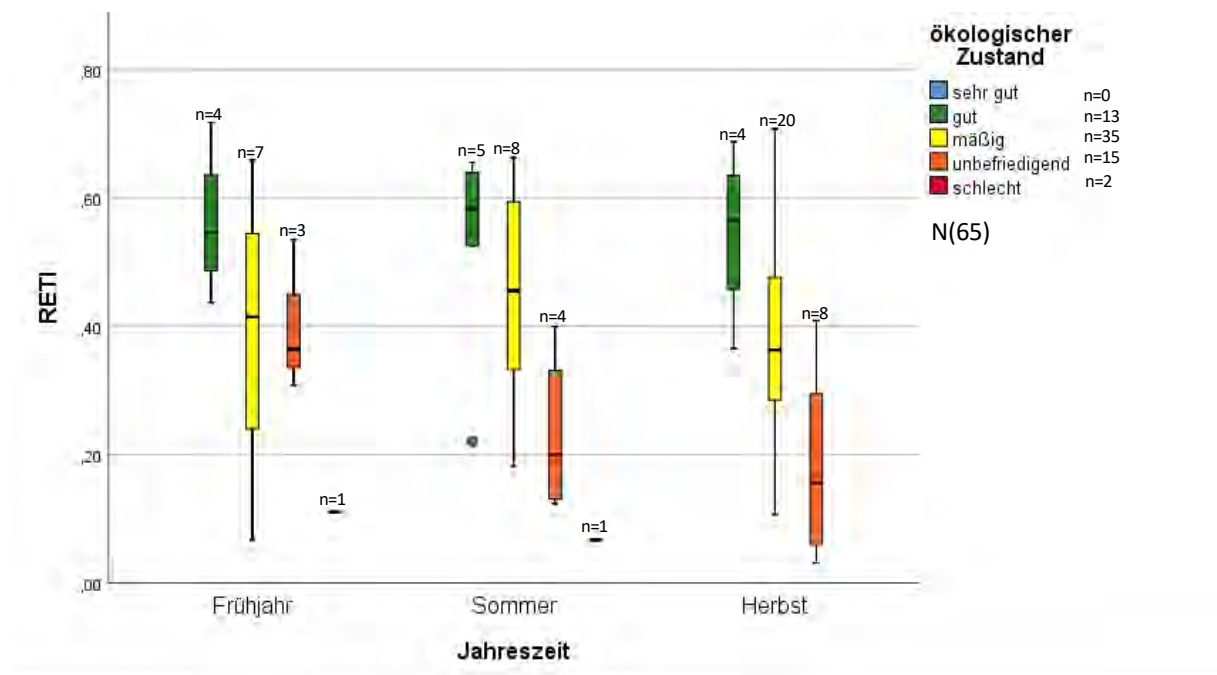


Figure 45: Box plot of the Rhithron nutrition type index (RETI) metric from different sampling sites in bioregion 14 with the assessed ecological status, grouped by sampling time (season).

The metric "total number of taxa" shows no significant differences within the ecological status classes over the course of the year (**Figure 46**). In summer, there are significant differences between the status classes good and unsatisfactory ($p=0.018$) and between good and poor ($p=0.040$). In autumn, there are significant differences between good and unsatisfactory ($p=0.047$).

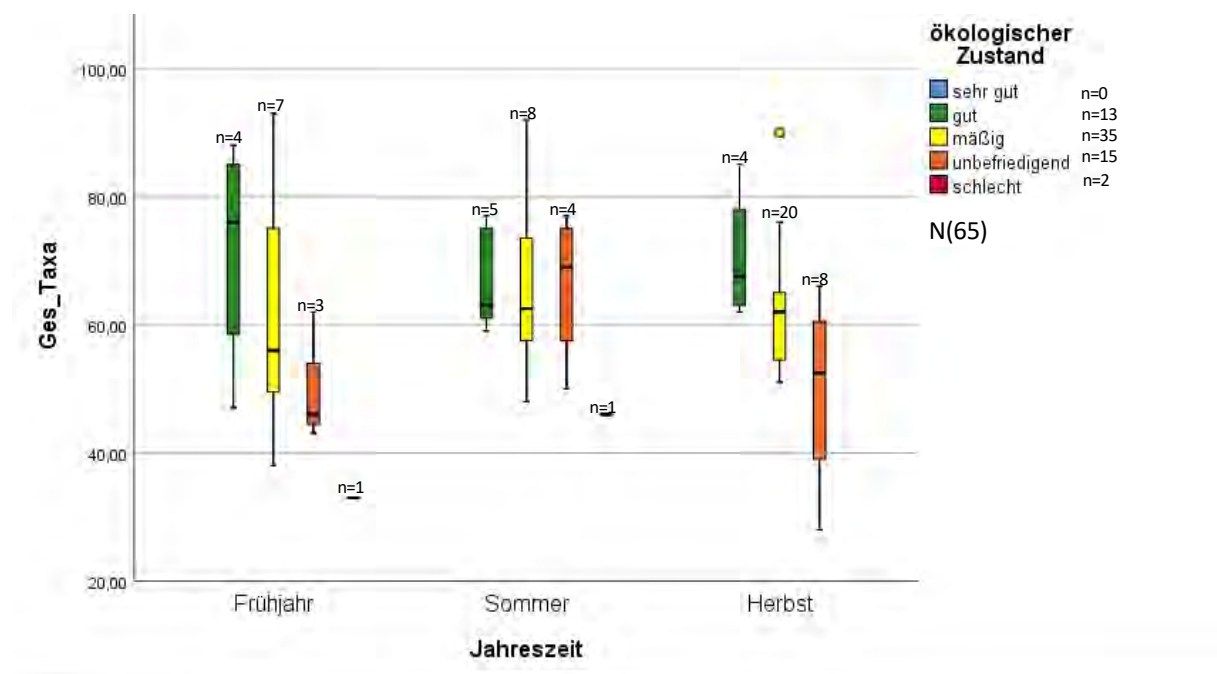


Figure 46: Box plot of the metric "Total number of taxa from different sampling sites in bioregion 14 with the assessed ecological status, grouped by sampling time (season).

The metric number of EPT taxa shows no significant differences within the ecological status classes, indicating seasonal stability (**Figure 47**). In spring, there are significant differences between the ecological status classes good and unsatisfactory ($p=0.044$) and between good and poor ($p=0.026$). In summer, there are significant differences between good and unsatisfactory ($p=0.038$) and between good and poor ($p=0.032$) as well as between moderate and unsatisfactory ($p=0.028$). In autumn, there are significant differences between the classes good and moderate ($p=0.049$), between good and unsatisfactory ($p<0.001$) and between moderate and unsatisfactory ($p=0.035$).

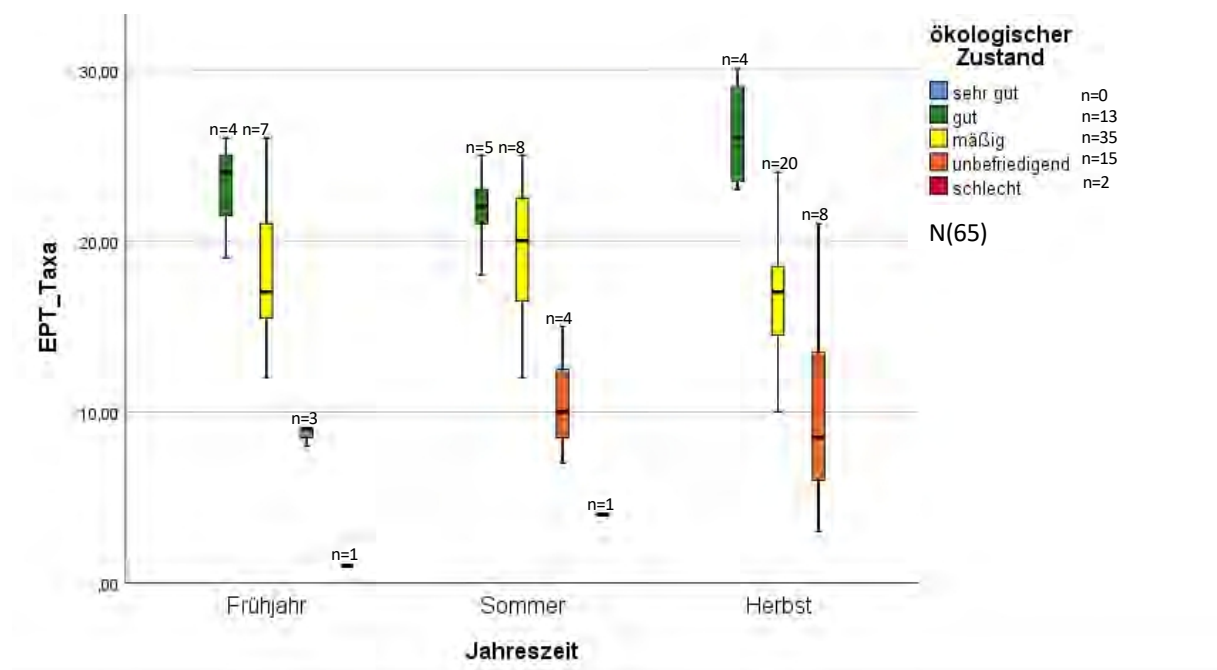


Figure 47: Box plot of the metric number of EPT taxa from different sampling sites in bioregion 14 with the assessed ecological status, grouped by sampling time (season). EPT=Ephemeroptera, Plecoptera, Trichoptera

The metric EPT taxa [%], see **Figure 48**, shows no significant differences within the ecological status classes over the course of the year. In spring, the status classes good and unsatisfactory ($p=0.016$) and good and poor ($p=0.040$) differ significantly, as do the classes moderate and unsatisfactory ($p=0.042$). In summer, there is a significant difference between good and unsatisfactory ($p=0.024$) and between good and poor ($p=0.029$), as well as a significant difference between moderate and unsatisfactory ($p=0.024$). In autumn, the condition class good differs significantly from moderate ($p=0.030$) and unsatisfactory ($p=0.002$).

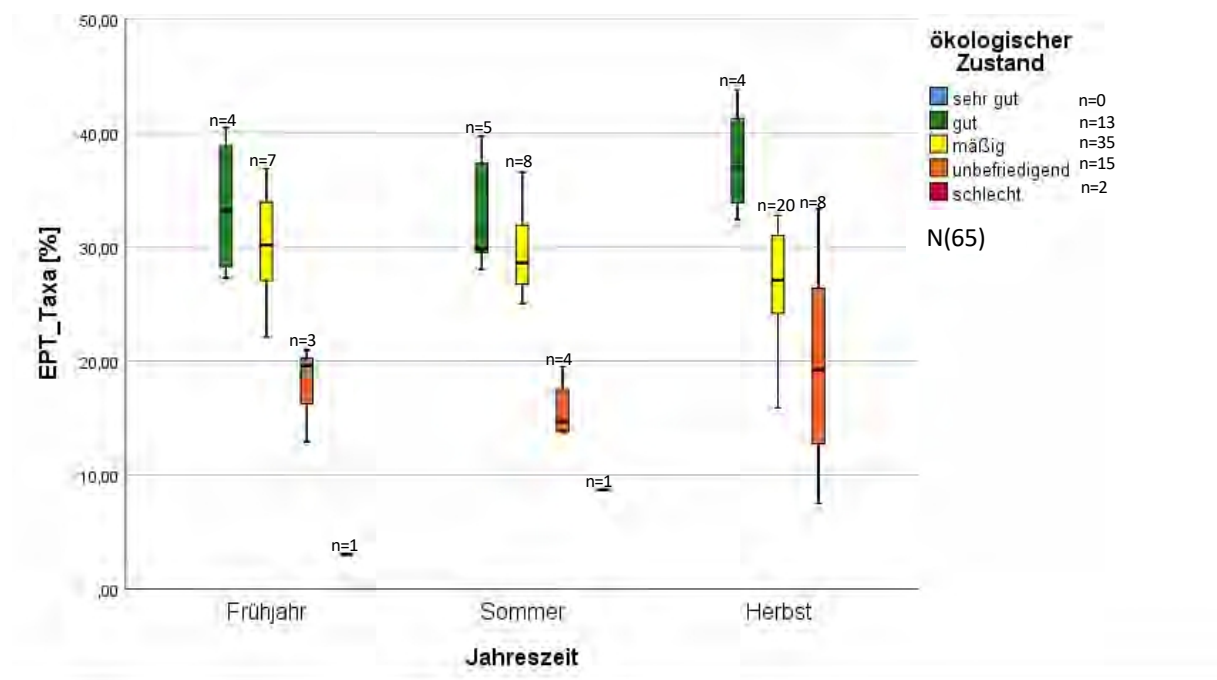


Figure 48: Box plot of the metric number of EPT taxa [%] from different sampling sites in bioregion 14 with the assessed ecological status, grouped by sampling time (season). EPT=Ephemeroptera, Plecoptera, Trichoptera

The littoral proportions show stability over the course of the year (**Figure 49**), as there are no significant differences within the ecological statuses. In spring, the status class good differs significantly from moderate ($p=0.012$), unsatisfactory ($p=0.045$) and poor ($p=0.012$). In summer, there are significant differences between the good and unsatisfactory classes ($p=0.012$) and between good and poor ($p=0.024$). In autumn, there is considerable variation within the study sites with unsatisfactory ecological status and moderate status in summer and autumn.

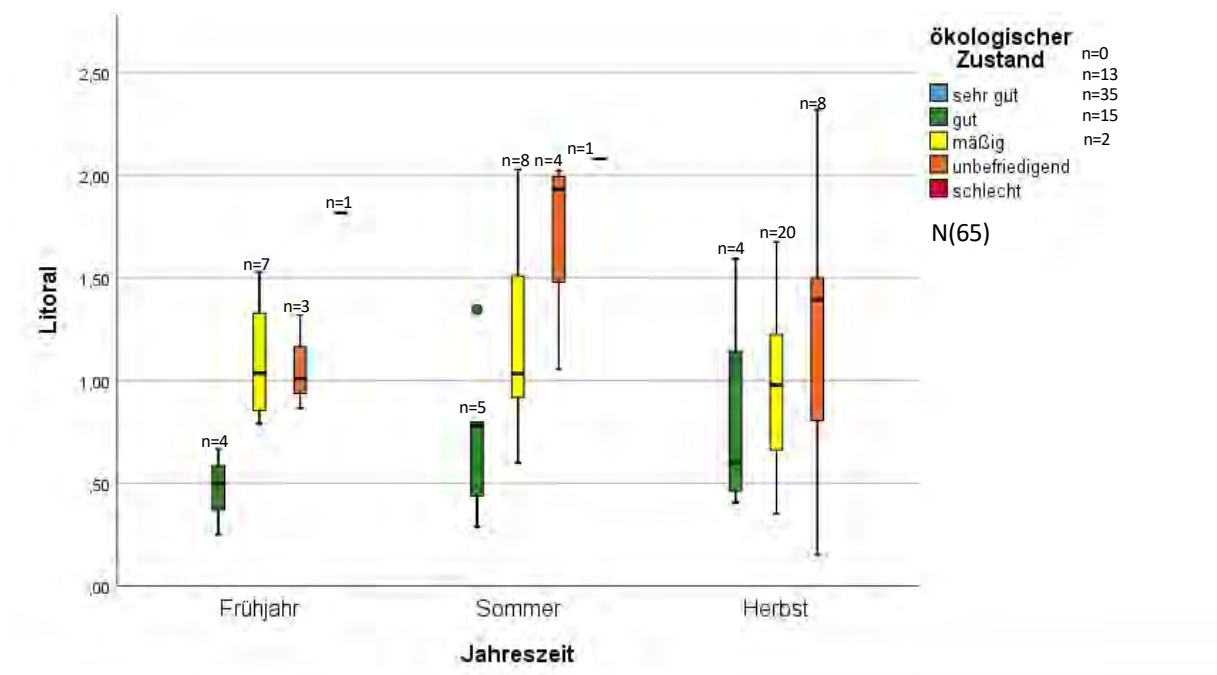


Figure 49: Box plot of the metric littoral proportions from different sampling sites in Bioregion 14 with the assessed ecological status, grouped by sampling time (season).

The metric Oligochaeta_Diptera_Taxa [%] remains stable throughout the seasons within the ecological conditions. In autumn, there are significant differences (**Figure 50**) between good and unsatisfactory ($p=0.031$) and between moderate and unsatisfactory ($p=0.010$). There are large variations within the moderate ecological statuses, especially in autumn.

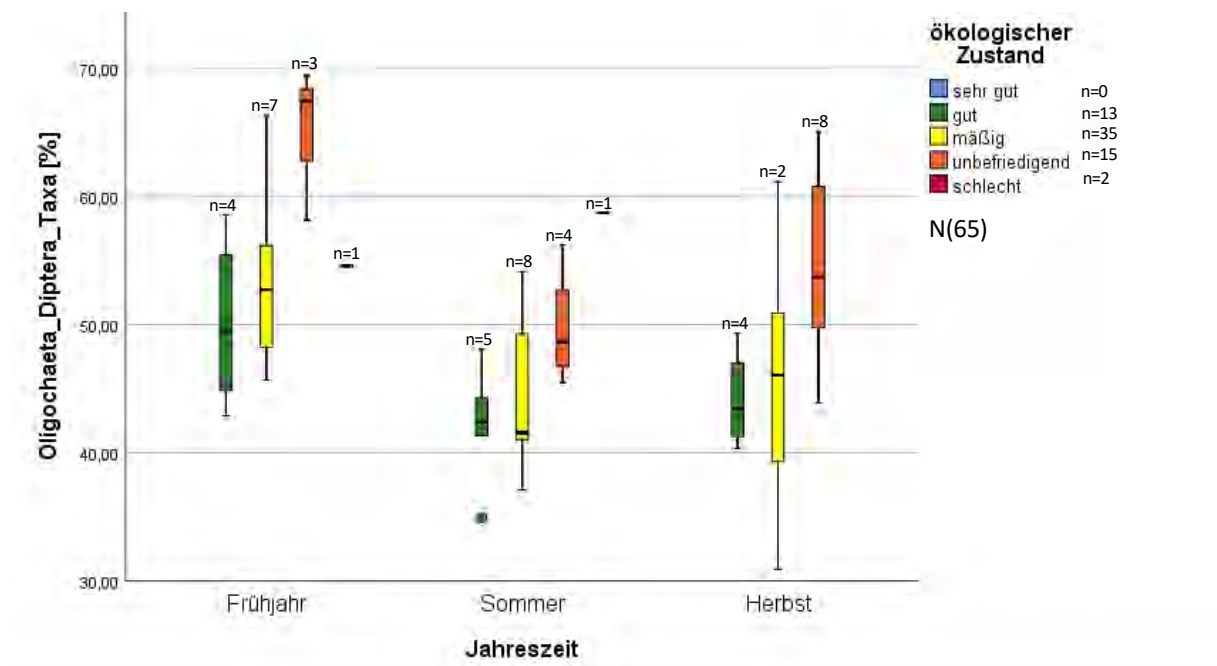


Figure 50: Box plot of the metric Oligochaeta&Diptera taxa [%] from different sampling sites in bioregion 14 with the assessed ecological status, grouped by sampling time (season).

The diversity index shows no significant differences (**Figure 51**) across seasons within the ecological status classes. There are also no significant differences between the ecological status classes within a given season. There are large variations within the study sites with moderate ecological status.

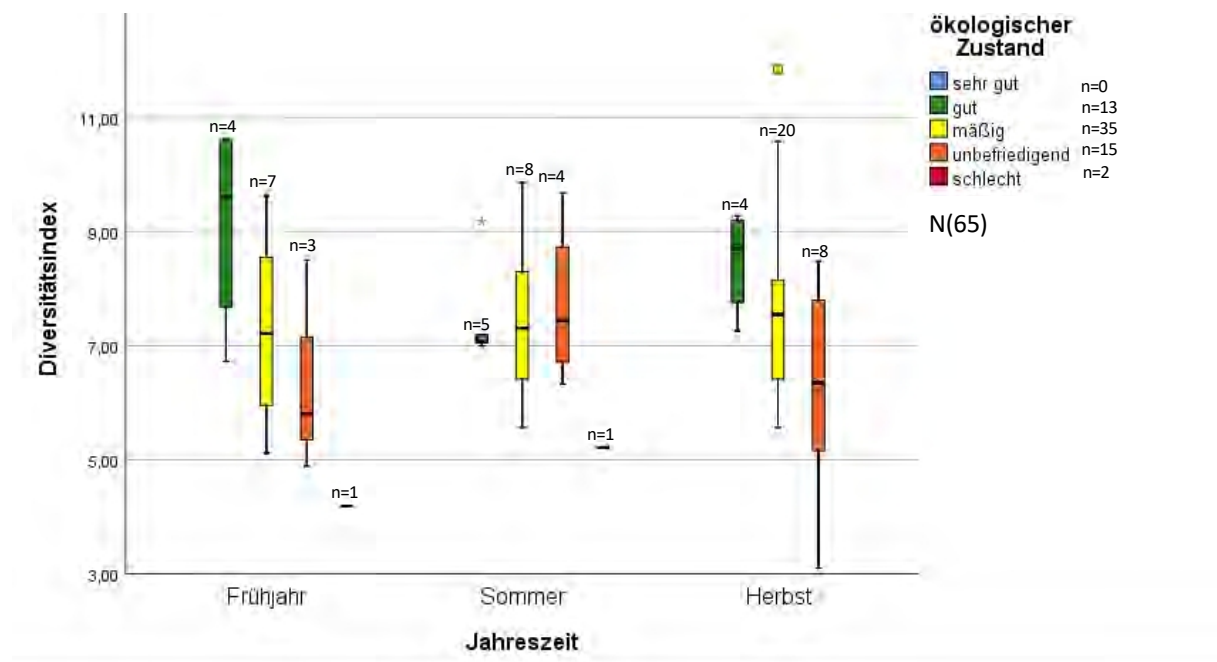


Figure 51: Box plot of the diversity index metric from different sampling sites in Bioregion 14 with the assessed ecological status, grouped by sampling time (season).

3.7.2 Analysis of the "best available sites"

Based on the "best available sites" (Chapter 2.5), the investigation sites were now analyzed using clusters 1 to 4 of the cluster analysis (**Figure 39**).

The degradation index (**Figure 52**) shows a very clear difference for cluster 4 (spring, intermittent) compared to the other investigation sites in spring (cluster 3). Clusters 2 and 1 show almost identical values within the moderate conditions with different variances.

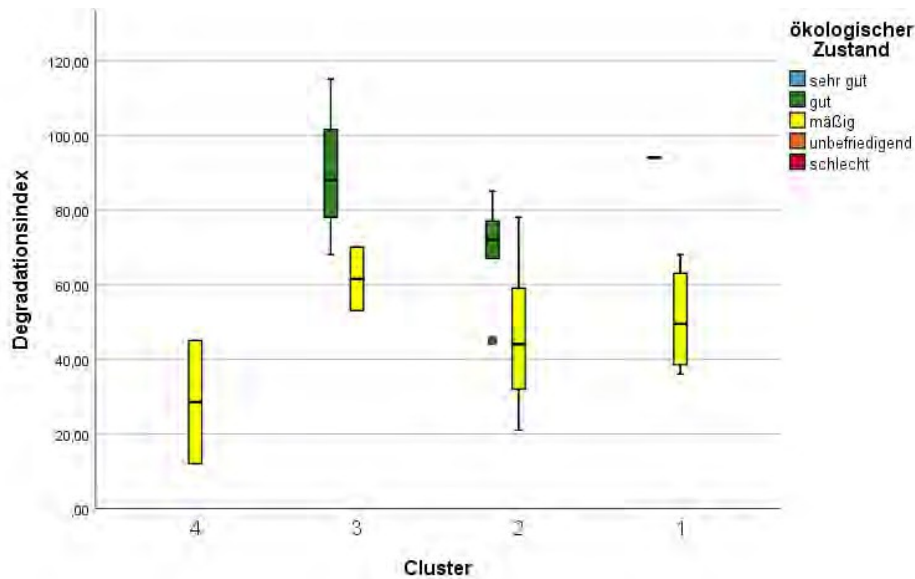


Figure 52: Box plot of the degradation index metric based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

The Rhithron nutrition type index (**Figure 53**) shows clear differences between cluster 4 and cluster 3. Clusters 2 and 1 show similar values within the ecological status classes, with the moderate status showing a large variance. The perennial waters in spring (cluster 3) show a very high index, which characterizes the rhithral.

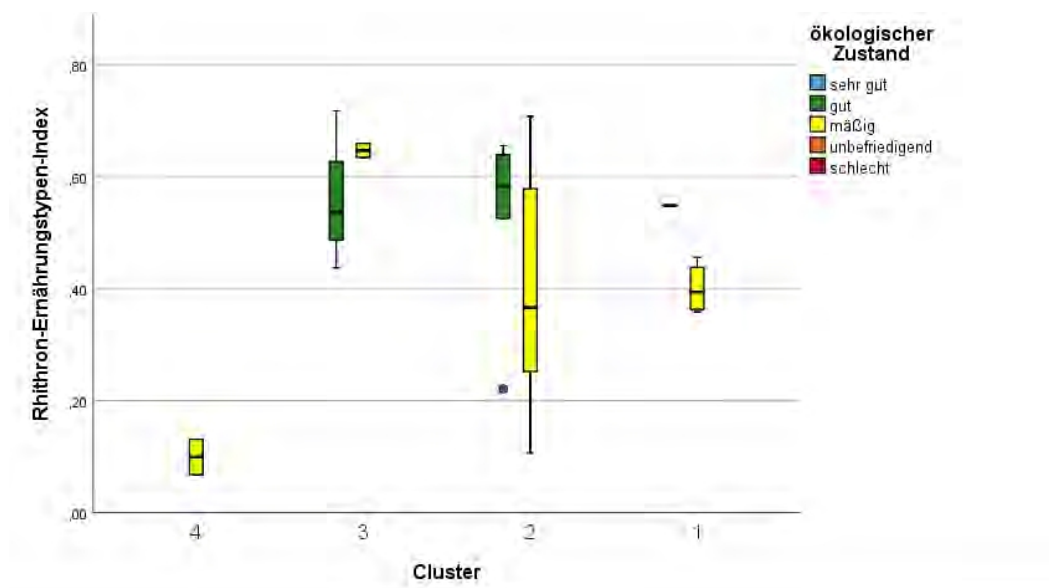


Figure 53: Box plot of the Rhithron nutrition type index (RETI) metric based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

The metric Total taxa shows a lower value within the intermittent study sites (cluster 4) compared to cluster 3, cluster 2, and cluster 1, which has the highest values.

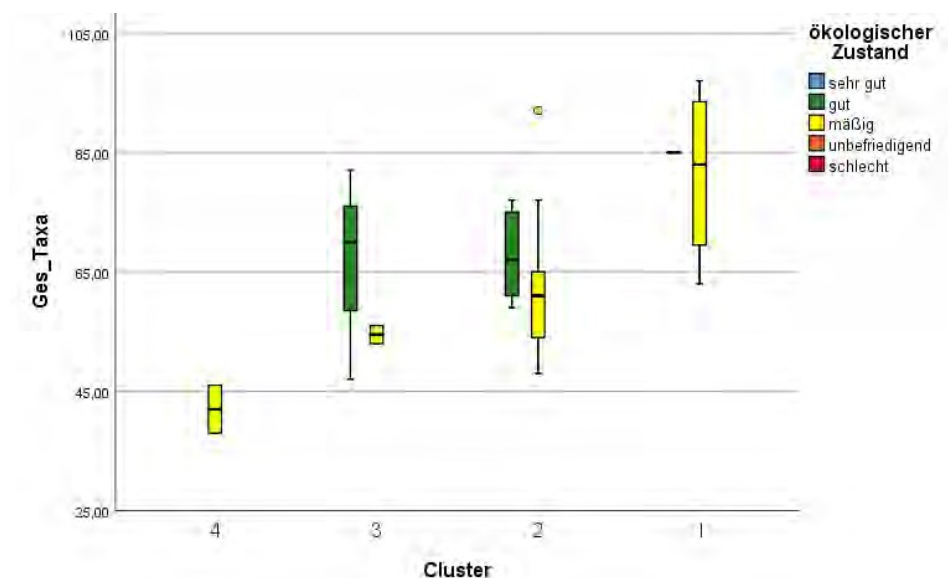


Figure 54: Box plot of the metric total number of taxa based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

The EPT taxa metric shows clear differences between intermittent (cluster 4) and perennial water bodies (cluster 3) (**Figure 55**). Cluster 2 shows a large variance within the moderate ecological status class.

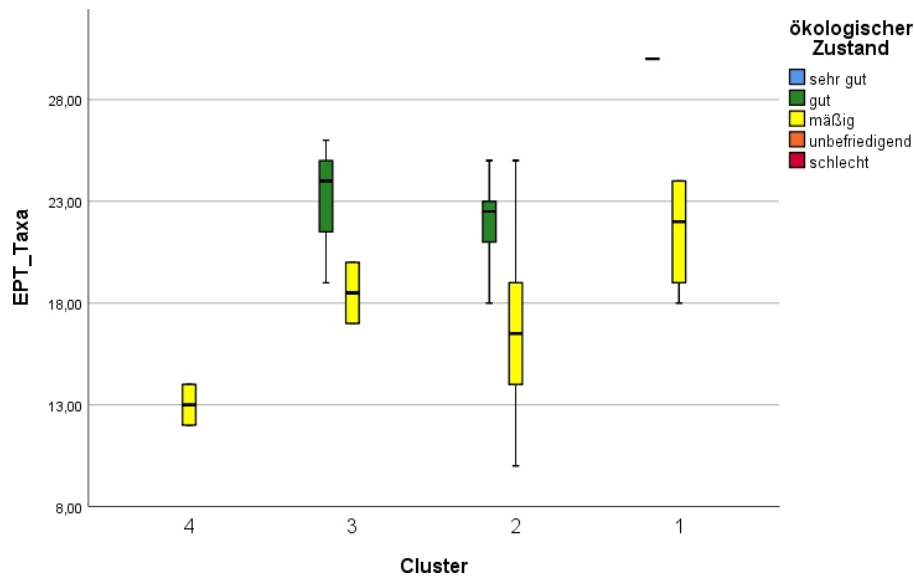


Figure 55: Box plot of the metric number of EPT taxa (Ephemeroptera, Plecoptera, Trichoptera) based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

The metric EPT taxa [%] (**Figure 56**) shows no clear differences between the clusters. Both within the good and moderate conditions, the median values of clusters 4 and 3 are slightly above those of clusters 2 and 1.

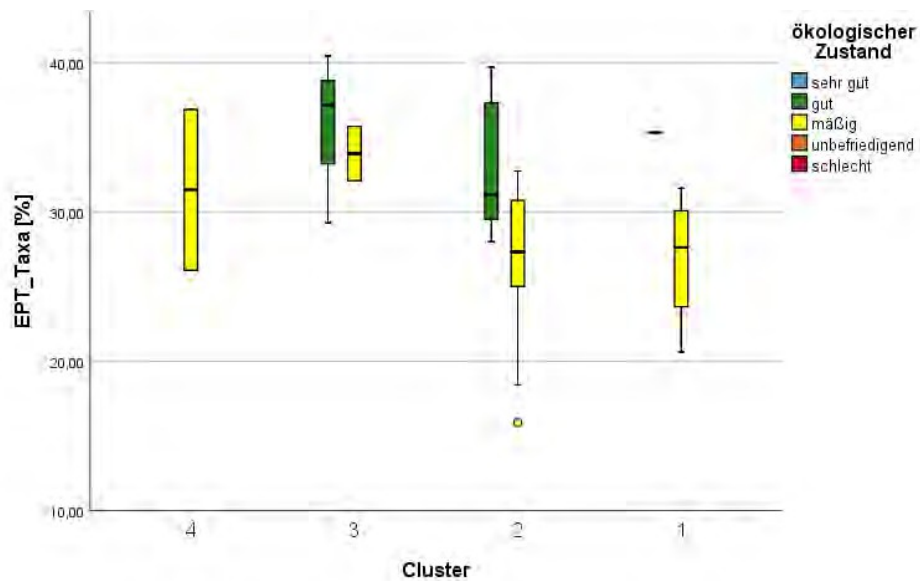


Figure 56: Box plot of the metric EPT taxa [%] (Ephemeroptera, Plecoptera, Trichoptera) based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

The metric "littoral proportion" shows large differences between intermittent (cluster 4) and perennial (cluster 3) water bodies in spring. The median littoral proportions of cluster 3 (spring) are slightly below the values of clusters 2 and 1 (summer/autumn).

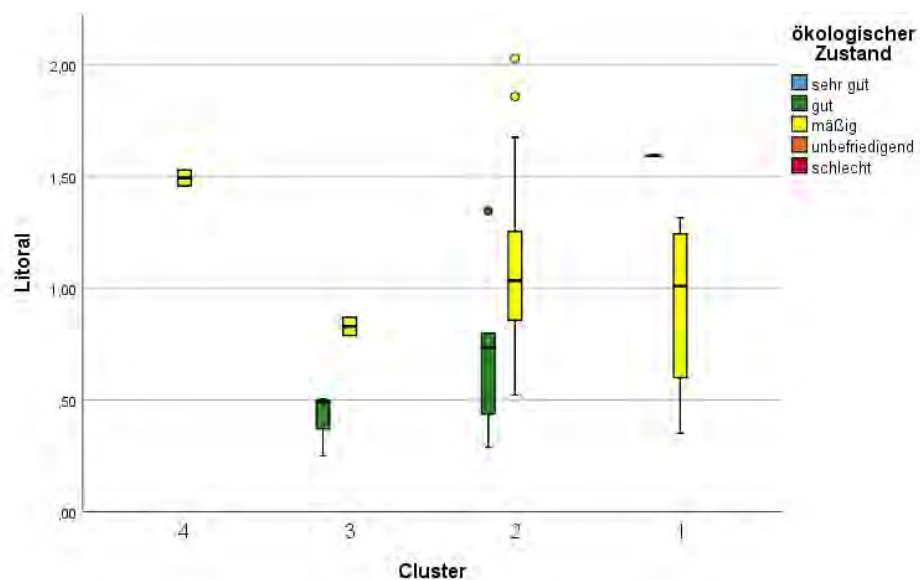


Figure 57: Box plot of the littoral proportion metric based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2= summer/autumn (n=28) 1= autumn (n=5); N(total)=40

The metric Oligochaeta & Diptera taxa [%] (**Figure 58**) shows no differences between the clusters.

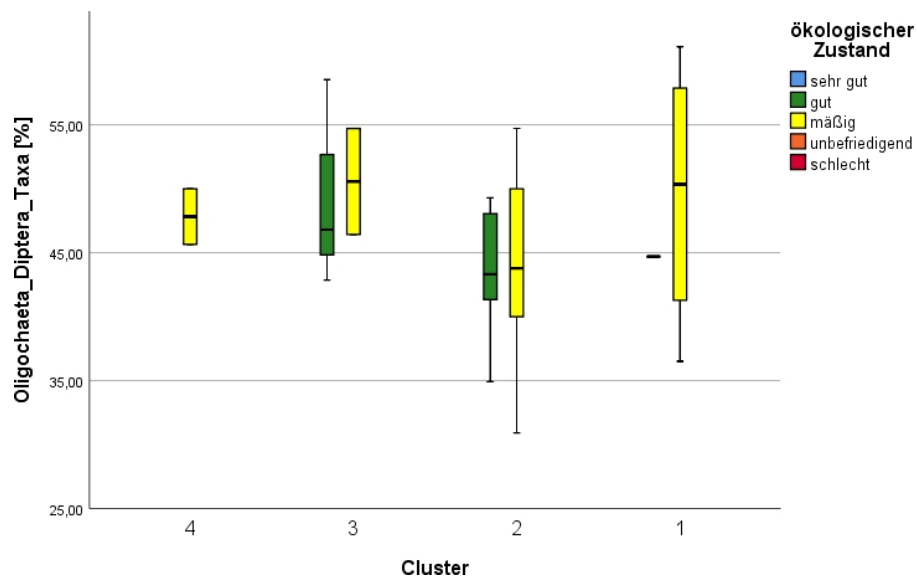


Figure 58: Box plot of the metric Oligochaeta & Diptera taxa [%] based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2= summer/autumn (n=28) 1= autumn (n=5); N(total)=40

The metric diversity index (**Figure 59**) shows clear differences between cluster 4 and cluster 3 within the spring. The median of cluster 1 is significantly higher than that of clusters 2 and 3, with the latter two behaving similarly.

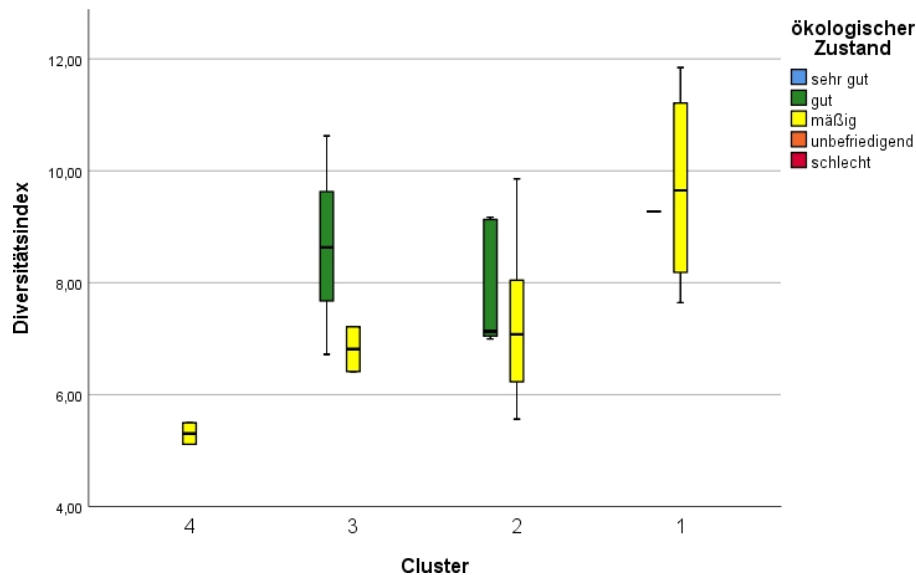


Figure 59: Box plot of the metric diversity index based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

3.7.3 Test of new metrics based on "best available sites"

3.7.3.1 Profundal proportion [%]

The profundal metric (**Figure 60**) shows a clear difference between intermittent and perennial in spring within the moderate condition. Cluster 2 and cluster 1 have a large variance, with the median of cluster 1 close to the median of cluster 4.

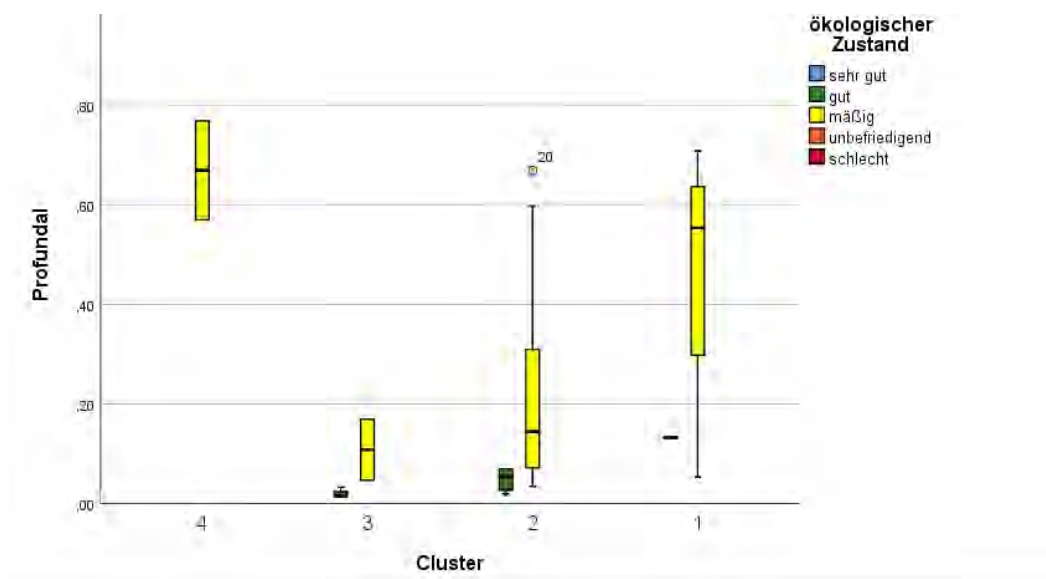


Figure 60: Box plot of the Profundal metric based on the "best available sites" from Bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

3.7.3.2 Littoral and profundal proportions

The combination of littoral and profundal proportions as a metric (**Figure 61**) shows a clear difference between intermittent (cluster 4) and perennial (cluster 3) watercourses within the moderate ecological status class.

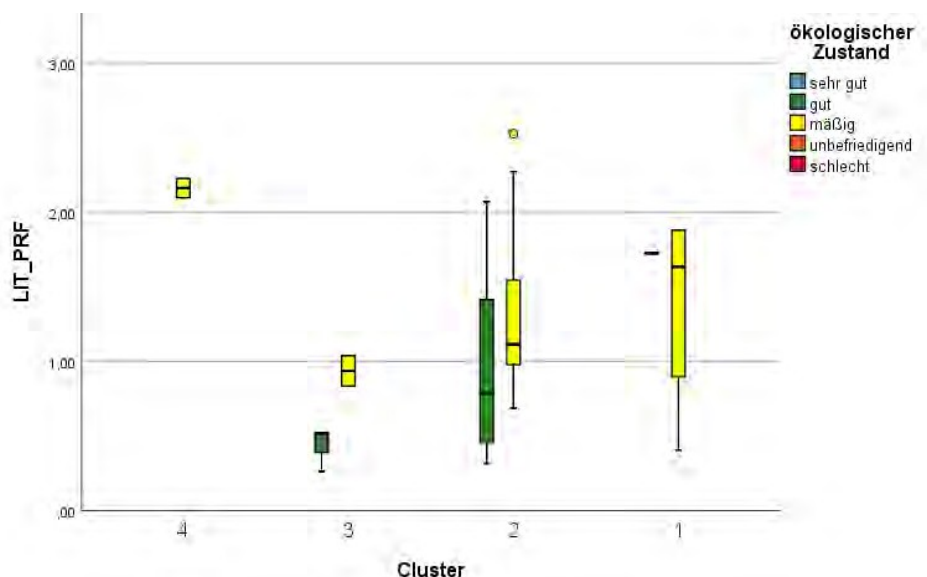


Figure 61: Box plot of the littoral-profundal metric based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39)

39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2= summer/autumn (n=28) 1= autumn (n=5); N(total)=40

3.7.3.3 Diptera taxa [%]

The metric Diptera taxa [%] (**Figure 62**) shows lower dominance in intermittent watercourses. The values for cluster 3 are higher than those for cluster 2 (summer/autumn).

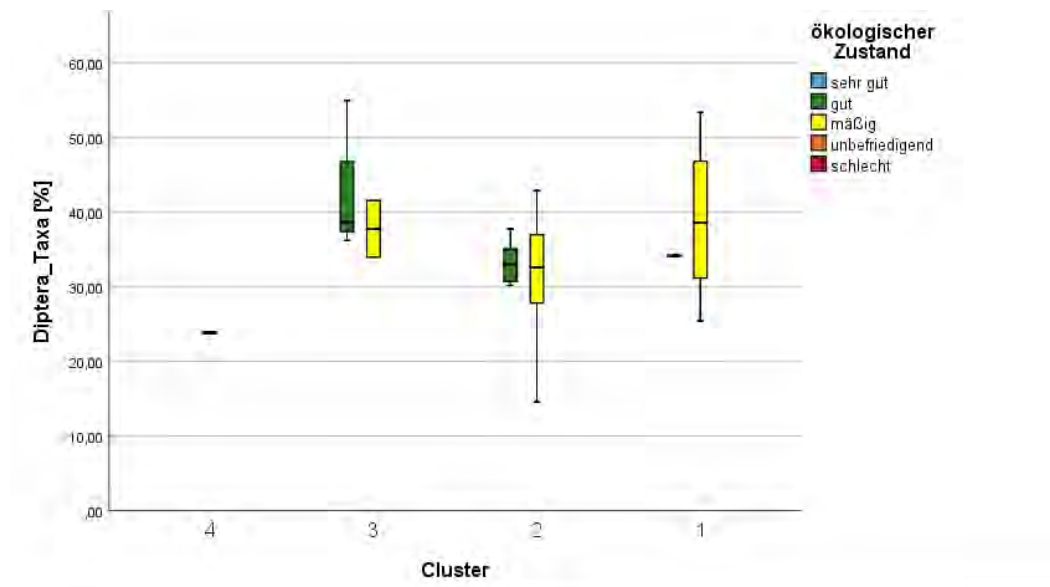


Figure 62: Box plot of the metric Diptera taxa [%] based on the "best available sites" from bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2= summer/autumn (n=28) 1= autumn (n=5); N(total)=40

3.7.3.4 Longitudinal Zonation Index (LZI)

The Longitudinal Zonation Index (LZI) metric also reacts within the spring between intermittent and perennial watercourses. Clusters 2 (summer/autumn) and Cluster 1 (autumn) react similarly in the median to Cluster 4 (spring, perennial).

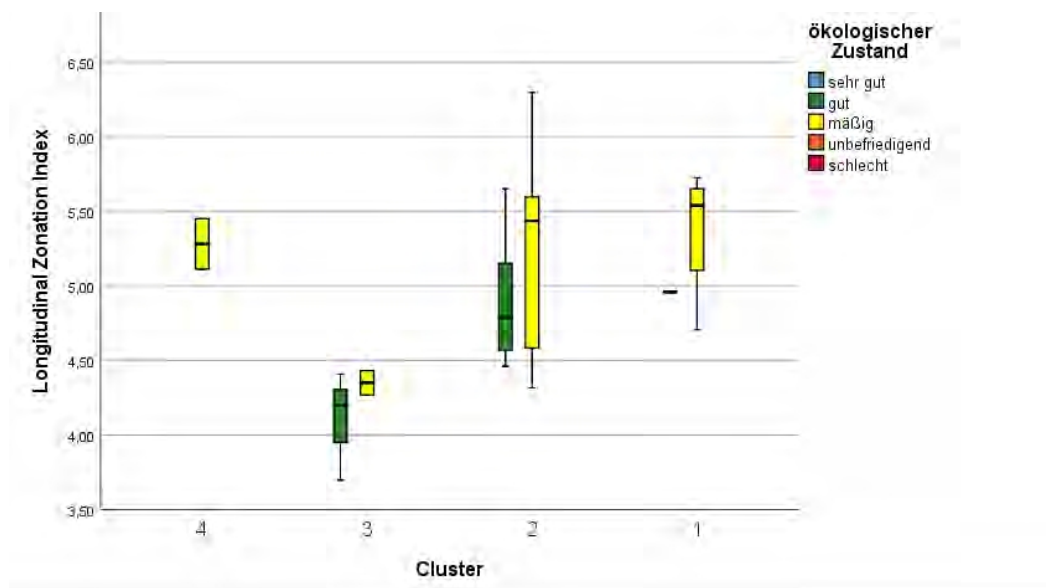


Figure 63: Box plot of the Metric Longitudinal Zonation Index (LZI) based on the "best available sites" from Bioregion 14 with the assessed ecological status, grouped according to clustering (Figure 39). 4=spring, intermittent (n=2); 3=spring, perennial (n=5); 2=summer/autumn (n=28) 1=autumn (n=5); N(total)=40

3.8 Genetic investigations and phylogenetic analysis

The phylogenetic analysis of the species *Ironoquia dubia* (Trichoptera) from the Pfaffenbach (AUT) shows, as expected, a grouping with the same taxa from Finland and Estonia, although there are small differences between the individuals from Austria and those from Finland and Estonia (0.9% divergent). Within the Limnephilidae family, the species *Ironoquia punctatissima* (USA, Canada) is more similar to *I. dubia* (12% divergent) than the species *Dicosmoecus atripes* (14% divergent, USA, Canada) from the genus *Dicosmoecus*. As expected, the species *Phryganea bipunctata* (Austria, Germany) from the family Phryganeidae is most different from the species *Ironoquia dubia*.

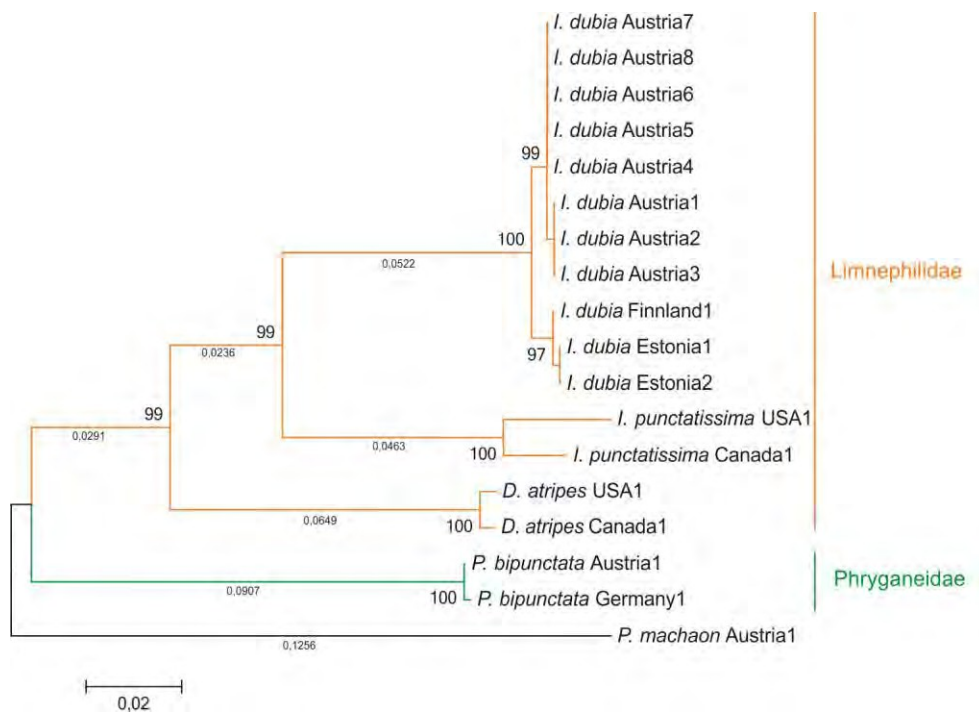


Figure 64: Neighbor-joining tree (p-distance) of *Ironoquia dubia* based on 650 bp of the COI gene (DNA barcode). The numbers in percent near the branches (if ≥ 80) correspond to the bootstrap values at 1000 replications. The values below the branches correspond to the genetic distances (p-distances); alignment length 650 bp; outgroup is the species *Papilio machaon* (swallowtail, Lepidoptera).

All individuals of the species *Siphonurus* cf. *armatus* and *S.* cf. *aestivalis* from Pfaffenbach (AUT) cluster very closely with the same species from Germany. Thus, the morphological determination *S.* cf. *aestivalis* can be genetically refuted as *S. armatus*. The species *S. aestivalis* (Norway) and *S. croaticus* (France, Germany) are more similar to the species *S. armatus* (15% divergent) than the species *Parameletus chelifer* (17% divergent) from Canada and Norway. The species *Tasmanophlebia lacustris* from the family Oniscigastridae differs most from *Siphonurus armatus*. Within the species *S. croaticus*, differences can be seen between individuals from France and Germany (0.3% divergent).

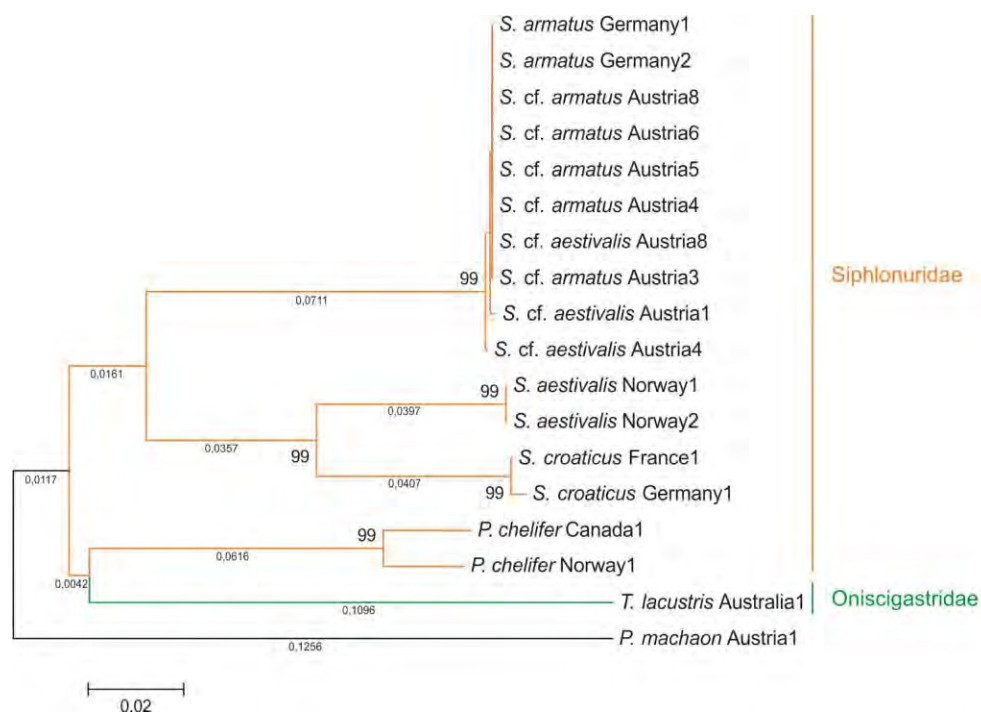


Figure 65: Neighbor-joining tree (p-distance) of *Siphonurus* cf. *armatus* and *S.* cf. *aestivalis* based on 626 bp of the COI gene (DNA barcode). The numbers in percent near the branches (if ≥ 80) correspond to the bootstrap values at 1000 replications. The values below the branches correspond to the genetic distances (p-distances); alignment length 626 bp; outgroup is the species *Papilio machaon* (swallowtail, Lepidoptera).

The individuals from the Wellingbach and Vocherabach streams that were morphologically identified as *Ecdyonurus subalpinus* form a group with minor differences within the group. The maximum divergence between two individuals within this group is 1.5%, so it is not possible to speak of multiple species. However, this group is clearly distinct from the group of the species *Ecdyonurus subalpinus* from Germany (16% divergent). Based on genetic analyses, it can therefore be assumed that the individuals from the Graz field and Grabenland (Austria) and the individuals from Germany are different species.

The individuals of the different morphological color type "red" cluster based on mtDNA (COI gene) without differences from the remaining "dark" morphs.

With the exception of the species *Ecdyonurus helveticus*, the species *E. venosus* and *E. torrentis* form the third cluster within the genus *Ecdyonurus*, alongside *E. cf. subalpinus* from Austria and *E. subalpinus* from Germany. The species *E. venosus* from Bulgaria does not cluster with the other two species from Germany and Italy. Here, too, the identification may be morphologically unclear, although the sequences have been loaded from bold.system.

There are slight differences between individuals of the species *E. torrentis*, *E. venosus*, and *E. helveticus* depending on the country from which the sample originates. The species *E. ujhelyii* from the closely related genus *Electrogena* differs most from the Austrian species *Ecdyonurus cf. subalpinus* within the family Siphonuridae, along with the species *Siphonurus armatus* and *Siphonurus aestivalis*.

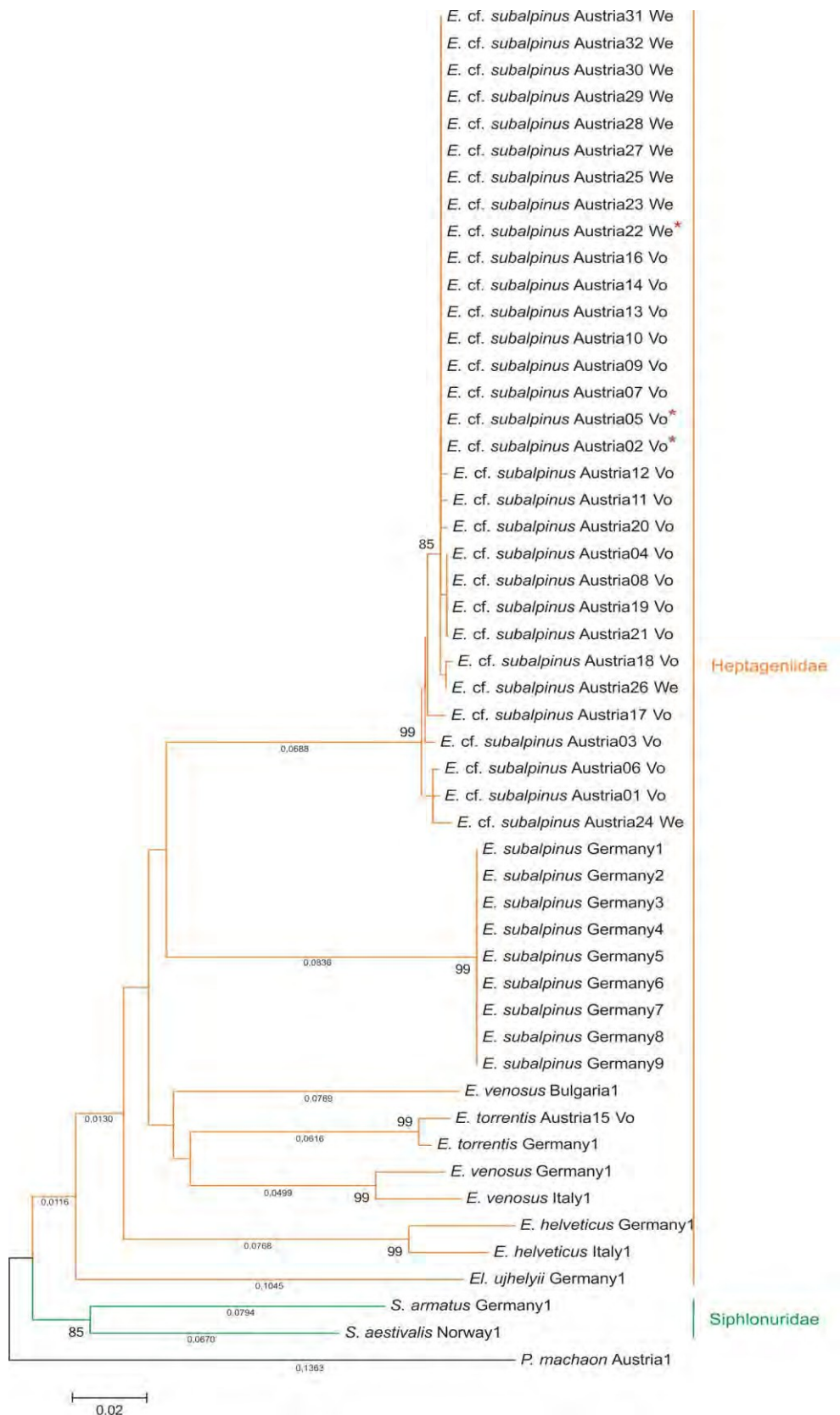


Figure 66: Neighbor-joining tree (p-distance) of *Ecdyonurus* cf. *subalpinus* and *E. torrentis* based on 621 bp of the COI gene (DNA barcode). The numbers in percent near the branches (if ≥ 80) correspond to the bootstrap values at 1000 replications. The values below the branches correspond to the genetic distances (p-distances); alignment length 621 bp; * = color morph red; outgroup is the species *Papilio machaon* (swallowtail, Lepidoptera).

4 Discussion

4.1 Typological classification

The EU Water Framework Directive (WFD) requires the ecological status of our waters to be determined, including the biological quality element macrozoobenthos. The current Austrian assessment method is based on a typological classification within 15 bioregions and three categories of "large rivers" (OFENBÖCK et al., 2010). Previous assessments of watercourses in the "Grazer Feld und Grabenland" bioregion often led to implausible results, as better ecological status was expected due to the morphology. The causes of these results may lie in the current assessment methodology, as it may be based on an insufficient data set. Alternatively, the results may be caused by strong seasonal fluctuations in the biotic communities. For this reason, ten streams were surveyed at near-natural sites in spring 2016 and investigated using existing data based on similarity analyses.

Based on this work, the current typological designation of bioregion 14 "Grazer Feld and Grabenland" (MOOG et al. 2001, SCHMIDT-KLOIBER et al. 2001) can be confirmed as an independent bioregion. Within bioregion 14, there are seasonal differences in the community structure, with the special type "intermittent water bodies" being detected in spring based on indicator species for intermittent water bodies.

The typological designation of bioregion 14 "Grazer Feld and Grabenland" according to (MOOG et al. 2001, SCHMIDT-KLOIBER et al. 2001) as a separate region from bioregion 3

"Bergrücken" and bioregion 13 "östliche Flach- und Hügelländer" forms the basis for classification in accordance with the requirements of the EU Water Framework Directive (WFD) and the associated Austrian methodological guidelines for determining ecological status based on macrozoobenthos. Based on a similarity analysis of the coenoses, bioregion 3 "mountain ridge landscape" forms a clearly defined region in contrast to bioregion 13 "eastern flat and hilly lands" and bioregion 14 "Grazer Feld und Grabenland." This separation reflects the taxa characteristic of mountainous areas found in Bioregion 3, including *Baetis alpinus*-Gr., *Protonemura* sp., *Ephemerella mucronata*, *Rhyacophila tristis*, and *Epeorus assimilis* (MOOG, Ed., 2017). These

The Kainach, Sulm, and Laßnitz water bodies in Bioregion 14 also exhibit mountain characteristics, as they originate in Bioregion 3, "Ridge Landscape." These rivers cluster away from Bioregion 14 and should therefore not be used as a basis for assessment. In addition, as has been done in Germany (POTTGIESSER et al., 2004), a new type assignment for transboundary water bodies such as Kainach, Sulm and Laßnitz in bioregion 14 should be discussed.

According to the results of this study, the typological designation of bioregions 13 and 14 is seasonal due to the macrozoobenthos community. In spring and summer, the communities separate and indicator species for the respective bioregion can also be identified. In autumn, these designations are not possible. This is the result of a fundamental methodological problem, which ultimately also applies to autumn studies in general, as the presence of juvenile larval stages in autumn leads to an increase in determinations at the genus level (personal communication, Graf Wolfram). This leads to macrozoobenthos communities that appear to become increasingly similar within the autumn season due to the lack of taxonomic resolution. Apart from this, according to OFENBÖCK et al. (2010), investigations in autumn are not in line with the guidelines.

Within bioregion 14 "Grazer Feld and Grabenland," there are seasonal differences in the community structure that affect the assessment of ecological status. Since the community ultimately reflects the ecological status based on macrozoobenthos, all ecological statuses from autumn studies must be viewed critically, as they are not in accordance with the guidelines and often the necessary taxonomic resolution cannot be achieved.

In the "Grazer Feld und Grabenland" bioregion 14, the indicator species *Siphonurus armatus*, *S. aestivalis*, or the taxon *Habrophlebia* sp. and the Trichoptera indicator species *Isonychia dubia*, *Oligostomis reticulata*, and *Glyptotendipes pallidus* were also confirmed for intermittent streams with the help of indicator species analyses. *S. aestivalis* and the taxon *Habrophlebia* sp., and the Trichoptera indicator species *Isonychia dubia*, *Oligostomis reticulata*, and *Glyptotendipes pallidus*. These indicator species for intermittent streams have adapted their life cycle to the drying up of a water body in summer (BOHLE & POTABGY, 1992; FIEDLER & BOHLE, 1994; SOMMERHÄUSER et al., 1997). The current typological classification of bioregion 14 in the internal differentiation (OFENBÖCK et al., 2010) needs to be re-discussed, as intermittent water bodies are present. An observation of the water supply in the Graz field

and Grabenland (LEITNER & GRAF, 2018) confirmed that the study sites Glauningbach and Pfaffenbach were dry. Thus, the special type "intermittent water bodies" for Bioregion 14 in the model catalog (WIMMER et al. 2012).

4.2 Development of an adapted assessment system

The different water body types "perennial" and "intermittent" as well as the seasonal differentiation of the zoenosis and hydrology in the

"Grazer Feld and Grabenland" make it questionable whether they can be assessed using the current method based on the EU WFD (OFENBÖCK et al., 2010) and the associated typological classification.

In contrast to the seasonal faunal differences due to the benthic community, the metrics react stably within the seasons and status classes, except for large variances in the metrics degradation index, Rhithron feeding type index (RETI), total taxa, and diversity index within the spring season and the moderate ecological status class.

Intermittent water bodies appear to be the cause of the large variations in spring, especially within moderate sites. The metrics degradation index, Rhithron feeding type index (RETI), total taxa, littoral proportion, and diversity index integrated into the current assessment react uniformly differently in intermittent water bodies than in perennial water bodies.

Thus, intermittent sites are rated worse with the current assessment method, as the metric thresholds are only defined for perennial waters. Furthermore, the thresholds of the current assessment method must be questioned, as they were calculated on the basis of an insufficient data set, which may not reflect the characteristics of bioregion 14 (e.g., Kainach).

The metrics currently not included in the assessment—profundal shares, littoral & profundal shares, Diptera taxa [%], and longitudinal zonation index (LZI)—respond to intermittent water bodies and could be used to develop a new assessment method for intermittent water bodies based on the results.

Based on seasonal analyses, the sampling period for bioregion 14 should be adjusted to spring according to the guidelines (OFENBÖCK et al., 2010), as the indicator taxa for intermittent water bodies could not be found in the remaining water-bearing sampling sites.

indicator taxa for intermittent water bodies can still be found at the water-bearing sampling sites

. The metrics values for perennial water bodies in spring may be more suitable for adapting the assessment method, as the current assessment system leads to implausible results.

Further investigations are necessary for all these adaptations in order to obtain a more comprehensive data set on intermittent and perennial sites in the Graz field and Grabenland. In addition, observations of water allocations and the recording of physicochemical data are necessary for the designation of reference sites, especially for intermittent sites (STUBBINGTON et al., 2018).

4.3 WRRL-relevant questions

- Can the Graz field and Grabenland be designated as a separate bioregion based on macrozoobenthos?

Yes, Bioregion 14 can be identified as a separate typological bioregion based on its macrozoobenthos.

- Is the current assessment system for streams in the Graz field and Grabenland region plausible, or does it need to be adapted to the specific conditions of this bioregion?

The current assessment system needs to be adapted to the seasonal conditions and to the special type of "intermittent water bodies."

- Can the designation of watercourse landscapes and watercourse nature areas according to FINK et al. (2000) be used for a typological classification of Bioregion 14 based on macrozoobenthos?

No, watercourse landscapes and natural watercourse areas do not form separate typological regions based on macrozoobenthos.

- Are there seasonal differences in the community structure within bioregion 14 due to hydrological differences?

Yes, there are seasonal differences in the community structure.

- Can indicators for intermittent streams be identified?

Indicator taxa of the order Ephemeroptera (mayflies) and Trichoptera (caddisflies) were identified.

- Is it possible to develop a data-based and plausible assessment methodology that includes the characteristics and typology of the streams in the Grazer Feld and Grabenland bioregion?

Yes, it is possible, taking into account further investigations of streams with regard to hydrology and physicochemical conditions.

4.4 Genetic identification and phylogenetic analysis of selected taxa

A DNA barcode (HEBERT et al., 2005) was created for the species *Ironoquia dubia* and *Siphonurus armatus*, which are indicators of intermittent streams, thus contributing to the development of a reference database (RATNASINGHAM & HEBERT, 2007) for the genetic taxonomic tools barcoding (HEBERT et al., 2003, 2005) and eDNA (DEJEAN et al., 2011).

Analysis of the species *Ecdyonurus cf. subalpinus* showed that, due to different color morphs, these are not two different species. However, this assumption must be confirmed with population genetic data. Further investigations are necessary to determine the species of *E. cf. subalpinus*, including the suspected species *E. siveci* (JACOB & BRAASCH, 1984).

Furthermore, the taxonomic tool "barcoding" helped to verify or falsify the determinations of morphologically difficult to differentiate species (*Siphonurus armatus* and *S. aestivalis*) made in the course of the work.

5 References

- AQEM Consortium (2002): Manual for the application of the AQEM system. A comprehensive method to assess European streams using benthic macroinvertebrates, developed for the purpose of the Water Framework Directive. Version 1.0, February 2002
- BMLFUW (2007): EU Water Framework Directive 2000/60/EC, Austrian report on the IST inventory, methodology for water bodies 10-100 km²
- BARBOUR, M.T., GERRITSEN, J., SNYDER, B.D. and STRIBLING, J.B. (1999) Rapid Bioassessment Protocols for Use in Streams and Wadeable Rivers: Periphyton, Benthic Macroinvertebrates and Fish. 2nd Edition, Report EPA 841-B-99-002, US Environmental Protection Agency, Office of Water, Washington DC.
- BOHLE, H.W. & POTABGY, G. (1992): *Metreletus balcanicus* (ULMER 1920), *Siphonurus armatus* (EATON 1870) (Ephemeroptera, Siphonuridae) and the fauna of summer-dry streams
- BOHLE, H.W. (2001): The role of autecology in understanding biocenotic and ecosystem relationships, using the example of the benthos of flowing waters. German Society for Limnology (DGL) – Conference Report 2000: 9-23.
- DEJEAN T, VALENTINI A, DUPARC A, PELLIER-CUIT S, POMPANON F, et al. (2011): Persistence of Environmental DNA in Freshwater Ecosystems. PLoS ONE 6(8):e23398. doi:10.1371/journal.pone.00233
- DUFRENE, M. & LEGENDRE, P. (1997): Species assemblages and indicator species: the need for a flexible asymmetrical approach. Ecological Monographs 67: 345-366.
- FIEDLER, A., & BOHLE, H.W. (1994): Ephemeroptera sommertrockener Bäche in Oberhessen. Development cycles and population structure
- FINK, M. H., MOOG, O., & WIMMER, R. (2000): Fließgewässer-Naturräume Österreichs (Natural watercourses in Austria). Federal Environment Agency.
- FOLMER, O., BLACK, M., HOEH, W., LUTZ, R., VRIJENHOEK, R. (1994): DNA primers for amplification of mitochondrial cytochrome c oxidase subunit I from diverse metazoan invertebrates. Molecular Marine Biology and Biotechnology (1994) 3(5), 294-299
- GATTOLLIAT, J.-L., CAVALLO E., VUATAZ L., SARTORI M. (2015): DNA barcoding of Corsican mayflies (Ephemeroptera) with implications on biogeography, systematics and biodiversity. Arthropod Systematics Phylogeny, 2015, vol. 73, no. 1, pp. 3-18.
- GERRITSEN, J., BARBOUR, M.T. & KING, K. (2000): Apples, oranges and ecoregions: on determining pattern in aquatic assemblages. Journal of North American Benthological Society, 2000, 19(3): 487-496
- GZÜV: Ordinance of the Federal Ministry of Agriculture, Forestry, Environment and Water Management on the monitoring of the status of water bodies; Water Status Monitoring Ordinance including annexes. Federal Law Gazette II No. 479/2006 as amended.

- HEBERT, PAUL D. N., S. RATNASINGHAM, J. R. DEWAARD. (2003): Barcoding animal life: cytochrome c oxidase subunit 1 divergences among closely related species. *Proceedings of the Royal Society of London B*, 70: 96-99.
- HEBERT, PAUL D. N. and GREGORY, T. RYAN (2005): The Promise of DNA Barcoding for Taxonomy. *Systematic Biology*, 54: 852-859.
- ILLIES, J. (Ed.) (1978): *Limnofauna Europaeae*. – 2nd revised and expanded edition, G. Fischer Verlag, Stuttgart, New York; Swets & Zeitlinger B.V., Amsterdam.
- JACOB, U., & BRAASCH, D. (1984). New and revised taxa of the *Ecdyonurus helveticus* group (Ephemeroptera, Heptageniidae). *Entomological Treatises State Museum of Zoology Dresden*, 48(6), 53-61.
- LEITNER, P., & GRAF, W., (2018): SURVEY OF THE ECOLOGICAL STATUS AT SELECTED LOCATIONS IN BIOREGION 14 – "GRAZER FELD UND GRABENLAND" BASED ON MACROZOOBENTHOS – INVESTIGATION YEAR 2018
- MARCINEK, J. & ROSENKRANZ, E. (1996): *Das Wasser der Erde: Eine geographische Meeres- und Gewässerkunde* (The Water of the Earth: A Geographical Study of the Seas and Waterways). 2nd revised and expanded edition. 1996. Klett Verlag, 328 pp.
- MCCUNE, B., & MEFFORD, M. J. (2006). PC-ORD 5.0. *Multivariate analysis of ecological data*. Gleneden Beach, Oregon, USA.
- MOOG, O., CHOVANEC A., HINTEREGGER H. & A. RÖMER (1999): Guidelines for saprobiological assessment of water quality of watercourses. Water Management Register, Federal Ministry of Agriculture and Forestry, Vienna.
- MOOG, O. & OFENBÖCK, T. (2000): Austria's share of European ecoregions. Chapter 3 in: Fink et al. (2000): *Fließgewässer-Naturräume Österreichs* (Watercourses – Natural Areas of Austria). UBA Monographs No. 128, Vienna
- MOOG, O., SCHMIDT-KLOIBER, A., OFENBÖCK, J. & GERRITSEN, J. (2001): Aquatic ecological regions and watercourse bioregions of Austria: A classification according to geoecological environmental factors and macrozoobenthos communities. www.lebensministerium.at. Federal Ministry of Agriculture, Forestry, Environment and Water Management.
- MOOG, O. (Ed.) (2002): *Fauna Aquatica Austriaca*, 2002 edition. Water Management Register, Federal Ministry of Agriculture, Forestry, Environment and Water Management, Vienna.
- MOOG, O. (2004): Standardization of habitat-weighted macrozoobenthos sampling in flowing waters (multi-habitat sampling; MHS). Federal Ministry of Agriculture, Forestry, Environment, and Water Management, 20 pp.
- OFENBÖCK, T., MOOG, O., GERRITSEN, J. & BARBOUR, M. (2004): A stressor specific multimetric approach for monitoring running waters in Austria using benthic macroinvertebrates. *Hydrobiologia*, 516, 251-268.
- OFENBÖCK T., MOOG O., HARTMANN A., STUBAUER I. (2010): Guidelines for the assessment of biological quality elements, Part A2 - Macrozoobenthos

- OFENBÖCK G., MAUTHNER-WEBER R. & F. H. WAGNER (2016): Guidelines for the assessment of biological quality elements – Introduction, BMLFUW
- POTTGIESSER, T., KAIL, J., SEUTER, S., HALLE, M. (2004): Final work on the classification of watercourses in accordance with the requirements of the EU Water Framework Directive – Part II Final report
- QZV - Ecology OG (2010): Quality Objectives Regulation for Surface Water Ecology: Regulation of the Federal Minister for Agriculture, Forestry, Environment, and Water Management on the determination of the ecological status of surface waters. Federal Law Gazette II No. 99/2010, as amended.
- RATNASINGHAM S., P. D. N. HEBERT. 2007. Bold: The Barcoding of Life Data System (www.barcodinglife.org). Molecular Ecology Notes, 7: 355-364.
- SAMBROOK, J., FRITSCH, E., & MANIATIS, T. (1989). Molecular Cloning: a Laboratory Manual (2nd ed.). New York: Cold Spring Harbor Laboratory Press.
- STUBAUER, I. (2002): Classification of saprobic basic conditions of Austrian watercourses. Dissertation, University of Natural Resources and Applied Life Sciences, Vienna
- STUBAUER, I. & MOOG, O. (2003): Saprobic basic conditions of Austrian watercourses. – Water Management Register, Federal Ministry of Agriculture, Forestry, Environment and Water Management, Vienna)
- SCHMIDT-KLOIBER, A. (2002): Deduction of watercourse bioregions based on multivariate analyses of invertebrate soil fauna. Dissertation, University of Natural Resources and Applied Sciences, Vienna
- SCHMIDT-KLOIBER, A., MOOG, O. & GERRITSEN, J. (2001): The aquatic ecoregions of Austria – Results of multivariate analyses of macrozoobenthos communities. – Austrian Fisheries, Vol. 54/2001: 154-161.
- SOMMERHÄUSER, M., ROBERT, B., & SCHUHMACHER, H. (1997). Flight periods and life history strategies of caddisflies in temporary and permanent woodland brooks in the Lower Rhine area (Germany). In *Proceedings of the 8th International Symposium on Trichoptera* (pp. 425-433).
- STUBBINGTON, R., CHADD, R., CID, N., CSABAI, Z., MILIŠA, M., MORAIS, M., ... & VERDONSCHOT, R. C. (2018). Biomonitoring of intermittent rivers and ephemeral streams in Europe: current practice and priorities to enhance ecological status assessments. *Science of the total environment*, 618, 1096-1113.
- TAMURA, K., STECHER, G., PETERSON, D., FILIPSKI, A., & KUMAR, S. (2013). MEGA6: molecular evolutionary genetics analysis version 6.0. *Molecular biology and evolution*, 30(12), 2725-2729.
- TER BRAAK, C.J.F. (1995): Ordination. In: JONGMAN, R.H.G., TER BRAAK, C.J.F. & van TONGEREN, O.F.R. (eds.) (1995): Data analysis in community and landscape ecology. New corrected edition. Cambridge University Press. 299 pp.

- WAGNER F. H., MAUTHNER-WEBER R. & OFENBÖCK G. (2010): Guidelines for the assessment of biological quality elements – Introduction p. 27
- WIMMER, R. & MOOG, O. (1994): River order numbers of Austrian watercourses. Federal Environment Agency, Monographs 51, 581 pp.
- WIMMER, R., & CHOVANEC, A. (2000): Types of watercourses in Austria as a basis for the development of a monitoring network in accordance with Annex II of the EU Water Framework Directive. Federal Ministry of Agriculture, Forestry, *Environment, and Water Management, Water Management Register*.
- WIMMER, R., CHOVANEC, A., GRUBER, D., FINK, M.H., MOOG, O. (2000a): Implementation of the EU Water Framework Directive – Classification of watercourses in Austria based on abiotic parameters. – Austrian Fisheries 53: 13-21.
- WIMMER, R., WINTERSBERGER, H., & PARTHL, G. (2012): Hydromorphological guidelines for the classification of watercourses in Austria. Federal Ministry of Agriculture, Forestry, Environment and Water Management – Dept. VII/1.
- WIMMER, R., WINTERSBERGER, H., & PARTHL, G. (2012): Hydromorphological guidelines for the classification of watercourses in Austria. Volume 4: Special types

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8 Appendix

8.1 Complete taxonomic list with frequencies of the 10 streams sampled in 2016

		Lieberbach	Glauningb.	Labillbach	Poppend. B.	Gnasb. Old arm	Aubachl (Fruttnerb.)	Vocherab.	Wellingb.	Pfaffenb.	Stiefing
		06.04.2016	06.04.2016	05.04.2016	06	06	04/06/2016	05	05	05	04.05.2016
		MHS	MHS	MHS	MHS	MHS	MHS	MHS	MHS	MHS	MHS
Class	Type	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]	[Ind/m²]
TURBELLARIA			7			0.8			4		
[Kl:Turbellaria]									4		
Turbellaria	Gen. sp.								4.8		
PLANARIIDAE			7.2			0.8					
Planaria	sp.		7.2			0					
NEMATODA					0.8	20		0		0	
[Kl:Nematoda]					0	20		0		0	
Nematoda	Gen. sp.				0.8	20		0.8		0	
GASTROPODA		1.6		5.6	4		1.6	0	1.6	3	
LYMNAEIDAE										0	
Radix	ovata/peregra									0	
PHYSIDAE				5.6			1.6	0			
Physella	sp.			5.6			1.6				
Physidae	Gen. sp. juv.							0			
PLANORBIDAE		1.6			4.8				1.6	2	
Ancylus	fluviatilis	1.6			4.8				1.6		
Planorbidae	Gen. sp.									2	
BIVALVIA				16	1.6	1040.8	1.6	16.8	2	503.2	0
PISIDIIDAE				16	1	1040.8	1	16.8	2	503.2	0
Pisidium	sp.			16	1.6	1040.8	1.6	16.8	2	503.2	0
OLIGOCHAETA		44.8	92.8	681.6	55.2	227.2	150.4	4.8	37.6	3,740.8	124.8
[Kl:Oligochaeta]				25.6	6.4				12.8		6
Oligochaeta	Gen. sp.			25.6	6.4				12.8		6.4
HAPLOTAXIDAE										28	
Haplotaxis	gordioides									28	
LUMBRICIDAE		0.8	9.6	0.8	0		1.6		4.8	8.8	0
Eiseniella	tetraedra	0.8	9.6	0.8	0.8		1.6		4.8	8.8	0
NAIDIDAE				627.2		6					
Nais	elinguis			627.2							
Stylaria	lacustris					6.4					
TUBIFICIDAE		14.4	55.2	27.2	41.6	180.8	145.6			540	8
Aulodrilus	japonicus					119.2					
Embolecephalus	velutinus		18.4							450.4	
Limnodrilus	claparedeianus		12.8		24		22.4				
Limnodrilus	hoffmeisteri		19.2	27.2	5.6	3.2	38.4				
Potamothenix	moldaviensis				5.6	12.8					
Potamothenix	sp. new						9.6				0
Rhyacodrilus	coccineus	0.8	4				26.4			89.6	

Tubifex	ignotus		0							
Tubifex	tubifex				45.6	48.8				
Tubificidae	Gen. sp. juv.	13.6			6.4					7.2
ENCHYTRAETIDAE				0.8			1.6		89.6	
Enchytraetidae	Gen. sp.			0.8			0		64	
Fridericia	sp.						0.8		25.6	
PROPAPPIDAE										6
Propappus	volki									6.4
LUMBRICULIDAE		29.6	28		6.4	40	3.2	3	20	3074.4
Bythonomus	lemanii		5							103.2
Lumbriculus	variegatus		10.4			8				1.6
Rhynchelmis	limosella									19.2
Stylodrilus	brachystylus		7.2		6.4		3.2			
Stylodrilus	heringianus	16.8				32			20	3028
Stylodrilus	sp.	12.8	4.8					3.2		84
HIRUDINEA				0.8		38.4	1.6			25.6
GLOSSIPHONIIDAE				0.8		32				19.2
Glossiphonia	complanata					22.4				
Glossiphonia	sp.					8				
Helobdella	stagnalis			0.8		1.6				
HAEMOPIIDAE						0.8				1.6
Haemopsis	sanguisuga					0.8				1.6
ERPOBDELLIDAE						5.6	1.6			3
Erpobdella	octoculata					5.6				
Erpobdella	sp.									1.6
Erpobdellidae	Gen. sp.						1.6			1.6
COPEPODA										25.6
CYCLOPIDAE										25.6
Cyclops	sp.									25.6
AMPHIPODA		167.2	8.8	56	191.2	11.2	659.2		404.8	262.4
GAMMARIDAE		167.2	8.8	56	191.2	11.2	659.2		404.8	262.4
Gammarus	fossarum	167.2	8.8	56	170.4	11.2	659.2		404.8	262.4
Gammarus	roeseli				20.8					1022.4
ISOPODA			1.6			289.6				
ASELLIDAE			1.6			289.6				51.2
Asellus	aquaticus		1.6			289.6				51.2
HYDRACHNIDIA		1.6			0.8		0	17.6	12.8	
[Ph:Hydrachnidia]		1.6			0.8		0	17.6	12.8	
Hydrachnidia	Gen. sp.	1.6			0.8		0	17.6	12.8	
EPHEMEROPTERA		367.2	332	652	175.2		426.4	616.8	907.2	487.2
SIPHONURIDAE			19.2				1			939.2
Siphonurus	aestivalis		3.2							460.8
Siphonurus	armatus		16				1.6			199.2
										261.6

BAETIDAE		4	0	30.4	173.6		265.6	371.2	124.8	0	332.8
Baetis	niger							12.8			
Baetis	rhodani	2.4	0	30.4	39.2		96	229.6	54.4	0.8	127.2
Baetis	sp. juv.				134.4		169.6	128	70.4		204.8
Centroptilum	luteolum	1.6						0.8			0
HEPTAGENIIDAE		302.4	0.8	7.2	1.6		152.8	172	256.8		575.2
Ecdyonurus	sp. juv.	3.2			1.6		6.4	6.4	6.4		
Ecdyonurus	subalpinus	22.4					9.6	33.6	5.6		4
Electrogena	ujhelyii						68.8	8.8	15.2		104.8
Rhithrogena	picteti	273.6		4			12	72	172		123.2
Rhithrogena	semicolorata						0.8				26.4
Rhithrogena	sp. juv.	3.2	0.8	3.2			55.2	51.2	57.6		316.8
LEPTOPHLEBIIDAE		56.8	311.2	614.4			5.6	73.6	478.4	25.6	
Habroleptoides	confusa								425.6		
Habrophlebia	fusca	21.6	4								
Habrophlebia	lauta								20.8		
Habrophlebia	sp. juv.	35.2	307.2	614.4			5.6	52.8	32	25.6	
Paraleptophlebia	submarginata							20.8			
EPHEMERIDAE							0.8		46.4		31.2
Ephemera	danica						0.8		46.4		31.2
EPHEMERELLIDAE									0.8		
Ephemerella	mucronata								0		
CAENIDAE		4									
Caenis	luctuosa	4									
ODONATA				1.6			3.2	2.4	3.2		4
CALOPTERYGIDAE								1.6	0.8		2
Calopteryx	sp. juv.							1.6			
Calopteryx	virgo								0.8		2.4
GOMPHIDAE								0.8			
Gomphidae	Gen. sp. juv.							0			
CORDULEGASTRIDAE				1.6			3.2				1.6
Cordulegaster	heros			1.6			3.2				1.6
CORDULIIDAE									2.4		
Somatochlora	flavomaculata								2.4		
PLECOPTERA		8.8	124	12.8			7.2	71.2	223.2	144	1.6
PERLODIDAE		8.8	109.6				6.4	23.2	10.4		
Isoperla	sp.	8.8	109.6				6.4	2			
Isoperla	tripartite							20.8	10.4		
TAENIOPTERYGIDAE								10.4	16		
Brachyptera	risi							10.4	16		
NEMOURIDAE			14.4	12.8			0	37.6	190.4	144	1.6
Amphinemura	sp.		10.4	12.8				28	6.4		1.6
Nemoura	cinerea							2.4		144	

Nemoura	sp.		4									
Protonemura	sp.					0.8	7.2	184				
LEUCTRIDAE								6.4				
Leuctra	sp.							6.4				
HETEROPTERA			0.8			1.6		0				
NEPIDAE						0.8						
Nepa	cinerea cinerea					0						
VELIIDAE			0.8									
Velia	sp.		0.8									
GERRIDAE						0.8		0.8				
Aquarius	najas					0						
Gerridae	Gen. sp.							0.8				
MEGALOPTERA						4		0.8				
SIALIDAE						4		0.8				
Sialis	lutaria					4		0.8				
COLEOPTERA		33.6	3.2	0	10.4	1.6	100.8	10.4	317.6	3.2	6.4	
DYTISCIDAE					0.8					0		
Dytiscidae	Gen. sp.									0		
Platambus	maculatus				0.8							
ELMIDAE		11.2	3.2	0.8	3.2		84.8	1.6	64		6.4	
Elmis	sp.	8.8		0			71.2		10.4		5.6	
Esolus	sp.								12.8			
Limnius	sp.						0.8	1	27.2			
Limnius	volckmari	2.4									0	
Oulimnius	tuberculatus		3.2		3.2		12.8		13.6			
GYRINIDAE								2.4				
Orectochilus	villosus							2.4				
HYDRAENIDAE		22.4			6.4		16	3.2	253.6	0		
Hydraena	sp.	22.4			6.4		16	3.2	253.6	0		
HYDROPHILIDAE						1.6				1.6		
Hydrophilidae	Gen. sp.									1.6		
Hydrophilidae	Gen. sp. Ad.					1.6						
SCIRTIDAE								3				
Scirtidae	Gen. sp.							3.2				
TRICHOPTERA		83.2	29.6	21.6	38.4	3.2	29.6	36	104.8	40	35.2	
RHYACOPHILIDAE								2.4	14.4		2	
Rhyacophila	s. str. sp.							2.4	14.4		2.4	
GLOSSOSOMATIDAE		53.6	23.2				0.8					
Agapetus	fuscipes	53.6	23.2				0					
HYDROPSYCHIDAE		1.6					0.8	24.8	1.6		2.4	
Hydropsyche	angustipennis	0.8					0.8					
Hydropsyche	bulbifera							0.8				
Hydropsyche	saxonica							12				

Hydropsyche	sp.	0.8						1.6			
Hydropsyche	sp. juv.							10.4	1.6		2.4
POLYCENTROPODIDAE								0.8	1.6	0	
Plectrocnemia	consersa									0.8	
Plectrocnemia	sp.								0.8		
Polycentropus	irroratus							0.8			
Polycentropus	sp. juv.								0.8		
PSYCHOMYIIDAE									4		
Lype	sp.								4		
PHRYGANEIDAE										0.8	
Oligostomis	reticulata									0.8	
LIMNEPHILIDAE		10.4	6.4	21.6	20.8	3.2	26.4	8	83.2	37.6	29.6
Anabolia	furcata	2.4	0.8	4.8	5.6				0.8		0
Glyptotaelius	pellucidus									1.6	
Halesus	sp.	0.8							2.4		0
Ironoquia	dubia		2.4		2.4					34.4	
Limnephilidae	Gen. sp.				2.4	3.2		0	65.6		25.6
Limnephilidae	Gen. sp. juv.	6.4		0			17.6	4	12.8		
Limnephilinae	Gen. sp.		0								
Limnephilus	rhombicus									0.8	
Limnephilus	sp.			2.4			0.8				
Limnephilus	cf. stigma						1.6				
Potamophylax	rotundipennis	0.8	2.4	13.6	10.4		6.4	3.2	1.6	0.8	2.4
GOERIDAE		5.6			2.4		1.6			0.8	0
Goera	pilosa				2.4						
Lithax	obscurus	5.6					1.6			0.8	0
LEPTOCERIDAE		12			15.2						
Athripsodes	sp.	12			15.2						
DIPTERA		164.8	70.39999	6506.398	840.8001	481.6	256.8	1023.2	346.4	45.6	524.7999
PEDICIIDAE		36.8	14.4	84	41.6		28	49.6	13.6		5.6
Dicranota	sp.	36.8	13.6	84	41.6		28	49.6	13.6		5.6
Pediciidae	Gen. sp.		0								
CHIRONOMIDAE		127.2	52	6419.199	799.2001	461.6	222.4	799.2002	307.2	31	509.6
Ablabesmyia (Ablabesmyia)	longistyla	1.6						4			
Brillia	bifida			12.8			3.2	3.2	6.4		
Brillia	flavifrons							3.2			
Chironomidae	Gen. sp.							25.6			
Chironomini	Gen. sp.							19.2			
Chironomus	sp.					40.8					
Cladotanytarsus	mancus-Gr.				12.8						
Cladotanytarsus	vanderwulpi	9.6		102.4				9.6			
Conchapelopia	sp.			12.8		8	1.6	48.8	19.2		
Corynoneura	sp.			102.4		6.4					

Cricotopus (Cricotopus)	bicinctus				0.8					
Cricotopus (Cricotopus)	triannulatus						6.4			
Diamesa	cinerella/zernyi group									0.8
Diamesa	insignipes				36	0.8				0.8
Epoicocladius	ephemerae							12.8		6.4
Eukiefferiella	devonica/ilkleyensis						6.4	6.4		
Eukiefferiella	minor						6.4			
Eukiefferiella	sp.							6.4		
Harnischia	sp.						20			
Heleniella	sp.									12.8
Heterotrissocladius	marcidus						2.4		0.	
Kiefferulus	tendipediformis				6.4					
Micropsectra	atrofasciata			6.4		7.2	1.6			
Micropsectra	atrofasciata-Agg.			38.4			9.6			
Micropsectra	sp.					64				
Microtendipes	cf. chloris									2.4
Microtendipes	cf. pedellus									6.4
Microtendipes	sp.	0.8			8		0.8			
Natarsia	nugax/punctata	1.6								0
Orthoclaadiinae	Gen. sp.	0.8	3.2	25.6	19.2	6.4	27.2	6.4		7.2
Orthoclaadiini	COP			115.2	12.8		52.8			
Orthocladus	sp.	0			8		1.6			
Orthocladus (Euorthocladus)	ashei			12.8			7.2			
Orthocladus (Euorthocladus)	frigidus						13.6			
Orthocladus (Euorthocladus)	rivicola			102.4	32.8			6.4		
Orthocladus (Euorthocladus)	rivulorum				6.4				6.4	
Orthocladus (Euorthocladus)	thienemanni							38.4		
Orthocladus (Orthocladus)	rubicundus			12.8	6.4		84			
Orthocladus (Orthocladus)	sp.	71.2	28	54	500	131.2	162.4	64		134.4
Parachironomus	sp.								0	
Paracladopelma	mikiana			25.6			0.8			
Parakiefferiella	bathophila						6.4			
Parakiefferiella	triquetra						6.4			
Parametriocnemus	stylatus							6.4		
Paratanytarsus	dissimilis						6.4			
Paratendipes	albimanus	3.2				12.8				
Paratendipes	cf. albimanus				32					
Paratrachocladus	rufiventris			182.4		3.2				
Paratrachocladus	skirwithensis/veronicae			38.4						
Paratrissocladius	excerptus							19.2		
Pentaneurini	Gen. sp.		8	12.8		3				
Phaenopsectra	sp.						6.4			
Polypedilum (Polypedilum)	pedestre	0.8								

Polypedilum (Tripodura)	aegyptium	6.4					6.4			
Polypedilum (Tripodura)	pullum/scalaenum	6.4		0.8	19.2	6.4	20.8	9.6		0
Polypedilum (Uresipedilum)	convictum	6.4		12.8	66.4		6.4	42.4	12.8	6.4
Procladius	sp.					6.4				
Procladius (Holotanypus)	sp.					316				0.8
Prodiamesa	olivacea						0.8			1.6
Prodiamesa	rufovittata							3		
Psectrocladius (Psectrocladius)	psilopterus									7.2
Psectrotanypus (Psectrotanypus)	varius					0.8				
Rheocricotopus (Psilocricotopus)	chalybeatus				6.4					
Rheocricotopus (Rheocricotopus)	fuscipes		9.6	105.6			3.2	19.2	6.4	7.2
Rheosmittia	spinicornis							6.4		
Rheotanytarsus	sp.	3.2	3.2		19.2		13.6	27.2	12.8	79.2
Stempellinella	brevis	3								32
Stictochironomus	maculipennis	8						8.8		2.4
Stictochironomus	sp.						1.6			
Synorthocladius	semivirens							19.2		
Tanytarsini	Gen. sp.						19.2	52	6.4	
Tanytarsus	eminulus	3.2								
Tanytarsus	sp.			12.8	6.4				70.4	
Thienemanniella	sp.									6.4
Tvetenia	calvescens			12.8				32.8		
Tvetenia	verralli							28.8		
Zavreliomyia	sp.								6.4	1.6
SIMULIIDAE			0.8	0				131.2	8.8	0
Prosimulium	hirtipes/tomosvaryi							8.8		2.4
Prosimulium	tomosvaryi								2.4	
Simulium	sp. juv.		0.8					64.8	3.2	0.8
Simulium (Nevermannia)	vernum-Gr.								1.6	0.8
Simulium (Simulium)	noelleri									0.8
Simulium (Simulium)	ornatum group			0.8						
Simulium (Simulium)	variegatum group							57.6	1	
ATHERICIDAE									12	
Ibisia	marginata								12.8	
CERATOPOGONIDAE			3.2					7.2	0.8	12.8
Ceratopogonidae	Gen. sp.		3.2					7.2	0.8	12
LIMONIIDAE				1.6		1.6	4	36	2.4	0.8
Antocha	sp.							6.4	0.8	
Hexatoma	sp.			0.8			1.6	29.6		6.4
Limoniidae	Gen. sp.			0		1.6	3.2		1.6	0.8
PSYCHODIDAE						16			0	
Psychodidae	Gen. sp.					16			0.8	
TABANIDAE				0.8		2.4	1			

Tabanidae	Gen. sp.			0.8		2.4	1.6				
TIPULIDAE		0									
Tipulidae	Gen. sp.	0									
Total		872.7999	670.4	7955.199	1319.2	212	1,639.2	1,800.8	2368	5311.999	2660
Total number of taxis	213	48	38	46	44	34	56	83	70	46	53

8.2 Rating Detailed MZB method and screening method

Section	Ditch			Ditch land			Ditch land			Ditch land			Ditch land		
Water	Lieberbach			Glauningbach			Labillbach			Poppendorfer stream			Gnasbach Old arm		
Date from	06			06			05			06			06		
Partial life space (TLR)	MHS			MHS			MHS			MHS			MHS		
Bioregion	GF (14)			GF (14)			GF (14)			GF (14)			GF (14)		
Basic condition MMI	1.75			1.75			1.75			1.75			1.75		
SI (Zelinka & Marvan)	1.35	Very good (hig	very good (hig	2.1	good	good	2.31	moderate (mod	moderate (mod	2.02	good	good	2.82	unsatisfactor	unsatisfactory
Multimetric index 1	0.63	good	good	0.44	moderate (mod	moderate (mod	0.4	moderate (mod	moderate (mod	0.46	moderate (mod	moderate (mod	0.29	unsatisfactor	unsatisfactory
Multimetric index 2	0	good	good	0	moderate (mod	moderate (mod	0.34	unsatisfactory	unsatisfactory	0.36	unsatisfactory	unsatisfactory	0.14	poor (ba	poor (ba
Individual density [Ind/m²]	866.4			670.4			7928.8			1310.4			2116.8		
Ecological status class	Good			Moderate			unsatisfactory (poor)			unsatisfactory (poor)			bad		
Metrics	Is	BW	Score	Actual	BW	Score	Actual	BW	Score	Actual	BW	Score	Actual	BW	Score
Total number of taxis	44	77	0.57	34	77	0.44	40	77	0.52	37	77	0.48	30	77	0.39
EPT taxa	18	36.5	0.49	14	36.5	0.38	8	36.5	0.22	8	36.5	0.22	0	36.5	0
% EPT taxa	40.91	58.96	0	41.18	58.96	0.7	20	58.96	0.34	21.62	58.96	0.37	0	58.96	0
% Oligochaeta & Diptera taxa	54.5	75.7	0	55	75.7	0.74	35	75.7	0.46	45.95	75.7	0.61	43.33	75.7	0.57
Diversity index (Margalef)	6.16	9.45	0.65	4.76	9.45	0.5	4.13	9.45	0.44	4.46	9.45	0.47	3.68	9.45	0.39
Degradation index	68	175.5	0.39	45	175.5	0.26	7	175.5	0.04	25	175.5	0.14	0	175.5	0
RETI	0.71	0.88	0.81	0.13	0.88	0.14	0.36	0.88	0.41	0.53	0.88	0.6	0.11	0.88	0.13
Littoral	4.51	5.81	0.78	3.46	5.81	0.6	4.14	5.81	0.71	3.99	5.81	0.69	3.19	5.81	0.55
Section	Trench			Trench			Trench			Grabenland			Ditch land		
Water	Aubachl (Fruttnerbach)			Vocherabach			Wellingbach			Pfaffenbach			Stiefing		
Date from	06.04.2016			05.04.2016			05			05			05		
Partial life area (PLA)	MHS			MHS			MHS			MHS			MHS		
Bioregion	GF (14)			GF (14)			GF (14)			GF (14)			GF (14)		
Basic condition MMI	1.75			1.75			1.75			1.75			1.75		
SI (Zelinka & Marvan)	2.09	good	good	1.91	good	good	1.62	very good (hig	very good (hig	1.92	good	good	1.78	good	good
Multimetric index 1	0.6	good	good	0.69	good	good	0.75	good	good	0.4	moderate (mod	moderate (mod	0.59	moderate (mod	moderate (mod
Multimeter index 2	0.53	Moderate (moderate	moderate (mod	0.62	good	good	0.71	good	good	0.39	unsatisfactory	unsatisfactory	0.53	moderate (mod	moderate (mod
Individual density [Ind/m²]	1620			1768.8			227			5284.8			262		
Ecological status class	Moderate			Good			good			moderate			moderate		
Metrics	Is	BW	Score	Actual	BW	Score	Actual	BW	Score	Actual	BW	Score	Actual	BW	Score
Total number of taxis	50	77	0.65	70	77	0.91	66	77	0.86	41	77	0.53	47	77	0.61
EPT taxa	17	36.5	0.47	21	36.5	0.58	25	36.5	0.68	12	36.5	0.33	15	36.5	0.41
% EPT taxa	34	58.96	0.58	30	58.96	0.51	37.88	58.96	0.64	29.27	58.96	0.5	31.91	58.96	0.54
% Oligochaeta & Diptera taxa	52	75.7	0	4	75	0.55	59.09	75.7	0.78	56.1	75.7	0.74	46.81	75.7	0.62
Diversity index (Margalef)	6.21	9.45	0.66	8.96	9.45	0.95	8.18	9.45	0.87	4.32	9.45	0.46	5.68	9.45	0.6
Degradation index	53	175.5	0.3	88	175.5	0.5	115	175.5	0.66	12	175.5	0.07	70	175.5	0
RETI	0.61	0.88	0.7	0.4	0.88	0.5	0.53	0.88	0.6	0.07	0.88	0.08	0.66	0.88	0.75
Littoral	4.13	5.81	0.71	4.75	5.81	0.82	4.5	5.81	0.77	3.54	5.81	0.61	4.21	5.81	0.72

Water	Lieberbach			Glauningbach			Labillbach			Poppendorfer Bach			Gnasbach Old arm		
Date from	06			06			05			06			06		
Partial life space (TLR)	MHS			MHS			MHS			MHS			MHS		
Bioregion	GF (14)			GF (14)			GF (14)			GF (14)			GF (14)		
Basic condition	1.75			1.75			1.75			1.75			1.75		
Metrics "still very good condition" (EQR I/II)	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR
Screening taxa	29	-	-	25	-	-	24	-	-	24	-	-	23	-	-
Sensitive taxa	9	20	0.45	7	20	0.35	5	20	0.25	4	20	0.2	1	20	0.05
Degradation score	64	105	0.61	41	105	0.39	13	105	0.12	30	105	0.29	0	105	0
Screening – General stress (AB-EQR I/II)			0			0			0			0			0.03
Saprobic score	82.74	83.5	0.99	99.25	83.5	1.19	103.64	83.5	1.24	96.64	83.5	1.16	123.11	83.5	1.47
Screening - Organic Burden (OB-EQR I/II)			0.99			1.19			1.24			1.16			1.47
Metrics "still in good condition" (EQR II/III)	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR
Screening taxa	29	-	-	25	-	-	24	-	-	24	-	-	23	-	-
Sensitive taxa	9	13	0.69	7	13	0.54	5	13	0.38	4	13	0.31	1	13	0.08
Degradation score	64	65	0.98	41	65	0.63	13	65	0.2	30	65	0.46	0	65	0
Screening - General stress (AB-EQR II/III)			0			0			0			0			0.04
Saprobic score	82.74	115	0.72	99.25	115	0.86	103.64	115	0.9	96.64	115	0.84	123.11	115	1.07
Screening - Organic burden (OB-EQR II/III)			0.72			0			0.9			0.84			1.07
Screening - General stress (AB-EQR I/II)															
Screening – General stress (AB-EQR II/III)	< 1	Action required		< 1	Action required		< 1	Action required		< 1	Action required		< 1	Action required	
Screening – Organic stress (OB-EQR I/II)	<= 1	Very good (high)		> 1	Good		> 1	good		> 1	good		> 1	good	
Screening - Organic burden (OB-EQR II/III)				<= 1	good		<= 1	good		<= 1	good		> 1	Action required	
Reductions (knockout criterion)															
Result of screening method (T-EQR)	Action required			Action required			Action required			Action required			Action required		
Water	Aubachl (Fruttnerbach)			Vocherabach			Wellingbach			Pfaffenbach			Stiefing		
Date from	06.04.2016			05.04.2016			05			05			05		
Partial life area (PLA)	MHS			MHS			MHS			MHS			MHS		
Bioregion	GF (14)			GF (14)			GF (14)			GF (14)			GF (14)		
Basic condition	1.75			1.75			1.75			1.75			1.75		
Metrics "still very good condition" (EQR I/II)	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR
Screening taxa	36	-	-	36	-	-	49	-	-	32	-	-	30	-	-
Sensitive taxa	15	20	0.75	13	20	0.65	19	20	0.95	5	20	0.25	10	20	0.5
Degradation score	56	105	0.53	78	105	0.74	101	105	0.96	13	105	0.12	59	105	0.56
Screening - General stress (AB-EQR I/II)			0.64			0.7			0			0.19			0.5
Saprobic score	94.6	83.5	1.13	78.3	83.5	0.94	76.03	83.5	0.91	110.42	83.5	1.32	82.66	83.5	0.99
Screening - Organic Burden (OB-EQR I/II)			1.13			0.94			0			1.32			0.99
Metrics "still in good condition" (EQR II/III)	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR	Observed	Expected	EQR
Screening taxa	36	-	-	36	-	-	49	-	-	32	-	-	30	-	-
Sensitive taxa	15	13	1.15	13	13	1	19	13	1.46	5	13	0.38	10	13	0.77
Degradation score	56	65	0.86	78	65	1.2	101	65	1.55	13	65	0.2	59	65	0.91
Screening – General stress (AB-EQR II/III)			1.01			1.1			1.51			0			0.84
Saprobic score	94.6	115	0.82	78.33	115	0.68	76.03	115	0.66	110.42	115	0.96	82.66	115	0.72
Screening - Organic burden (OB-EQR II/III)			0.82			0			0			0.96			0.72
Screening – General stress (AB-EQR I/II)	< 1	Good		< 1	good		< 1	good		< 1			< 1		
Screening - General stress (AB-EQR II/III)	>= 1	good		>= 1	good		>= 1	good		< 1	Action required		< 1	Action required	
Screening - Organic Pollution (OB-EQR I/II)	> 1	good		<= 1	very good (high)		<= 1	very good (high)		> 1	good		<= 1	Very good (high)	
Screening - Organic pollution (OB-EQR II/III)	<= 1	good								<= 1	good				
Reductions (elimination criterion)															
Result of screening method (T-EQR)	good			good			Good			Action			Action required		

8.3 EPT evaluations of the streams examined

EPT evaluation									
Section:		GF & ditch land			Water body:		Lieberbach		
Investigation site:		Lieberbach							
Date/time:		April 6, 2016							
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	9	19.15	1	2.13	9	19.15	19	40.43	47
Genre	6	15.38	1	2.56	8	20.51	15	38.46	39
Family	4	19.05	1	4.76	5	23.81	10	47.62	21
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	17	15	1.12	1.27					
Partial habitat:	MHS	Substrate			Area:	1.25 m	Sample location:		1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	9	19.15	1	2.13	9	19.15	19	40.43	47
Genre	6	15.38	1	2.56	8	20.51	15	38.46	39
Family	4	19.05	1	4.76	5	23.81	10	47.62	21
Individuals/flight	459	42.07	11	1.01	104	9.53	574	52.61	1091
Biomass [g/fl.]	0		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	17	15	1.12	1.27					
Individuals/fl.	206	159	2.79	3.61					
Biomass [g/fl.]	0	0.0000							
Only sampling sites with taxa are taken into account.									
EPT evaluation									
Section:		GF & ditch land			Water bodies:		Glauningbach		
Investigation site:		Glauningbach							
Date/time:		April 6, 2016							
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	6	15.79	3	7.89	5	13.16	14	36.84	38
Type	4	11.76	3	8.82	5	14.71	12	35.29	34
Family	4	20	2	10	2	10	8	40	20
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	9	5	1.56	2.80					
Partial habitat:	MHS	Substrate			Area:	1.25 m	Sample location:		1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	6	15.79	3	7.89	5	13.16	14	36.84	38
Type	4	11.76	3	8.82	5	14.71	12	35.29	34
Family	4	20.00	2	10.00	2	10	8	40	20
Individuals/flight	415	49.52	155	18.50	37	4.42	607	72.43	838
Biomass [g/fl.]	0		0		0		0		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	9	5	1.56	2.80					
Individuals/fl.	8	65	6.90	9.34					
Biomass [g/plant]	0	0							
Only sampling sites with taxa taken into account.									

EPT evaluation									
Section:	GF & ditch land				Water bodies:	Labillbach			
Investigation site:	Labillbach								
Date/time:	05.04.2016								
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	4	8.70	1	2.17	4	8.70	9	19.57	46
Type	3	7.69	1	2.56	4	10.26	8	20.51	39
Family	3	14.29	1	4.76	1	4.76	5	23.81	21
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	26	21	0.35	0.43					
Partial habitat:	MHS	Substrate			Area:	1.25 m		Sample location:	1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	4	8.70	1	2.17	4	8.70	9	19.57	46
Type	3	7.69	1	2.56	4	10.26	8	20.51	39
Family	3	14.29	1	4.76	1	4.76	5	23.81	21
Individuals/flight	815	8.20	16	0.16	27	0.27	858	8.63	9944
Biomass [g/fl.]	0		0		0		0		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	26	21	0.35	0.43					
Individuals/fl.	813	8024	0.11	0.11					
Biomass [g/plant]	0	0							
Only sampling sites with taxa taken into account.									
EPT evaluation									
Section:	GF & ditch land				Water bodies:	Poppendorfer Bach			
Investigation site:	Poppendorfer Bach								
Date/time:	April 6, 2016								
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	3	6.98	0	0.00	6	13.95	9	20.93	43
Genre	2	5.56	0	0.00	6	16.67	8	22.22	36
Family	2	10.53	0	0	3	15.79	5	26.32	19
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	18	17	0.50	0.53					
Partial habitat:	MHS	Substrate			Area:	1.2 m		Sample location:	1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	3	6.98	0	0.00	6	13.95	9	20.93	43
Genre	2	5.56	0	0.00	6	16.67	8	22.22	36
Family	2	10.53	0	0	3	15.79	5	26.32	19
Individuals/flight	219	13.28	0	0	48	2.91	267	16.19	1649
Biomass [g/plant]	0		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	18	17	0.50	0.53					
Individuals/bottle	1051	999	0.25	0.27					
Biomass [g/plant]	0	0							
Only sampling sites with taxa taken into account.									

EPT evaluation									
Section:	GF & ditch land			Water body:	Gnasbach oxbow lake				
Investigation site:	Gnasbach Altarm								
Date/time:	06.04.2016								
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	0	0.00	0	0.00	1	3.03	1	3.03	33
Type	0	0	0	0	1	3.13	1	3.13	32
Family	0	0	0	0	1	5.0	1	5	20
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	11	8	0.09	0.13					
Partial habitat:	MHS	Substrate			Area:	1.25 m		Sample location:	1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	0	0.00	0	0.00	1	3.03	1	3.03	33
Type	0	0	0	0	1	3.13	1	3.13	32
Family	0	0	0	0	1	5.0	1	5	20
Individuals/bottle	0	0	0	0	4	0.15	4	0.15	2650
Biomass [g/fl.]	0		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	11	8	0.09	0.13					
Individuals/fl.	602	577	0.01	0.01					
Biomass [g/plant]	0	0.0000							
Only sampling sites with taxa taken into account.									
EPT evaluation									
Section:	GF & ditch land			Water body:	Aubachl (Fruttnerbach)				
Investigation site:	Aubachl (Fruttnerbach)								
Date/time:	06.04								
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	11	19.64	2	3.57	7	12.50	20	35.71	56
Type	7	14.29	2	4.08	6	12.24	15	30.61	49
Family	5	19.23	2	7.69	4	15.38	11	42.31	26
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	19	15	1.05	1.33					
Partial habitat:	MHS	Substrate			Area:	1.25 m		Sample location:	1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	11	19.64	2	3.57	7	12.50	20	35.71	56
Type	7	14.29	2	4.08	6	12.24	15	30.61	49
Family	5	19.23	2	7.69	4	15.38	11	42.31	26
Individuals/flight	533	26.01	9	0.44	37	1.81	579	28.26	2049
Biomass [g/fl.]	0		0		0		0		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	19	15	1.05	1.33					
Individuals/fl.	321	278	1.80	2.08					
Biomass [g/fl.]	0	0.0000							
Only sampling sites with taxa are taken into account.									

EPT evaluation									
Section:	GF & ditch land				Water bodies:		Vocherabach		
Investigation site:	Vocherabach								
Date/time:	05.04.2016								
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	11	13.41	6	7.32	7	8.54	24	29.27	82
Type	7	10.94	5	7.81	5	7.81	17	26.56	64
Family	3	11.11	3	11.11	4	14.81	10	37.04	27
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	45	38	0.53	0.63					
Partial habitat:	MHS	Substrate			Area:	1.25 m	Sample location:	1	
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	11	13.41	6	7.32	7	8.54	24	29.27	82
Type	7	10.94	5	7.81	5	7.81	17	26.56	64
Family	3	11.11	3	11.11	4	14.81	10	37.04	27
Individuals/flight	771	34.25	89	3.95	45	2.00	905	40.20	2251
Biomass [g/bottle]	0		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	45	38	0.53	0.63					
Individuals/fl.	1279	999	0.71	0.91					
Biomass [g/plant]	0	0.0000							
Only sampling sites with taxa were taken into account.									
EPT evaluation									
Section:	GF & ditch land				Water bodies:		Wellingbach		
Test site:	Wellingbach								
Date/time:	05.04.2016								
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	12	17.14	5	7.14	9	12.86	26	37.14	70
Type	8	13.11	5	8.20	9	14.75	22	36.07	61
Family	5	14.29	4	11.43	5	14.29	14	40.00	35
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	27	17	0.96	1.53					
Partial living space:	MHS	Substrate			Area:	1.25 m	Sample location:	1	
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	12	17.14	5	7.14	9	12.86	26	37.14	70
Type	8	13.11	5	8.20	9	14.75	22	36.07	61
Family	5	14.29	4	11.43	5	14.29	14	40.00	35
Individuals/bottle	1134	38.31	279	9.43	131	4.43	1544	52.16	2960
Biomass [g/fl.]	0.00		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	27	17	0.96	1.53					
Individuals/fl.	433	384	3.57	4.02					
Biomass [g/plant]	0	0							
Only sampling sites with taxa taken into account.									

EPT evaluation									
Section:		GF & ditch land			Water bodies:		Pfaffenbach		
Investigation site:		Pfaffenbach							
Date/time:		05.04.2016							
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	4	8.70	1	2.17	7	15.22	12	26.09	46
Type	3	6.82	1	2.27	7	15.91	11	25.00	44
Family	3	10.34	1	3.45	4	13.79	8	27.59	29
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	11	8	1.09	1.50					
Partial habitat:	MHS	Substrate			Area:		1.25 m	Sample location:	1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	4	8.70	1	2.17	7	15.22	12	26.09	46
Type	3	6.82	1	2.27	7	15.91	11	25.00	44
Family	3	10.34	1	3.45	4	13.79	8	27.59	29
Individuals/flight	609	9.17	180	2.71	50	0.75	839	12.64	6640
Biomass [g/fl.]	0		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiroprac tic.					
Type	11	8	1.09	1.50					
Individuals/bottle	57	39	14.72	21.51					
Biomass [g/plant]	0	0							
Only sampling sites with taxa taken into account.									
EPT evaluation									
Section:		GF & ditch land			Water bodies:		Stiefing		
Test site:		Stiefing							
Date/time:		05.04.2016							
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	9	16.98	1	1.89	7	13.21	17	32.08	53
Type	6	13.33	1	2.22	7	15.56	14	31.11	45
Family	3	13.04	1	4.35	4	17.39	8	34.78	23
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	22	16	0.77	1.06					
Partial habitat:	MHS	Substrate			Area:		1.25 m	Sample location:	1
	Ephemeroptera		Plecoptera		Trichoptera		EPT		Total
	Number	%	Number	%	Number	%	Number	%	Number
Type	9	16.98	1	1.89	7	13.21	17	32.08	53
Type	6	13.33	1	2.22	7	15.56	14	31.11	45
Family	3	13.04	1	4.35	4	17.39	8	34.78	23
Individuals/flight	1174	35.31	2	0.06	44	1.32	1220	36.69	3325
Biomass [g/fl.]	0		0		0		0.0000		0.0000
	Diptera	Chiro.	EPT/Dip.	EPT/Chiro.					
Species	22	16	0.77	1.06					
Individuals/fl.	656	637	1.86	1.92					
Biomass [g/plant]	0	0							
Only sampling sites with taxa are taken into account.									

8.4 Indices & distribution

Projekt: Grabenland

Abschnitt: Grabenland

Untersuchungsstelle: Lieberbach

Teillebensraum: MHS

Gewässer: Lieberbach

Datum/Zeit: 06.04.2016

Diversität		Indices		Fresstyp	Valenz	HauptFT	kum	Region	Valenz	kum
Taxa (Gesamt)	48	Taxa	25	Taxa	41			Taxa	23	
Diversität W&D	3,65	SI Zelinka&Marvan	1,35	ZKL	1,68	18,55	1,68	EUK	0,90	0,90
Diversität S&W	2,53	Streuung	± 0,028	WEI	4,78	52,94	6,46	HYK	0,72	1,62
Evenness	0,65	SI Pantle&Buck	1,37	aFIL	0,05	1,06	6,51	ER	3,99	5,60
Margalef	6,73	Streuung	± 0,028	pFIL	0,04		6,56	MR	2,25	7,85
				DET	2,48	27,45	9,04	HR	0,94	8,79
		Saprobie	Valenz	MIN	0,00		9,04	EP	0,58	9,37
		xeno	1,75	HOL	0,00		9,04	MP	0,09	9,47
		oligo	3,72	RÄU	0,96		10,00	HP	0,01	9,47
		beta	3,83	PAR	0,00		10,00	LIT	0,49	9,97
		alpha	0,70	SON	0,00		10,00	PRO	0,03	10,00
		poly	0,00							
				Index				Index	ungew.	gew.
				RETI	0,71			LZI	3,70	3,37
				PETI	0,29			RIZI	3,44	3,26

Abschnitt: Grabenland

Gewässer: Glauningbach

Untersuchungsstelle: Glauningbach

Datum/Zeit: 06.04.2016

Teillebensraum: MHS

Diversität		Indices		Fresstyp	Valenz	HauptFT	kum	Region	Valenz	kum
Taxa (Gesamt)	37	Taxa	18	Taxa	31			Taxa	16	
Diversität W&D	3,17	SI Zelinka&Marvan	2,11	ZKL	0,39	4,66	0,39	EUK	1,29	1,29
Diversität S&W	2,20	Streuung	± 0,216	WEI	0,66	7,88	1,05	HYK	0,94	2,23
Evenness	0,61	SI Pantle&Buck	2,00	aFIL	0,02	0,83	1,06	ER	1,02	3,25
Margalef	5,35	Streuung	± 0,195	pFIL	0,05		1,12	MR	1,29	4,53
				DET	7,24	86,63	8,36	HR	1,33	5,86
		Saprobie	Valenz	MIN	0,00		8,36	EP	1,17	7,04
		xeno	1,00	HOL	0,00		8,36	MP	0,50	7,53
		oligo	2,19	RÄU	1,64		10,00	HP	0,34	7,88
		beta	3,07	PAR	0,00		10,00	LIT	1,54	9,42
		alpha	2,17	SON	0,00		10,00	PRO	0,58	10,00
		poly	1,57							
				Index				Index	ungew.	gew.
				RETI	0,13			LZI	5,10	4,43
				PETI	0,87			RIZI	4,22	3,68

Water body: Lablllbach
Date/time: 05. 04.2016

Section: Grabenland Investigation site: Poppendorfer Bach Sub-habitat: MHS

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Section: Grabenland Investigation site:

Water body: Gnasbach Altarm

Gnasbach Altarm Subhabitat: MHS:

Date/time: 06.04.2016

TaxB (total)	3	Taxa	1	Taxa	2s	Taxa	15
Diversity WSS	2.64	SI	Bz	W	0.47	0.47	0.21
German 5&W	1.83	SI	0.415	wEl	0.50	5.e7	0.97
Evenness	0.52	SI	2.7	aFIL	5.14	5B,81	8.11
Margalef	4.19	SI	0.389	pFIL	0.00	8.11	MR
				DET	2.66	30.34	8.77
				MIN	0	8.37	EP
				HOL	0.00	6.77	MP
				RAU	1.23	40	HP
				PAR	0.00	10.00	LIT
				sON	0.00	10,00	PRO
				Index			Indgx
				RETI	0.11		LZI
				PETI	0.05		RIZI

Section Grabenlafid

Water body: Auba?hl (Fruttn.e.rbach)

Investigation site: Aubachl (Fruttn.e.rbach) Tetra life

Date/time: 06.04.2016

area: MHS

Diversität	Indic	Fresstyp	Valence	Main FT	cum	Region	Vajenz	hum
Taxa (Gesarrt)	54	Taxa	25	Taxa	47	Taxa	23	
Diversity W8D	3.45	SI	2.09	ZKL	3.42	35.8 9	3.42	EUK
Diversity 56W	2.39	Dispersion	0.072	WEI	2.44	1	r,ss	HYK
Evenness	0.50	SI	2.7	aFIL	0.04	2S,57	5.90	ER
Marsai+f	e,ss	strevuny	0.389	priL	0.05	1.28	s,sa	un
				DET	3.55	37.2 6	9.53	HR
				MIN	0.00	9.53	EP	0.96
				HOL	0.00	9.53	MP	0.12
				RAU	0.46	10.05	HP	0.11
				PAB	0.00	0.00	LIT	0.00
				SDN	0.00	10.00	PRO	0.17
				Index			Index	ungew.
				RETI	0.61		LZI	4.41
				PETI	0.29		RIZI	3.90

Section: GrabenJand Investigation
 site: Vocherabach Teilebensraum:
 MHS

Water body: Vocherabach
 Date/time: 05.04.2016

diversity	Index	Frasstyp	Valence	Main FT	k u m	Region	Valence	cum
Taxa (Geaam)	83	Taxa	37	Taxa	6g	Taxa	35	
Diversity W&D	5.21	SI Zelinka&Marvan	1.91	ZhL	0.16	1.83	0.16	0.06
Diversity S&W	3.5t	Dispersion	0.002	WEI	5.80	1.83	0.16	0.06
Evenness	0.82	SI Pantle&Buck	1.89	aFIL	0.22	16.24	1.98	2.85
Margalef	0.62	Dispersion	0.061	pFIL	1.1	5.15		5.49
		DET	3.45		40.11	8.61	HR	2.4B
		SEP	0.00			8.6	EP	1.46
		HOL	0.00			8.6	MP	0
		RAU	1.39		10.0D		HP	0.02
		PAR	0.00		10.00		LIT	0.2S
		SON	0.00		10.0		PRO	0.01
		Index					Index	gaw.
		RET	0.44				LZI	4.41
		Phi	0.58				RZI	4.31

Section: Grabenland Study site:
 Wellingbach Sub-habitat: MHS

Water body: Wellingbach
 Date/time: DN.04.2016

diversity	Index	Frasstyp	Valence	Main FT	k u m	Region	Valence	kum
Taxa (GeBamg)	69	Taxa	28	Taxa	61	Taxa	26	
Diversity W&D	4.25	SI Zelinka&Marvan	1.02	ML	2.21	23.5	2.21	0.33
Diversity S&W	2.95	Stagflation	0.041	WEI	2.78	20.64	4.99	0.79
Evenness	0.70	SI Pantle&Buofi	0.62	aFIL	0.26	3.83	5.25	3.34
Margalef	8.6	Dispersion	0.041	pFIL	0.10	5.35		6.64
		DET	4.04		43.01	9.40	HR	1.92
		SEP	0.00			9.40	EP	0.90
		HOL	0.00			9.40	MP	0.03
		RAU	0.80		0		UP	0.00
		PAR	0.00		10.00		LIT	0.50
		SON	0.00		10.00		PRO	0.02
		Index					Index	ungen.
		RET	0.53				LZI	4.19
		PET	0.47				RZI	3.93

Section: Grabenland Investigation
 site: Pfaffenbach Sub-region: MHS

Water: Pfaffenbach
 Date/time: 05.04.2016

Diversität		Indices		Fresstyp	Valenz	HauptFT	kum	Region	Valenz	kum
Ta&s(s(Sea mt)	B	Texa 2g		* a	36			Taxa	23	
Oiversität W&D	2.48	SI 2alinka&Marvan 1.8 2		SKL	0.58	s,19	0,5 B	eUK	0.17	0.17
OIV*sk9t S&W	1.72	Standard deviation z 0.051		WEJ	0.10	0.99	0.6 8	HYK	0.s0	1.07
Evennese	0.45	SI Panlle6Buck 1 .6B		aFIL	B,96	9.70	1.6 4	ER	0.68	2.04
Maigaler	4.e9	Std. dev. E D		pFIL	0.00	1.64		MR	1.90	5.84
				DET	B,32	9.96		HR	1.80	5.84
		Sapiobie	Valenz	kum	um	q,00		9,95	EP	1.84 7.66
		xeno	0.74	0.74	HOL	0.00		9,85	MP	0.0B 7.78
		oligo	2.16	2.90	RAU	0,04		10,00	HP	0.01 7.7
		bela	4,M	7.23	PAR	0,00		10,00	LIT	1.46 8.23
		alpha	2,74	8.st	soN	0.00		10,00	PRO	D,77 10.00
		poly	0.03	30						
				Index				Index	Unspecifi ed	gew
				RETI	0.07			LZü	4.45	5.62
				Pol	0.93			RE2]	4.39	d, 47

Section: Grabenland Survey
 location: Stiefing Habitat: MHS

Water body: Stiefing
 Date/time: 05.04.2016

Diversity		Indices		Feeding type	Valehz	Main FT	k u m	Region	Valence	cum
Taxa (total)	s3	[23 t		Taxa	45	t		Taxa	21	
Drversltgt W8O	3.32	SI ZellInka&Marvan 1.78		ZKL	2.96	2.9 0		EUK	0.59	0.50
Diversit8L S&W	2.30	Dispersion t o,049		WEI	3.55	z5,B O a ,5 j		HYK	g,94	t.53
Evennese	0.5B	dsI Pantle&Bucc 7 1		aFIL	0.22	5.0 1 6.7 Z		ER	2.12	3.65
Mafg8lelf	6.41	Dispersion 0.045)		pFIL	D,27	7.00		MR	2.35	6.00
				DET	2.88	9.89		OR	2.12	8.11
		Saprobie	Valence **	MIN	0,00	9,89		«	1.03	9.15
		xano	0.63 0.63	riOL	0,00	9,99		MP	0.01	9.16
		oligo	2.88 3.52	RAU	0,11	10,00		EP	0.01	9.1t
		bete	4.53 8.06	PAR	0,00	10,00		LIT	0.79	9.B5
		alpha	1.94 10.00	SON	0,00	10,00		ID	0.05	10.00
		poly	0.00 10.0							
				Index				Index	unqew.	sew.
				RETI	0.66			LZI	4.27	4.16
				PETI	0.34			RIZI	3.82	3.74

Auswertung quantitativ, standard

8.5 Data set for faunistic analyses

ID	Bioregion	Water body name	Project
54993	3	Burggrabenbach	2010_GZÜV-Steiermark
25594	3	Feistritz	2007_GZÜV-Lower Austria
55064	3	Griffenbach	2010_GZÜV-Carinthia
19506	3	Gurk	MMI_WGEV_Kärnten_2003
19509	3	Gurk	MMI_WGEV_Kärnten_2003
19511	3	Gurk	MMI_WGEV_Kärnten_2003
23358	3	Gurk	2005_WGEV_Carinthia
23360	3	Gurk	2005_WGEV_Carinthia
55004	3	Kleinsemmeringbach	2010_GZÜV-Styria
1995	3	Lafnitz	MMI_WGEV_Steiermark_2003
24241	3	Lafnitz	2005_WGEV_Styria
19520	3	Metnitz	MMI_WGEV_Kärnten_2003
23364	3	Metnitz	2005_WGEV_Carinthia
24251	3	Mürz	2005_WGEV_Styria
24252	3	Mürz	2005_WGEV_Styria
24253	3	Mürz	2005_WGEV_Styria
25017	3	Mürz	2007_GZÜV-Styria
55011	3	Mürz	2010_GZÜV-Styria
54893	3	Reißenbach	2010_GZÜV-Lower Austria
24264	3	Schäffernbach	2005_WGEV_Styria
13460	3	St. Oswaldbach	AQEM Austria
19960	3	Tauchenbach	MMI_WGEV_Steiermark_2003
24268	3	Tauchenbach	2005_WGEV_Styria
19519	3	Wimitz	MMI_WGEV_Kärnten_2003
23365	3	Wimitz	2005_WGEV_Carinthia
50102	3	Wimitz	2007_GZÜV-Carinthia
58063	3	Wimitzbach	2013_GZÜV-Carinthia
13458	3	Winklbach	AQEM Austria
57998	3	Zöbernbach	2013_GZÜV-Lower Austria
57878	13	Arany patak	2013_GZÜV/GGK-Burgenland
54967	13	Drumlingbach	2010_GZÜV-Burgenland
55095	13	Edelbach, cold water spring	2011_GZÜV-Burgenland
55096	13	Edlesbach	2011_GZÜV-Burgenland
54975	13	Eisbach	2010_GZÜV-Burgenland
57875	13	Eisbach [Wulka]	2013_GZÜV/GGK-Burgenland
54965	13	Erlbach	2010_GZÜV-Burgenland
24235	13	Feistritz	2005_WGEV_Styria
25008	13	Feistritz	2007_GZÜV-Styria
54982	13	Feistritz	2010_GZÜV-Styria
24234	13	Feistritz	2005_WGEV_Styria
19623	13	Fischa	WGEV-MHS Fischa 26.05.04
19625	13	Fischa	WGEV-MHS Fischa 26.05.04
54972	13	Frauenbrunnbach	2010_GZÜV-Burgenland

47205	13	Goldbach	2007_GZÜV-Burgenland
54958	13	Goldbach	2010_GZÜV-Burgenland
55107	13	Goldbach	2011_GZÜV/GGK-Burgenland
54980	13	Golser Canal	2010_GZÜV-Burgenland
54989	13	Grazbach	2010_GZÜV-Styria
54963	13	Grieselbach	2010_GZÜV-Burgenland
54992	13	Gschmaierbach	2010_GZÜV-Styria
57874	13	Güns	2013_GZÜV/GGK-Burgenland
54966	13	Guribach	2010_GZÜV-Burgenland
25011	13	Hartberger Safen	2007_GZÜV-Styria
55002	13	Hartberger Safen	2010_GZÜV-Styria
55003	13	Hartberger Safen	2010_GZÜV-Styria
57876	13	Hirmer Bach	2013_GZÜV/GGK-Burgenland
55098	13	Hornstein Stream	2011_GZÜV-Burgenland
55097	13	Hornsteiner Bach	2011_GZÜV-Burgenland
1994	13	Ilzbach	MMI_WGEV_Steiermark_2003
24237	13	Ilzbach	2005_WGEV_Styria
54979	13	Johannesbach	2010_GZÜV-Burgenland
24239	13	Kainach	2005_WGEV_Styria
54986	13	Kalkbach	2010_GZÜV-Styria
19633	13	Kamp	WGEV-MHS Kamp
19635	13	Kamp	WGEV-MHS Kamp
54985	13	Kornbach	2010_GZÜV-Styria
54988	13	Kötschmanngrabenb.	2010_GZÜV-Styria
13439	13	Lafnitz	AQEM Austria
13489	13	Lafnitz	AQEM Austria
23029	13	Lafnitz	2005_WGEV_Burgenland
24242	13	Lafnitz	2005_WGEV_Styria
25009	13	Lafnitz	2007_GZÜV-Styria
47207	13	Lafnitz	2007_GZÜV-Burgenland
54960	13	Lafnitz	2010_GZÜV-Burgenland
54983	13	Lafnitz	2010_GZÜV-Styria
55109	13	Lafnitz	2011_GZÜV/GGK-Burgenland
54999	13	Lahn	2010_GZÜV-Styria
57872	13	Lapincs	2013_GZÜV/GGK-Burgenland
55006	13	Lehenbach	2010_GZÜV-Styria
47200	13	Leitha	2007_GZÜV-Burgenland
54781	13	Leitha	2009_GZÜV/GGK-Burgenland
54953	13	Leitha	2010_GZÜV-Burgenland
55103	13	Leitha	2011_GZÜV/GGK-Burgenland
57886	13	Leitha	2013_GZÜV/GGK-Burgenland
55099	13	Lukabach	2011_GZÜV-Burgenland
54974	13	Marzer Bach	2010_GZÜV-Burgenland
54971	13	Nikitschbach	2010_GZÜV-Burgenland
54977	13	Nodbach	2010_GZÜV-Burgenland
54978	13	Parndorfer Stream	2010_GZÜV-Burgenland

54987	13	Petersdorf Stream	2010_GZÜV-Styria
13435	13	Pinka	AQEM Austria
13436	13	Pinka	AQEM Austria
19969	13	Pinka	WGEV Burgenland 2003
23027	13	Pinka	2005_WGEV_Burgenland
47204	13	Pinka	2007_GZÜV-Burgenland
54783	13	Pinka	2009_GZÜV/GGK-Burgenland
54957	13	Pinka	2010_GZÜV-Burgenland
55106	13	Pinka	2011_GZÜV/GGK-Burgenland
57885	13	Pinka	2013_GZÜV/GGK-Burgenland
54995	13	Prötisbach	2010_GZÜV-Styria
54984	13	Prüfingbach	2010_GZÜV-Styria
24257	13	Raab	2005_WGEV_Styria
24258	13	Raab	2005_WGEV_Styria
24517	13	Raab	2005_WGEV_IWG/Burgenland
24519	13	Raab	2005_WGEV_IWG/Burgenland
25007	13	Raab	2007_GZÜV-Styria
47201	13	Raab	2007_GZÜV-Burgenland
47202	13	Raab	2007_GZÜV-Burgenland
54954	13	Raab	2010_GZÜV-Burgenland
55005	13	Raab	2010_GZÜV-Styria
55104	13	Raab	2011_GZÜV/GGK-Burgenland
57881	13	Raab	2013_GZÜV/GGK-Burgenland
57882	13	Raab	2013_GZÜV/GGK-Burgenland
19973	13	Rabnitz	WGEV-Burgenland 2003
23023	13	Rabnitz	2005_WGEV_Burgenland
57879	13	Rabnitz	2013_GZÜV/GGK-Burgenland
54970	13	Raidingbach	2010_GZÜV-Burgenland
54996	13	Rauschbach	2010_GZÜV-Styria
47206	13	Rechnitzbach	2007_GZÜV-Burgenland
54959	13	Rechnitzbach	2010_GZÜV-Burgenland
55108	13	Rechnitzbach	2011_GZÜV/GGK-Burgenland
57877	13	Rechnitzbach	2013_GZÜV/GGK-Burgenland
54994	13	Rettenbach	2010_GZÜV-Styria
55101	13	Ribicabach	2011_GZÜV-Burgenland
55000	13	Rittschein	2010_GZÜV-Styria
55001	13	Rittschein	2010_GZÜV-Styria
25010	13	Rittscheinbach	2007_GZÜV-Styria
57888	13	Rodlingbach	2013_GZÜV/GGK-Burgenland
54998	13	Römerbach	2010_GZÜV-Styria
19957	13	Schäffernbach	MMI_WGEV_Steiermark_2003
54990	13	Schirnitzbach	2010_GZÜV-Styria
54991	13	Schirnitzbach	2010_GZÜV-Styria
54973	13	Sieggrabenbach	2010_GZÜV-Burgenland
19958	13	Stögersbach	MMI_WGEV_Steiermark_2003
24265	13	Stögersbach	2005_WGEV_Styria

54968	13	Stögersbach	2010_GZÜV-Burgenland
57873	13	Stoobar Bach	2013_GZÜV/GGK-Burgenland
19975	13	Stoobarbach	WGEV-Burgenland 2003
23021	13	Stoobarbach	2005_WGEV_Burgenland
47203	13	Strem	2007_GZÜV-Burgenland
54782	13	Strem	2009_GZÜV/GGK-Burgenland
54956	13	Strem	2010_GZÜV-Burgenland
55105	13	Strem	2011_GZÜV/GGK-Burgenland
57871	13	Strem	2013_GZÜV/GGK-Burgenland
54969	13	Stubenbach	2010_GZÜV-Burgenland
55100	13	Sulzbach	2011_GZÜV-Burgenland
1997	13	Tauchenbach	WGEV-Burgenland 2003
23025	13	Tauchenbach	2005_WGEV_Burgenland
57880	13	Tauchenbach	2013_GZÜV/GGK-Burgenland
54976	13	Tauscherbach	2010_GZÜV-Burgenland
57887	13	Teichbach	2013_GZÜV/GGK-Burgenland
54997	13	Werksbach	2010_GZÜV-Styria
19865	13	Vienna	MMI_WGEV_Vienna_2003
19866	13	Vienna	MMI_WGEV_Vienna_2003
23019	13	Wulka	2005_WGEV_Burgenland
47199	13	Wulka	2007_GZÜV-Burgenland
54952	13	Wulka	2010_GZÜV-Burgenland
55102	13	Wulka	2011_GZÜV/GGK-Burgenland
57883	13	Wulka	2013_GZÜV/GGK-Burgenland
57884	13	Wulka	2013_GZÜV/GGK-Burgenland
54964	13	Zickenbach	2010_GZÜV-Burgenland
57889	13	Zickenbach OW	2013_GZÜV/GGK-Burgenland
59175	14	Aubachl (Fruttnerb.)	Grabenland
55045	14	Auersbach	2010_GZÜV-Styria
55022	14	Piercing Mur	2010_GZÜV-Styria
55021	14	Erabach	2010_GZÜV-Styria
55035	14	Ferbersbach	2010_GZÜV-Styria
55036	14	Ferbersbach	2010_GZÜV-Styria
55023	14	Fernitzer mill canal	2010_GZÜV-Styria
59171	14	Glauningbach	Grabenland
25022	14	Gnasbach	2007_GZÜV-Styria
59174	14	Gnasbach Old arm	Grabenland
24238	14	Kainach	2005_WGEV_Styria
25018	14	Kainach	2007_GZÜV-Styria
55012	14	Kainach	2010_GZÜV-Styria
55018	14	Katzelbach	2010_GZÜV-Styria
55019	14	Katzelbach	2010_GZÜV-Styria
54962	14	Klausenbach	2010_GZÜV-Burgenland
55024	14	Kutschenitza	2010_GZÜV-Styria
55047	14	Kutschenitza	2010_GZÜV-Styria
55048	14	Kutschenitza	2010_GZÜV-Styria

59172	14	Labillbach	Grabenland
55027	14	Leibenbach	2010_GZÜV-Styria
55028	14	Leibenbach	2010_GZÜV-Styria
54961	14	Lendva	2010_GZÜV-Burgenland
55040	14	Lieberbach	2010_GZÜV-Styria
59170	14	Lieberbach	Grabenland
55025	14	Muggenbach	2010_GZÜV-Styria
55026	14	Muggenbach	2010_GZÜV-Styria
55020	14	Mühlgang-Rechts	2010_GZÜV-Styria
55034	14	Oisnitz Bach	2010_GZÜV-Styria
55041	14	Ottersbach	2010_GZÜV-Styria
55042	14	Ottersbach	2010_GZÜV-Styria
55043	14	Ottersbach	2010_GZÜV-Styria
59178	14	Pfaffenbach	Grabenland
59173	14	Poppendorfer Bach	Grabenland
19951	14	Rassachbach	MMI_WGEV_Steiermark_2003
24259	14	Rassachbach	2005_WGEV_Styria
55015	14	Rassachbach	2010_GZÜV-Styria
55030	14	Ratscherbach	2010_GZÜV-Styria
55031	14	Ratscherbach	2010_GZÜV-Styria
19954	14	Saßbach	MMI_WGEV_Steiermark_2003
1995	14	Saßbach	MMI_WGEV_Steiermark_2003
24262	14	Saßbach	2005_WGEV_Styria
24263	14	Saßbach	2005_WGEV_Styria
25021	14	Saßbach	2007_GZÜV-Styria
55044	14	Saßbach	2010_GZÜV-Styria
25020	14	Schwarzaubach	2007_GZÜV-Styria
55039	14	Schwarzaubach	2010_GZÜV-Styria
55037	14	Stiefing	2010_GZÜV-Styria
55038	14	Stiefing	2010_GZÜV-Styria
59179	14	Stiefing	Grabenland
54743	14	Strasser Mühlgang	2011_GZÜV-Styria
55029	14	Stullneggbach	2010_GZÜV-Styria
24266	14	Sulm	2005_WGEV_Styria
24267	14	Sulm	2005_WGEV_Styria
25019	14	Sulm	2007_GZÜV-Styria
55013	14	Sulm	2010_GZÜV-Styria
55032	14	Sulzbach	2010_GZÜV-Styria
55033	14	Sulzbach	2010_GZÜV-Styria
55016	14	Teipl Bach	2010_GZÜV-Styria
1996	14	Voherabach	MMI_WGEV_Steiermark_2003
24270	14	Voherabach	2005_WGEV_Styria
59176	14	Voherabach	Ditch country
59177	14	Wellingbach	Grabenland
54745	14	Zachenbach	2011_GZÜV-Styria
55014	14	Zirknitzbach	2010_GZÜV-Styria

8.6 Data set for genetic analyses

Table 23: List of sequences generated and downloaded from Bold/GenBank (COI)

Alignment	Bold	GenBankAccession	Family	Genus	Species	Subfamily	Water
Austria			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria2			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria3			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria4			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria5			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria6			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria7			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Austria8			Limnephilidae	Ironoquia	dubia	Dicosmoecinae	Pfaffenbach
Finland1	TRIF958-13	KX291838	Limnephilidae	Ironoquia	dubia	Dicosmoecinae	
Estonia1	TRIF923-13	KX294032	Limnephilidae	Ironoquia	dubia	Dicosmoecinae	
Estonia2	TRIF924-13	KX291191	Limnephilidae	Ironoquia	dubia	Dicosmoecinae	
USA1	PKCAD757-08	HM102849	Limnephilidae	Ironoquia	punctatissima	Dicosmoecinae	
Canada1	RRINV3865-15	KT707415	Limnephilidae	Ironoquia	punctatissima	Dicosmoecinae	
USA	UMNEA408-08	KX106638	Limnephilidae	<i>Dicosmoecus</i>	atripes	Dicosmoecinae	
Canada1	RMCAD495-08	HM382242	Limnephilidae	<i>Dicosmoecus</i>	atripes	Dicosmoecinae	
Austria1	HMCAD322-08	KX104764	Phryganeidae	Phryganea	bipunctata		
Germany	FBAQU273-09	GU713165	Phryganeidae	Phryganea	bipunctata		
Austria1	ABOLD047-16		Papilionidae	<i>Papilio</i>	machaon		

Alignment	Bold	GenBank Accession	Family	Genus	Species	Subfamily	Water
Austria1			Siphonuridae	Siphonurus	aestivalis		Pfaffenbach
Austria4			Siphonuridae	Siphonurus	aestivalis		Pfaffenbach
Austria8			Siphonuridae	Siphonurus	aestivalis		Pfaffenbach
Austria3			Siphonuridae	Siphonurus	armatus		Pfaffenbach
Austria4			Siphonuridae	Siphonurus	armatus		Pfaffenbach
Austria5			Siphonuridae	Siphonurus	armatus		Pfaffenbach
Austria6			Siphonuridae	Siphonurus	armatus		Pfaffenbach
Austria8			Siphonuridae	Siphonurus	armatus		Pfaffenbach
Germany1	FBAQU854-10	HQ563252	Siphonuridae	Siphonurus	armatus		
Germany	GBEPT874-14	KY262425	Siphonuridae	Siphonurus	armatus		
Norway1	FWONN047-16		Siphonuridae	Siphonurus	aestivalis		
Norway2	FWONN051-16		Siphonuridae	Siphonurus	aestivalis		
France1	GBMIN65420-17	MF458829	Siphonuridae	Siphonurus	croaticus		
Germany	GBEPT872-14	KY261537	Siphonuridae	Siphonurus	croaticus		
Norway1	EPHF096-12	KC158572	Siphonuridae	<i>Parameletus</i>	<i>chelifer</i>		
Canada1	CUMAY045-09	HM431544	Siphonuridae	<i>Parameletus</i>	<i>chelifer</i>		
Australia1	GBMIN65375-17	KX868576	Oniscigastridae	Tasmanophlebia	lacustris		
Austria1	ABOLD047-16	No entry	Papilionidae	<i>Papilio</i>	machaon		

Alignment	Bold	GenBank Accession	Family	Genus	Species	Subfamily	Water
Austria1	VoEc01		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria2	VoEc02		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria3	VoEc03		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria4	VoEc04		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria5	VoEc05		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria6	VoEc06		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria7	VoEc07		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria8	VoEc08		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria9	VoEc09		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria10	VoEc10		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria11	VoEc11		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria12	VoEc12		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria13	VoEc13		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria14	VoEc14		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria15	VoEc15		Heptageniidae	Ecdyonurus	torrentis		Vocherabach
Austria16	VoEc16		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria17	VoEc17		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria18	VoEc18		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria19	VoEc19		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria20	VoEc20		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria21	VoEc21		Heptageniidae	Ecdyonurus	subalpinus		Vocherabach
Austria22	WeEc01		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria23	WeEc02		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria24	WeEc03		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria25	WeEc04		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria26	WeEc05		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria27	WeEc06		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria28	WeEc07		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria29	WeEc08		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria30	WeEc09		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria31	WeEc10		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Austria32	WeEc11		Heptageniidae	Ecdyonurus	subalpinus		Wellingbach
Germany1	FBAQU806-10	HQ563217	Heptageniidae	Ecdyonurus	torrentis		
Germany1	GST008-15	KY262204	Heptageniidae	Ecdyonurus	subalpinus		
Germany	GST012-15	KY261832	Heptageniidae	Ecdyonurus	subalpinus		
Germany3	GST010-15	KY261431	Heptageniidae	Ecdyonurus	subalpinus		
Germany	GST019-15	KY261994	Heptageniidae	Ecdyonurus	subalpinus		
Germany	GST006-15	KY262296	Heptageniidae	Ecdyonurus	subalpinus		
Germany	GST004-15	KY261688	Heptageniidae	Ecdyonurus	subalpinus		
Germany	GST005-15	KY262199	Heptageniidae	Ecdyonurus	subalpinus		
Germany	GST007-15	KY262335	Heptageniidae	Ecdyonurus	subalpinus		
Germany9	GST009-15	KY262386	Heptageniidae	Ecdyonurus	subalpinus		
Bulgaria1	BGMAY044-10		Heptageniidae	Ecdyonurus	venosus		
Italy1	GBA22907-15	LN734727	Heptageniidae	Ecdyonurus	venous		
Germany1	FBAQU1104-12	KY262191	Heptageniidae	Ecdyonurus	venosus		
Germany	FBAQU801-10	HQ563212	Heptageniidae	Ecdyonurus	helveticus		
Italy1	GBMIN64741-17	LT626111	Heptageniidae	Ecdyonurus	helveticus		
Germany	GBEPT471-14	KY262361	Heptageniidae	Electrogena	ujhelyi		
Germany	GBEPT874-14	KY262425	Siphonuridae	Siphonurus	armatus		
Norway1	FWONN047-16		Siphonuridae	Siphonurus	aestivalis		
Austria1	ABOLD047-16		Papilionidae	<i>Papilio</i>	machaon		